

Topological Quantum Computing

Exploring Majorana Zero Modes or something subtitley

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Exploring Majorana Zero Modes or something
subtitley

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Abstract

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Chapter 1

Introduction

Quantum computing offers the possibility of solving classes of problems that are intractable on classical hardware, but this promise is limited by one central difficulty: quantum states are fragile. Decoherence, noise, and imperfect control make it challenging to preserve and manipulate quantum information long enough to perform useful computations. One of the most attractive proposed solutions to this problem is topological quantum computing, where information is stored non-locally in collective quantum states that are intrinsically less sensitive to local perturbations. In this picture, protection is not achieved only through better engineering, but through the structure of the quantum states themselves.

Majorana zero modes have therefore attracted broad interest as candidate building blocks for fault-tolerant quantum computing. In condensed-matter systems, they appear not as fundamental particles, but as emergent zero-energy quasiparticles in topological superconductors [2]. Their most important feature is that a pair of spatially separated Majorana modes can encode a non-local fermionic degree of freedom. Because local perturbations cannot easily access this information, such states offer a route toward more robust quantum operations. In addition, Majorana modes are predicted to obey non-Abelian exchange statistics, meaning that exchanging them can implement quantum gates through braiding rather than through purely local control pulses.

Although the basic theoretical picture is well established, realizing and controlling Majorana modes in realistic devices remains a major challenge. Much of the experimental and theoretical effort has therefore focused on engineered hybrid systems, where conventional superconductivity, tunneling, spin structure, and confinement can be combined to produce effective topological behavior [1]. Among the simplest of these platforms is the quantum-dot–superconductor–quantum-dot setup, often referred to as the Poor Man’s Majorana system. This device can be understood as a minimal analogue of the Kitaev chain: hopping between dots plays the role of the kinetic term, non-local pairing induced by the superconductor plays the role of effective p -wave pairing, and gate-controlled dot energies act as local chemical potentials.

The appeal of this platform is twofold. First, it is conceptually simple enough to connect directly to the Kitaev-chain picture, which makes it well suited for theoretical analysis. Second, it is experimentally tunable: dot energies, effective pairing strengths, and tunnel couplings can all be adjusted in a controlled way. At the same time, this simplicity comes with limitations. In finite and interacting systems, one should be careful not to overstate the meaning of low-energy degeneracies or Majorana-like observables. The relevant question is therefore not only whether a model displays signatures associated with Majorana physics, but whether those signatures survive under

interactions and whether they remain useful for braiding operations.

This thesis studies that question in interacting chains of coupled quantum dots. More specifically, I investigate whether an interacting n -dot Poor Man’s Majorana chain can be numerically optimized to produce low-energy states with the expected Majorana-like characteristics, and whether those optimized states support a useful braiding protocol. The goal is not to claim exact topological protection in a finite microscopic model. Rather, the aim is to identify parameter regimes where the low-energy structure is sufficiently close to the Majorana ideal to make the physical interpretation meaningful and the braiding behavior nontrivial.

To do this, I model the system with an interacting quantum-dot Hamiltonian containing on-site energies, nearest-neighbor hopping, induced pairing, and Coulomb repulsion. The search for promising parameter regimes is formulated as an optimization problem, where the loss function is built from several physically motivated Majorana criteria: near-degeneracy between parity sectors, finite low-energy gaps, small charge differences between parity partners, and a Majorana-polarization profile consistent with edge-localized modes. Once optimized configurations are found, I analyze their parity-resolved spectra and local diagnostics, reconstruct effective Majorana operators from the low-energy manifold, and compare an ideal four-Majorana braiding protocol with a braid generated by projected microscopic junction operators.

The main results show a clear distinction between weakly and strongly interacting regimes. The optimization reproduces the known non-interacting two-dot sweet spot and, for longer chains, finds more inhomogeneous parameter profiles with strong central structure in the on-site energies and pairing amplitudes. Among the interacting cases studied here, the three-dot system at weak interaction emerges as the most convincing candidate: it combines a very small ground-state splitting, a clear gap to excited states, and a spatial Majorana profile concentrated at the chain edges. In the lowest projected manifold, the corresponding microscopic braid agrees remarkably well with the ideal effective-model braid. However, this agreement deteriorates as stronger interactions and higher-energy sectors are included, indicating that useful braiding depends sensitively on how well the exchange remains confined to a clean low-energy subspace.

The broader motivation of the thesis is therefore not only to identify candidate Majorana-like configurations, but also to clarify the boundary between effective low-energy success and microscopic breakdown. In that sense, the work is positioned between abstract Majorana models and realistic device-level questions: the optimization tells us which interacting parameter regimes are promising, while the braiding analysis tests whether those regimes support the operation that ultimately matters for topological quantum computing.

The remainder of the thesis is organized as follows. Chapter 2 presents the theoretical background, beginning with superconductivity and the Bogoliubov-de Gennes formalism, then introducing the Kitaev chain, the quantum-dot realization of Poor Man’s Majoranas, and the basic logic of braiding. Chapter 3 describes the numerical methods used to optimize the interacting dot chains, analyze the resulting low-energy states, and construct the braiding protocol. Chapter 4 presents the optimization results, local Majorana diagnostics, and braiding benchmarks in both the idealized and projected microscopic models. Chapter 5 discusses the physical interpretation of these findings, their limitations, and the role of interactions in shaping the low-energy manifold. Finally, Chapter 6 summarizes the main conclusions and outlines possible directions for future work.

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Chapter 2

Theory

2.1 Majorana Zero Modes and Topological Motivation

Topological quantum computing is an approach to fault-tolerant quantum computation in which quantum gates are implemented by braiding non-Abelian anyons. The central idea is that information can be encoded non-locally in a degenerate subspace, so that local perturbations have limited access to the encoded quantum state. In this setting, the result of a gate depends on the topology of the braid rather than on microscopic details of the path.

Anyons are quasiparticles that, so far, have only been observed in effectively two-dimensional systems. In three spatial dimensions, identical particles are classified as either bosons or fermions, corresponding to symmetric and antisymmetric exchange statistics. In two dimensions, however, the topology of particle exchange allows for a richer possibility: exchanging two particles can produce statistics that are neither bosonic nor fermionic

<https://sciencex.com/wire-news/347971706/finally-anyons-reveal-their-exotic-quantum-properties.html>

. In particular, exchanging two identical particles twice is not necessarily topologically equivalent to doing nothing, because the exchange path cannot always be continuously shrunk to a point without crossing another particle. Particles with such exchange statistics are called anyons, and they are classified as Abelian or non-Abelian depending on how the quantum state transforms under exchange. Abelian anyons acquire only a phase factor, while non-Abelian anyons transform the state within a degenerate subspace. This latter property is what makes non-Abelian anyons especially interesting for quantum computation<https://www.nature.com/articles/nphys767>.

A collection of non-Abelian anyons with fixed positions and fixed local quantum numbers can possess a non-trivial topological degeneracy. Quantum information can then be encoded in this degenerate subspace, while braiding the anyons implements unitary transformations within it. Since the resulting transformation depends only on the braid topology, and not on small local deformations of the path, this provides a possible route toward fault-tolerant quantum gates. In suitable schemes, braiding operations can be combined with additional ingredients to implement the gates needed for quantum computation.

<https://www.nature.com/articles/npjqi20151>

One of the most promising candidates for realizing non-Abelian exchange statistics

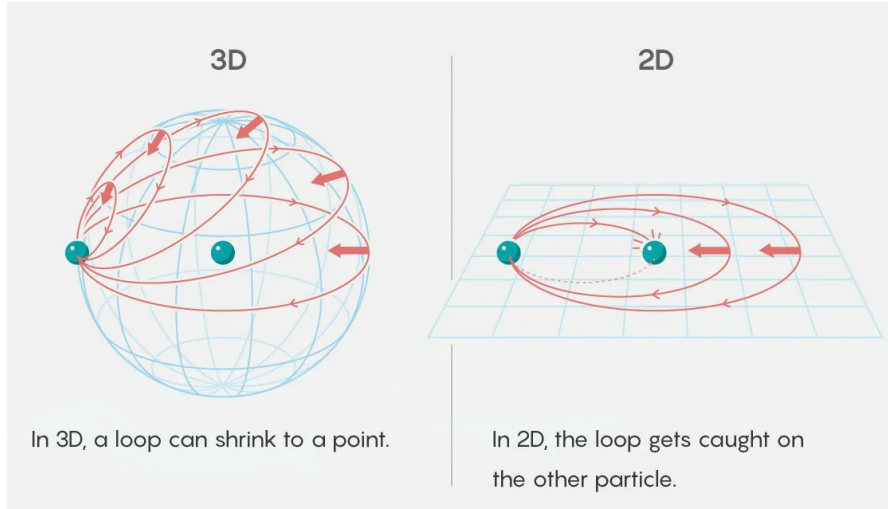


Figure 2.1: Visualization of the difference between particle exchange in two and three dimensions. In three dimensions, the exchange loop can be shrunk to a point, making a double exchange topologically equivalent to leaving the particles unchanged. In two dimensions, the corresponding loop cannot be shrunk to a point without crossing the particle worldlines. This topological distinction allows anyons to exhibit non-trivial exchange statistics.[https :
//media.wired.com/photos/5ebf0488d58e99fafced8087/master/w_1600%2Cc_limit/Science_Anyons-
graphic – FINAL.jpg](https://media.wired.com/photos/5ebf0488d58e99fafced8087/master/w_1600%2Cc_limit/Science_Anyons-graphic-FINAL.jpg)

in condensed-matter systems is the Majorana zero mode (MZM), predicted to occur in certain topological superconductors. A Majorana zero mode is a zero-energy quasiparticle described by an operator that is equal to its own Hermitian conjugate. In one-dimensional topological superconductors, such modes can appear at the ends of the system. Unlike an ordinary fermionic mode, a single MZM can be viewed as “half” of a Dirac fermion: two Majorana operators can be combined to form one conventional fermionic operator,

$$c = \frac{1}{2}(\gamma_1 + i\gamma_2) \quad (2.1)$$

with the inverse relations

$$\gamma_1 = c + c^\dagger, \quad \gamma_2 = -i(c - c^\dagger). \quad (2.2)$$

Here γ_1 and γ_2 are Majorana operators satisfying

$$\gamma_i = \gamma_i^\dagger, \quad \{\gamma_i, \gamma_j\} = 2\delta_{ij} \quad (2.3)$$

The non-Abelian character of MZMs is reflected in the fact that exchanging two of them produces a non-trivial unitary transformation on the degenerate ground-state manifold<https://www.nature.com/articles/npjqi20151>.

The primary motivation for studying MZMs and their braiding properties is their combination of non-local encoding and spectral protection. When a fermionic state is formed from two spatially separated MZMs, local environmental noise cannot easily determine or flip the encoded state, because such an error would require coupling to both separated Majorana components. Therefore, local perturbations are strongly suppressed as a source of decoherence<https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.80.1083>. MZMs also give rise to ground-state degeneracies. A system with $2n$ Majorana zero

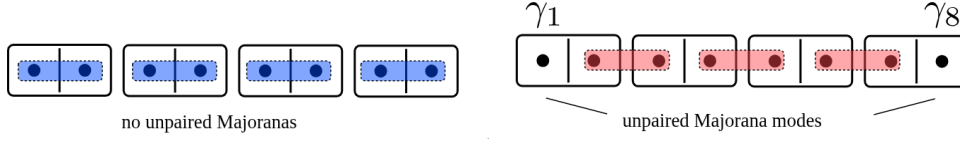


Figure 2.2: Schematic representation of the Kitaev chain in its two distinct phases. Left: In the trivial phase, Majorana modes on the same site pair to form conventional fermions, leaving no unpaired modes. Right: In the topological phase, Majoranas on adjacent sites couple, leaving unpaired modes at the chain ends, which combine into a non-local fermionic mode. **Topocondmat**

modes spans a 2^n -dimensional fermionic Hilbert space associated with the occupations of n non-local fermions. If the total fermion parity is fixed, the physical ground-state degeneracy is reduced to 2^{n-1} . In either case, the useful low-energy manifold must be separated from excited states by a finite spectral gap, which protects the encoded information against thermal excitations. <https://arxiv.org/abs/cond-mat/0010440>.

The remaining question is how such Majorana operators can emerge from an ordinary fermionic Hamiltonian. The Kitaev chain provides the minimal model where this mechanism can be seen explicitly, and it also gives the conceptual starting point for the Poor Man's Majorana systems studied in this thesis.

2.2 Kitaev Chain and Poor Man's Majorana Systems

Before discussing the Majorana models used in this thesis, we first introduce their theoretical origin: the Kitaev chain. The Kitaev chain, or Kitaev-Majorana chain, is a simplified model for a topological superconductor. First proposed by Alexei Kitaev in 2000 <https://arxiv.org/pdf/cond-mat/0010440>, the Kitaev chain is a one-dimensional superconducting system that can host Majorana zero modes at its ends under certain conditions, illustrated in Figure 2.2. The model is defined on a lattice of spinless fermions with nearest-neighbor hopping and superconducting pairing. The Hamiltonian of the Kitaev chain is

$$H = -\mu \sum_{j=1}^N \left(c_j^\dagger c_j - \frac{1}{2} \right) - t \sum_{j=1}^{N-1} (c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j) + \Delta \sum_{j=1}^{N-1} (c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger), \quad (2.4)$$

Each term in Eq. 2.4 has a clear physical interpretation. The first term represents the chemical potential μ , which controls the filling of the fermionic states. The second term describes the hopping of fermions between neighboring sites with amplitude t . The third term represents p -wave superconducting pairing, where pairs of fermions are created or annihilated on adjacent sites with amplitude Δ .

As shown schematically in Fig. 2.2, the Kitaev chain has two distinct phases depending on the parameters μ , t , and Δ . In the trivial phase, the Majorana modes on the same site pair up to form conventional fermions, leaving no unpaired modes. In the topological phase, Majoranas on adjacent sites couple, leaving unpaired modes at the chain ends. These unpaired Majoranas combine into a non-local fermionic mode that can be occupied or empty without costing energy, leading to a twofold degeneracy in the ground state. This degeneracy is protected as long as local perturbations do not close the bulk gap or directly hybridize the edge modes.

The topological phase of the Kitaev chain is the basic model for the Majorana zero modes studied in this thesis. The goal is then to understand how similar Majorana-like

physics can be engineered in more realistic systems.

2.2.1 Bulk Spectrum and Topological Phase

In order to determine when the Kitaev chain is topological, we need to consider the properties of the bulk spectrum.

For a translationally invariant chain, we can perform a Fourier transform to momentum space, which allows us to analyze the energy spectrum and identify the conditions for the topological phase. The Fourier transform is given by

$$c_j = \frac{1}{\sqrt{N}} \sum_p e^{ipaj} c_p \quad (2.5)$$

This transformation allows the Hamiltonian to be expressed in terms of momentum-space operators c_p and $c_p^\dagger$. The resulting Hamiltonian in momentum space is

$$H = \sum_p \xi_p \left(c_p^\dagger c_p - \frac{1}{2} \right) - \sum_p t \cos pa + \Delta \sum_p \left(c_p^\dagger c_{-p}^\dagger e^{ipa} + c_{-p} c_p e^{-ipa} \right), \quad (2.6)$$

with $\xi_p = -\mu - 2t \cos pa$ and a being the lattice spacing. In the Bogoliubov–de Gennes formalism, the quasiparticle spectrum becomes

$$E_{p,\pm} = \pm \sqrt{(\mu + 2t \cos pa)^2 + 4\Delta^2 \sin^2 pa}. \quad (2.7)$$

Even though the Bogoliubov–de Gennes formalism describes a non-interacting quasiparticle problem, it still provides the basic topological picture that the interacting models in this thesis build on. In Fig. 2.3, we can see how the spectrum changes as the chemical potential μ is tuned. The presence of a gap is crucial for the stability of the Majorana modes, and the closing and reopening of this gap signals a topological phase transition.

The quasiparticle spectrum is gapped for most parameter values, but the gap closes when

$$|\mu| = 2|t|.$$

This is also visible in Fig. 2.3. The gap closing marks the transition between the trivial and topological phases. When μ crosses this boundary, the ground state changes its topological character without requiring ordinary symmetry breaking [Qtech](#).

Quasiparticle Band Structure for Different Chemical Potentials

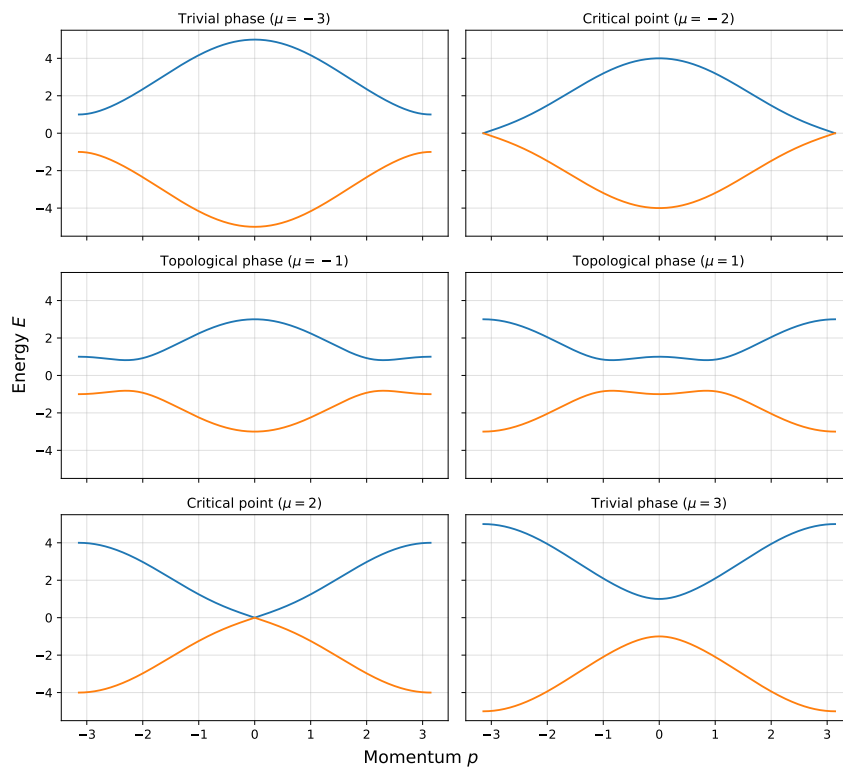


Figure 2.3: Quasiparticle energy spectrum of the Kitaev chain for different chemical potentials μ . Left: In the trivial phase ($|\mu| > 2|t|$), the spectrum is fully gapped with no zero-energy modes. Right: In the topological phase ($|\mu| < 2|t|$), the gap closes and reopens, signaling the emergence of zero-energy edge modes.

2.2.2 Majorana Representation and Zero Modes

Looking back at Fig. 2.2, we want a representation that makes the unpaired edge modes more explicit. To do so, we express the fermionic operators in terms of Majorana operators. For each site j , we define two Majorana operators γ_j^A and γ_j^B as follows:

$$c_j = \frac{1}{2}(\gamma_j^A + i\gamma_j^B), \quad c_j^\dagger = \frac{1}{2}(\gamma_j^A - i\gamma_j^B), \quad (2.8)$$

where $\gamma_j^{A,B}$ satisfy $\{\gamma_j^\alpha, \gamma_k^\beta\} = 2\delta_{jk}\delta_{\alpha\beta}$ and $(\gamma_j^\alpha)^\dagger = \gamma_j^\alpha$. In the trivial phase, the Majorana operators pair on the same site. In the topological phase, and especially at the ideal point $t = \Delta$ and $\mu = 0$, γ_j^B couples to γ_{j+1}^A . This leaves two edge modes unpaired, γ_1^A and γ_N^B , at the chain ends. These two unpaired Majoranas define a non-local fermionic mode,

$$f = \frac{1}{2}(\gamma_1^A + i\gamma_N^B), \quad (2.9)$$

which satisfies $\{f, f^\dagger\} = 1$ and costs zero energy to occupy. This non-local fermionic mode creates the basis for the exploration of Majorana physics in this thesis, and it is the key to understanding how Majorana modes can be used for quantum computation. The occupation of this non-local mode corresponds to a twofold degeneracy in the ground state, which is protected as long as local perturbations do not close the bulk gap or directly hybridize the two edge modes. This non-local encoding of information is what gives Majorana modes their potential for fault-tolerant quantum computation, as local errors cannot easily access or change the encoded state. [Qtech](https://arxiv.org/pdf/cond-mat/0010440)

2.2.3 Poor Man's Majorana Systems

The ideal Kitaev chain is a useful theoretical model, but it is not directly realizable in most physical systems, as it ideally prefers to be infinite, homogeneous, and non-interacting. The reason for this, is that the Majorana modes are localized at the edges of the chain, and their energy splitting decreases exponentially with the chain length $\exp(-\frac{L}{l_0})$ <https://arxiv.org/pdf/cond-mat/0010440>. This would make the Majorana modes robust against any local perturbation. However, in practice we have to work with finite systems, and the Majorana modes will generally have a small energy splitting due to their overlap. However, a Poor mans Majorana system is a finite system, which under the right parameter conditions, can mimic the low-energy physics of a Kitaev chain, including the presence of Majorana-like modes, in a much more compact and experimentally accessible setup.

While the Kitaev chain is the minimal theoretical model, it assumes spinless fermions and intrinsic p -wave pairing. Real superconductors are usually spinful and dominated by conventional s -wave pairing. The Poor Man's Majorana (PMM) platform is a compact engineered system whose low-energy behavior can mimic a short Kitaev chain. The simplest version is a double quantum dot coupled through a superconducting segment. Each dot acts as a site, elastic cotunneling produces effective hopping, and crossed Andreev reflection produces an effective non-local pairing term.

The effective two-dot Hamiltonian can be written as

$$H = \sum_{i=L,R} \epsilon_i n_i + t(d_L^\dagger d_R + d_R^\dagger d_L) + \Delta(d_L^\dagger d_R^\dagger + d_R d_L) + U n_L n_R, \quad (2.10)$$

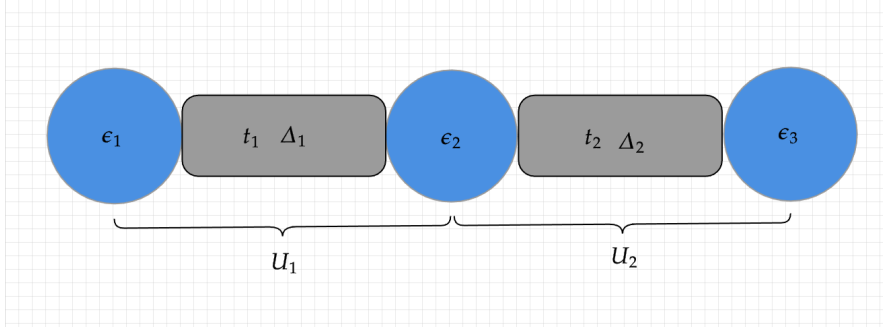


Figure 2.4: Schematic representation of the PMM setup. The three quantum dots are represented by the blue circles, and the superconducting segments are represented by the grey rectangles. The tunable parameters of the system, such as the dot energy levels, ECT, CAR and the Coulomb interaction, are indicated by their respective labels. By tuning these parameters, we can access different regimes of the system and identify the sweet spots where Majorana-like modes emerge.

where $d_{L/R}^\dagger$ and $d_{L/R}$ are creation and annihilation operators on the left and right dots, n_i is the number operator, $\epsilon_{L,R}$ are the dot energy levels, t is the elastic cotunneling amplitude, Δ is the crossed Andreev reflection amplitude, and U is the interdot Coulomb interaction energy. The parameters map directly onto the Kitaev-chain language:

- **Elastic cotunneling (ECT):** An electron tunnels coherently from one dot to the other through a virtual process in the superconductor. This produces the effective hopping amplitude t .
- **Crossed Andreev reflection (CAR):** A Cooper pair splits, with one electron entering each dot. This produces the non-local pairing amplitude Δ .
- **On-site energy:** The dot energies ϵ_L and ϵ_R play the role of local chemical potentials.

At the ideal non-interacting sweet spot, $|\epsilon_L| = |\epsilon_R| = 0$ and $|t| = |\Delta|$, the two-dot system develops Majorana-like states localized primarily on opposite dots. Since the system is finite and lacks a true bulk, these states are not topologically protected in the same sense as an infinite Kitaev chain. This is why they are referred to as Poor Man's Majoranas [Qtech](#).

For the purposes of this thesis, the PMM idea is generalized from two dots to interacting n -dot chains. A schematic representation is shown in Figure 2.4. Adding more dots introduces more tunable parameters and more possible low-energy structure, but it also makes the analytic sweet-spot problem much harder. This motivates the numerical optimization strategy discussed later in the thesis.

2.3 Diagnostics of Majorana-like States

In a finite PMM system, the presence of low-energy or near-zero-energy states is necessary, but not sufficient, evidence for Majorana physics. As discussed in the previous section, the ideal Kitaev chain has zero-energy Majorana modes at its ends, producing a degenerate low-energy manifold. In a short and finite PMM chain, however, the Majorana modes can overlap, hybridize, and split away from zero energy. Other ordinary low-energy states can also appear because the dot energies, tunnel couplings, pairing

amplitudes, and interactions are all tunable.

For this reason, the identification of Majorana-like states must rely on several complementary diagnostics rather than a single spectral feature. The diagnostics used here can be divided into two broad groups. Spectral diagnostics test whether the low-energy manifold has the expected parity degeneracy and excitation gap. Local diagnostics test whether the corresponding states are non-local, locally indistinguishable, and Majorana-like in their particle-hole structure. Together, these quantities define what is meant in this thesis by a promising Majorana-like configuration.

2.3.1 Fermion Parity

The first diagnostic is fermion parity. The mean-field Hamiltonian of a superconductor does not conserve particle number, because electrons can be added or removed through Cooper-pair processes. It does, however, conserve the parity of the total particle number. The parity operator can be written as

$$P = (-1)^N = \prod_n (1 - 2c_n^\dagger c_n),$$

where N is the total particle number. The eigenvalues of P are $+1$ for even parity and -1 for odd parity.

In a PMM system, the qubit is encoded in the global fermion parity of the non-local fermionic mode formed by the two Majorana components. At the ideal two-dot sweet spot, where the effective hopping and pairing satisfy $t = \Delta$, the ground state becomes twofold degenerate, with one state in the even-parity sector and one in the odd-parity sector <https://www.nature.com/articles/s41586-022-05585-1>; <https://doi.org/10.1103/PhysRevB.86.134528>. Classifying the spectrum by parity is therefore the first step in identifying candidate Majorana-like states. In this thesis, this parity classification is also used to ask a more detailed question: whether even-odd degeneracy can persist beyond the ground state in finite interacting chains, and what such higher-level degeneracy would imply for robustness.

2.3.2 Energy Degeneracy and Spectral Gap

Once the states have been separated into parity sectors, the next diagnostic is the energy splitting between parity partners. Majorana-like physics requires a near-degenerate pair of states, usually one even and one odd, associated with the occupation of the non-local fermionic mode. In an ideal infinite Kitaev chain, this degeneracy is exact. In a finite PMM chain, the edge modes can overlap, producing a small but nonzero splitting. The goal is therefore not only to find low-energy states, but to find parameter regimes where the even-odd splitting is small compared with all other relevant energy scales.

The excitation gap is equally important. The gap separates the low-energy parity manifold from quasiparticle excitations and suppresses thermal or dynamical leakage out of the computational subspace. In PMM and quantum-dot-chain proposals, the

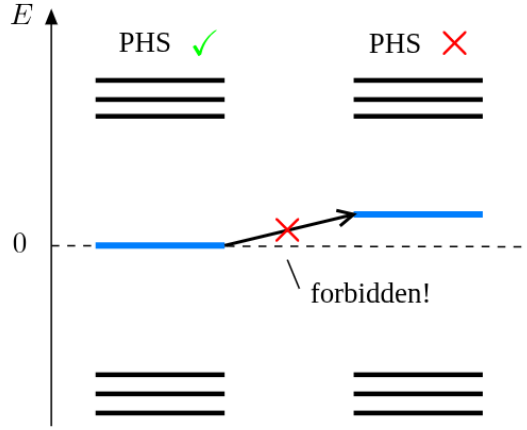


Figure 2.5: Energy spectrum showing how the energy eigenvalues of the system must be symmetric around zero energy due to particle-hole symmetry. [Topocondmat](#)

useful gap is controlled by the same physical couplings that create the effective Kitaev chain, especially the elastic cotunneling and crossed-Andreev pairing amplitudes. If the gap closes, the Majorana-like signature is no longer protected, and the low-energy interpretation becomes unreliable <https://www.nature.com/articles/ncomms1966>. A sizable gap is also essential for the braiding calculations in this thesis, because it justifies projecting the dynamics onto a low-energy subspace.

This thesis also explores whether finite interacting chains can display approximate even-odd degeneracy across several energy levels, not only in the ground state. If such a structure survives, then some Majorana-like features may remain meaningful even when excited states are included in the projected Hilbert space. Figure 2.5 illustrates the particle-hole symmetric structure expected in the idealized superconducting spectrum. In practice, the relevant diagnostic is whether the candidate parity partners are both nearly degenerate and well separated from the rest of the spectrum [Topocondmat](#); <https://www.nature.com/articles/s41586-022-05585-1>.

2.3.3 Local Distinguishability

A defining feature of spatially separated Majorana modes is that the encoded information is non-local. Local measurements should therefore not reveal whether the system is in the even or odd ground state. This principle is central to the topological motivation for Majorana qubits: the qubit state is stored globally, while local perturbations should have only limited access to it <https://doi.org/10.1103/RevModPhys.80.1083>. In the PMM setting, the same idea appears in a finite-system form: the parity qubit can only be fully read out by probing both Majorana components, rather than a single dot alone <https://doi.org/10.1103/PhysRevB.86.134528>.

One way to quantify this property is through the local distinguishability parameter

$$\text{LD} = \frac{1}{N} \sum_{j=1}^N \|\rho_j^o - \rho_j^e\| = \frac{1}{N} \sum_{j=1}^N \|\delta\rho_j\|.$$

Lund1; VikMar; <https://doi.org/10.1103/PhysRevB.110.155436>. Here, $\rho_j^{o/e}$ are the reduced density matrices of site j for the odd and even parity ground states, respectively, and $||\cdot||$ is the trace norm. A value close to zero indicates that the two states are locally indistinguishable, while a larger value indicates that local measurements can tell the two parity states apart.

2.3.4 Majorana Polarization

Local indistinguishability tests whether the information is hidden from local measurements, but it does not by itself say whether the operator connecting the two parity states has the particle-hole structure expected of a Majorana mode. For this, we use the Majorana polarization (MP). The MP is a local measure of how Majorana-like a state is: an ideal Majorana mode should be self-conjugate, with balanced particle and hole character, and should be localized near the ends of the chain. In an ideal PMM configuration, the polarization should therefore be concentrated at the edge dots, with opposite signs on the two ends and little or no weight in the bulk <https://doi.org/10.1103/PhysRevB.110.155436>.

One useful definition is

$$M_\alpha = \frac{W_\alpha^2 - Z_\alpha^2}{W_\alpha^2 + Z_\alpha^2},$$

where

$$W_\alpha = \sum_\sigma \langle o | c_{\alpha\sigma} + c_{\alpha\sigma}^\dagger | e \rangle, \quad Z_\alpha = i \sum_\sigma \langle o | c_{\alpha\sigma} - c_{\alpha\sigma}^\dagger | e \rangle.$$

Here, α denotes the site index, σ is the spin index, and $|o\rangle$ and $|e\rangle$ are the odd- and even-parity ground states. In an ideal sweet spot with two perfectly localized Majoranas, the edge values would satisfy $M_L = -M_R = \pm 1$ [Qtech](#). In the spinless model used later in this thesis, the same idea is applied without the spin sum: the diagnostic checks whether the parity-changing Majorana matrix elements are localized at the chain ends with the expected relative sign.

2.3.5 Charge Expectation and Non-locality

Another diagnostic is the local charge expectation value in the even and odd states,

$$\langle e | n_{L(R)} | e \rangle, \quad \langle o | n_{L(R)} | o \rangle.$$

An ideal Majorana mode has no ordinary local charge identity: it is an equal particle-hole superposition rather than a conventional electron or hole. In a PMM system, this means that the local charge distribution should not reveal the parity state of the non-local fermion. At the sweet spot, the relevant charge signatures become symmetric between the two parity states, corresponding to the “half-electron” or charge-neutral character of the Majorana components [4, 3].

For the diagnostics used here, the important quantity is not the absolute charge on a dot, but the difference between the even- and odd-parity partners. A small charge difference means that the two states share nearly the same local charge distribution,

Visualizing the Loss Landscape for Parameter Search

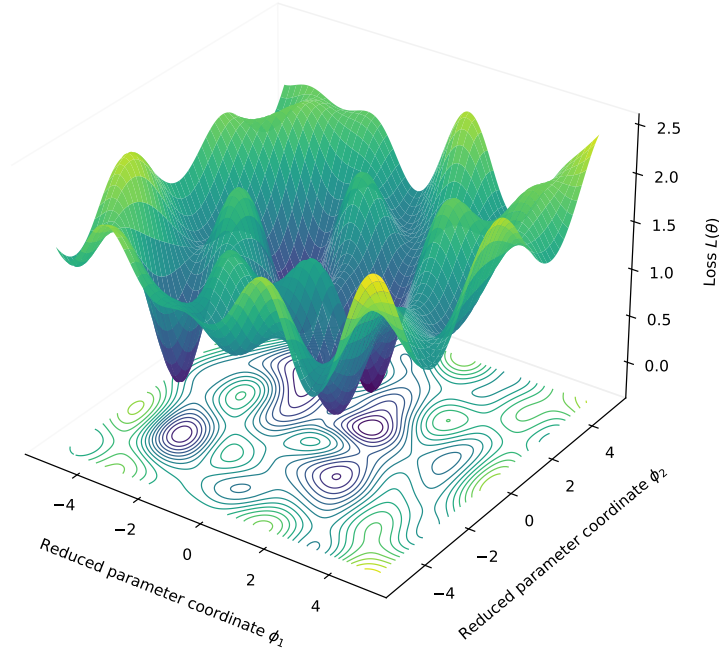


Figure 2.6: Visualization of the parameter space for an n-dot system, showing how multiple parameters and their combinations create a complex landscape with multiple local minima.

which is another way of saying that the parity information is stored non-locally rather than being visible on a single dot.

2.3.6 Optimization Motivation

Together, these diagnostics provide the physical ingredients used later to build the optimization loss function. For a simple two-dot system, the relevant tunable parameters are the dot energies, elastic cotunneling, crossed Andreev reflection, and interaction strength. For longer chains, the number of tunable parameters grows quickly, visualized in Figure 2.6. Since interactions and inhomogeneous parameters make it difficult to identify sweet spots analytically, the diagnostics above can be combined into a loss function and used to search numerically for promising Majorana-like regimes. The resulting loss landscape may contain many local minima, which motivates the repeated-restart and basin-hopping strategy described in the methods chapter.

2.4 Majorana Exchange and Braiding

Braiding is the operation that makes Majorana modes useful for topological quantum computation. The basic idea is to exchange two Majorana modes adiabatically so that the quantum state transforms according to their non-Abelian statistics. Because the transformation is tied to the topology of the exchange, rather than to the exact microscopic path, braiding offers a route to gates that are less sensitive to local control errors. Braiding in the right patterns, with the right systems, can lead to the implementation of useful quantum gates.[SOURCE](#).

In a system of Poor Man’s Majoranas, braiding does not come from physically moving the Majorana modes or any other physical pieces of the setup, but rather by tuning the parameters of the system in a way that mimics the effect of an exchange. What this does, is couple the Majorana modes in a way that produces the same unitary transformation on the low-energy subspace as a physical braid would. This is sometimes called “measurement-only” or “parameter-tuning” braiding, and it relies on the same underlying topological principles as physical braiding, but it is implemented through a sequence of parameter changes rather than through spatial motion.

2.4.1 From Qubit Gates to Fermionic Operators

In quantum computation, quantum gates are unitary operations that manipulate the state of qubits. Quantum gates, or quantum logic gates, are the building blocks of quantum circuits, the same way classical logic gates are for digital circuits. Unlike many classical logic gates, quantum logic gates are reversible, meaning that they can be undone by applying the inverse operation. This reversibility is a fundamental property of quantum mechanics and is essential for quantum computation.

Quantum gates are unitary operations and can be described as unitary matrices, relative to some orthonormal basis. For example, the Pauli gates X , Y , and Z are represented by the following matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2.11)$$

The Pauli gates (X, Y, Z) are fundamental single-qubit gates that correspond to rotations around the x , y , and z axes of the Bloch sphere, respectively. They are also examples of involutory gates, meaning that applying them twice yields the identity operation (e.g., $X^2 = Y^2 = Z^2 = I$).

In order to replicate these quantum logic gates using Majorana modes, we need to understand how to express these gates in terms of fermionic operators. This is because the Majorana modes are described by fermionic operators, and we want to find a way to translate the abstract quantum gates into operations that can be implemented in a Majorana system. By expressing the Pauli gates in terms of fermionic creation and annihilation operators, we can identify how to manipulate the Majorana modes to achieve the desired quantum gate operations. This connection is crucial for implementing quantum computation using Majorana-based qubits.

Let us start by introducing two quantum states $|0\rangle$ and $|1\rangle$, which we can simply represent

as column vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (2.12)$$

These two vectors represent the two possible states of a qubit, where $|0\rangle$ is the state where the qubit is in the "0" state and $|1\rangle$ is the state where the qubit is in the "1" state. These states form the computational basis for a single qubit. The Pauli gates act on these basis states as follows:

$$\begin{aligned} X|0\rangle &= |1\rangle, & X|1\rangle &= |0\rangle, \\ Y|0\rangle &= i|1\rangle, & Y|1\rangle &= -i|0\rangle, \\ Z|0\rangle &= |0\rangle, & Z|1\rangle &= -|1\rangle. \end{aligned}$$

To put these operations in the context of fermionic systems, we can see that the X gate corresponds to a bit-flip operation, or in other words, it annihilates the "0" state and creates the "1" state, and vice versa. The Y gate also flips the states but introduces a phase factor of i or $-i$, meaning it combines a bit-flip with a phase shift. The Z gate, on the other hand, does not change the state but applies a phase of -1 to the $|1\rangle$ state while leaving the $|0\rangle$ state unchanged.

Putting these operations in the context of fermionic operators, we can express the Pauli gates in terms of creation and annihilation operators for a two-site system. For example, if we have fermionic operators c_1 and c_2 corresponding to two sites, we can write:

$$\begin{aligned} X &= c_1^\dagger c_2 + c_2^\dagger c_1, \\ Y &= -i(c_1^\dagger c_2 - c_2^\dagger c_1), \\ Z &= c_1^\dagger c_1 - c_2^\dagger c_2. \end{aligned} \quad (2.13)$$

We can now see how easily the quantum logic gates can be transformed into fermionic language, which is the first step towards understanding how to implement these gates in a Majorana system. By expressing the gates in terms of fermionic operators, we can identify the necessary interactions and couplings in a Majorana-based setup that would allow us to perform the desired quantum gate operations. This is crucial for the development of Majorana-based quantum computation, as it provides a clear pathway for translating abstract quantum operations into physical manipulations of Majorana modes.

2.4.2 Jordan-Wigner Transformation

Jordan-Wigner transformation is a mathematical technique used to map spin operators to fermionic creation and annihilation operators. Proposed by Pascual Jordan and Eugene Wigner in 1928 <https://link.springer.com/article/10.1007/BF01331938>, where they originally introduced it to solve the one-dimensional spin-1/2 chain problem, but later was extended to two-dimensional systems and beyond. The transformation is particularly useful in the study of quantum many-body systems, as it allows for the analysis of spin systems using fermionic techniques.

The Jordan-Wigner transformation is useful here because it lets us represent qubit or spin operators in terms of fermionic operators while preserving the correct anticommutation

relations between different sites. This is not only a formal point: in the numerical braiding calculations, Majorana operators from different sites and subsystems must be embedded into a common Hilbert space with the correct fermionic algebra. The Jordan-Wigner strings provide this bookkeeping. The transformation allows us to express the spin-1/2 Pauli operators in terms of fermionic creation and annihilation operators [https : //futureofmatter.com/assets/fermions_and_jordan_wigner.pdf](https://futureofmatter.com/assets/fermions_and_jordan_wigner.pdf). For a chain of spin-1/2 particles, the transformation is given by:

$$\begin{aligned}\sigma_j^+ &= \frac{1}{2}(\sigma_j^x + i\sigma_j^y) \equiv f_j^\dagger, \\ \sigma_j^- &= \frac{1}{2}(\sigma_j^x - i\sigma_j^y) \equiv f_j, \\ \sigma_j^z &= 2f_j^\dagger f_j - 1,\end{aligned}$$

Leaving us with the correct same-site fermionic relations, but operators on different sites commute instead of anticommute. To fix this, we take a chain of fermions, and define a new set of fermionic operators:

$$\begin{aligned}a_j^\dagger &= e^{i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} \cdot f_j^\dagger \\ a_j &= e^{-i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} \cdot f_j, \\ a_j^\dagger a_j &= f_j^\dagger f_j\end{aligned}$$

The exponential term, can be simplified to a string of σ^z operators, giving us

$$e^{\pm i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} = \prod_{k=1}^{j-1} (-\sigma_k^z). \quad (2.14)$$

The transformed spin operators now satisfy the correct fermionic anticommutation relations:

$$\begin{aligned}\{a_j^\dagger, a_k\} &= \delta_{jk}, \\ \{a_j^\dagger, a_k^\dagger\} &= 0, \\ \{a_j, a_k\} &= 0.\end{aligned}$$

The Jordan-Wigner transformation is a powerful tool for analyzing spin systems using fermionic techniques, and it provides a clear example of how quantum gates can be expressed in terms of fermionic operators. By applying this transformation, we can gain insights into the behavior of spin systems and their connections to fermionic systems, which is essential for understanding the physics of Majorana modes and their applications in quantum computation.

2.4.3 Pauli Matrices in the Majorana Representation

The reason for rewriting the Pauli operators in terms of Majorana bilinears is that braiding acts naturally on Majorana operators, while the gate implemented by the braid is usually compared with a Pauli operation on an encoded qubit. By expressing the Pauli gates in terms of Majorana bilinears, we can directly relate the physical operations performed on the Majorana modes to the abstract quantum gates that we want to implement. This connection is important to both understand and implement quantum

gates in a Majorana-based quantum computing platform. We start by considering four Majorana operators, grouped into two ordinary fermions:

$$\begin{aligned} c_1 &= \frac{1}{2}(\gamma_0 + i\gamma_1), & c_1^\dagger &= \frac{1}{2}(\gamma_0 - i\gamma_1), \\ c_2 &= \frac{1}{2}(\gamma_2 + i\gamma_3), & c_2^\dagger &= \frac{1}{2}(\gamma_2 - i\gamma_3). \end{aligned}$$

These four Majorana operators γ_0 , γ_1 , γ_2 , and γ_3 span a four-dimensional Hilbert space, which can be thought of as the state space of two fermionic modes. By fixing the total fermion parity, we can select a two-dimensional subspace that serves as the computational basis for a qubit. Within such a fixed-parity subspace, we can represent the Pauli operators as bilinears of the Majorana operators. Specifically, we can write:

$$\begin{aligned} X &= i\gamma_0\gamma_3 = -i\gamma_1\gamma_2, \\ Y &= -i\gamma_0\gamma_2 = -i\gamma_1\gamma_3, \\ Z &= i\gamma_0\gamma_1 = i\gamma_2\gamma_3. \end{aligned}$$

These expressions can be derived by substituting the fermionic operators in Eq. 2.13 with the Majorana representation of the fermionic operators. The resulting bilinears of Majorana operators correspond to the Pauli gates acting on the encoded qubit.

Thus, controlled Majorana couplings of the form $i\gamma_i\gamma_j$ act as Pauli operations on the encoded qubit. This is the operator-level connection between Majorana braiding and gate implementation. <https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.80.1083>; <https://arxiv.org/abs/1202.1293>; <https://scipost.org/SciPostPhys.3.3.021>

2.4.4 Majorana Exchange Gates

An anticlockwise exchange of two Majoranas γ_i and γ_j is represented by the unitary

$$B_{ij} = \exp\left(-\frac{\pi}{4}\gamma_i\gamma_j\right) = \frac{1}{\sqrt{2}}(1 - \gamma_i\gamma_j),$$

up to orientation conventions and an overall phase. With this convention, the unitary acts on the Majorana operators as

$$\begin{aligned} B_{ij}\gamma_i B_{ij}^\dagger &= \gamma_j, \\ B_{ij}\gamma_j B_{ij}^\dagger &= -\gamma_i, \\ B_{ij}\gamma_k B_{ij}^\dagger &= \gamma_k, \quad k \neq i, j. \end{aligned}$$

The important point is that exchanges are generated by Majorana bilinears. Therefore, the same $i\gamma_i\gamma_j$ operators that represent Pauli operations in the encoded qubit space also determine the target braid gates used later in this thesis. For example, squaring an exchange produces a parity-dependent bilinear operation,

$$B_{ij}^2 = -\gamma_i\gamma_j,$$

up to phase and orientation conventions. This is why the numerical braid unitary can be checked by comparing its action on the Majorana operators and by comparing its projected parity blocks with the expected target gate.

<https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.80.1083>; <https://arxiv.org/abs/1202.1293>

The remaining question is how to compute the unitary generated by an adiabatic coupling protocol, which is the role of Kato transport in the next section.

2.5 Projected Subspaces, Kato Transport and Braid Unitaries

This section connects the high-level topological theory of braiding with the microscopic details of the PMM system. The goal is to understand how to compute the braid unitary generated by a time-dependent parameter protocol, and how to compare it with the ideal target gate. In these systems, braiding is not performed by physically moving particles through space, but rather by tuning Hamiltonian parameters in a way that mimics the effect of an exchange.

An adiabatic parameter protocol defines a path in the space of Hamiltonians. If the relevant low-energy subspace remains isolated along this path, the geometric transport of that subspace gives the braid unitary. The Kato formulation provides a convenient way to compute this unitary without having to track individual eigenstates and their arbitrary phases. By projecting the dynamics onto the low-energy subspace defined by the Majorana-like states, we can extract the effective braid unitary and compare it with the ideal target gate expressed in terms of Majorana bilinears.

2.5.1 Kato Transport in a Degenerate Subspace

The Berry phase is a geometric phase acquired by a quantum state during adiabatic evolution around a closed loop in parameter space. For a non-degenerate eigenstate, the state returns to itself up to a phase. This phase has a dynamical part, which depends on the time scale and the energy of the state, and a geometric part, which depends only on the path in parameter space. The geometric contribution is the Berry phase.

For a degenerate subspace, the situation is richer. The adiabatic evolution can mix states within the degenerate manifold, so the Berry phase becomes a matrix rather than a scalar phase. In this thesis, this non-Abelian Berry matrix is what determines the braid unitary acting on the Majorana qubit space.

The Kato formulation provides a convenient way to compute this adiabatic transport. Instead of tracking individual eigenvectors and their phases, one works directly with the projector $P(t)$ onto the relevant low-energy subspace. In the convention used in the numerical implementation, the Kato evolution satisfies

$$\partial_t U_t = [P, \partial_t P] U_t, \quad U_0 = I.$$

<https://journals.jps.jp/doi/epdf/10.1143/JPSJ.5.435> The final unitary U_T is the geometric transport associated with the path in the chosen subspace. A full physical time evolution would also contain dynamical phases depending on the instantaneous energies and total protocol time. The Kato construction isolates the adiabatic geometric part, which is the component relevant for identifying the braid gate. In numerical calculations, $P(t)$ is obtained by diagonalizing the Hamiltonian at each time step and constructing the projector onto the chosen low-energy manifold.

This unitary matrix is the braid unitary generated by the parameter protocol. By comparing U_T with the ideal target gate expressed in terms of Majorana bilinears, we can assess how well the parameter-tuning braid approximates the desired quantum gate operation. This comparison can be done both at the operator level, by checking how U_T

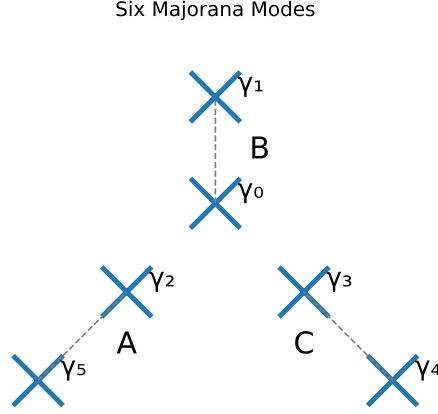


Figure 2.7: Schematic representation of a system hosting six majoranas. The illustration shows pairs of two and two Majoranas coupling together, in a Y-junction, each pair form a subsystem that can be described by a Poor Man's Majorana setup, A, B and C.

transforms the Majorana operators, and at the level of the projected parity blocks, by checking how U_T acts on the encoded qubit states.

2.5.2 Effective Braiding Hamiltonian

As a reference model, assume that the system contains four Majorana modes γ_0 , γ_1 , γ_2 , and γ_3 involved in the braid. A useful physical picture is a Y-junction made from three PMM subsystems, where one central Majorana is coupled sequentially to different neighboring Majoranas.

In Fig. 2.7, the four Majoranas γ_0 , γ_1 , γ_2 , and γ_3 are the modes actively used in the braid, while γ_4 and γ_5 form a spectator pair. The ideal effective Hamiltonian can be written as

$$\begin{aligned}
 H = & \Delta_{01} i \gamma_0 \gamma_1 + \\
 & \Delta_{02} i \gamma_0 \gamma_2 + \\
 & \Delta_{03} i \gamma_0 \gamma_3,
 \end{aligned} \tag{2.15}$$

where Δ_{ij} is a tunable coupling between Majoranas γ_i and γ_j . By turning these couplings on and off in a controlled sequence, the system can implement an effective exchange operation. This model is not meant to describe all microscopic details of the dot device directly. Instead, it serves as a reference model where the intended Majorana exchange can be tested without additional complications from the full interacting Hamiltonian.

2.5.3 Projected Microscopic Junctions

In a microscopic PMM device, the tunable quantities are not abstract Majorana bilinears, but physical hopping and pairing couplings between dots or subsystems. A junction between two PMM subsystems can be modeled by

$$H_{JuncAB} = t_{AB}(d_A^\dagger d_B + d_B^\dagger d_A) + \Delta_{AB}(d_A^\dagger d_B^\dagger + d_B d_A), \tag{2.16}$$

which is the same type of elastic cotunneling and crossed-Andreev pairing structure appearing in the PMM Hamiltonian. After such microscopic junction operators are projected into the retained low-energy subspace, they can be compared with the effective Majorana bilinears of Eq. 2.15. This motivates comparing two related braid constructions: one built from ideal or energy-level Majorana operators, and one built from projected microscopic local operators.

2.5.4 Energy-Level Versus Local-Operator Braids

The central comparison in the braiding analysis is between two ways of replacing the ideal operators γ_2 and γ_3 in Eq. 2.15. The effective Hamiltonian assumes that the Majorana operators are ideal and perfectly localized, while the optimized PMM system only provides approximate Majorana-like degrees of freedom. To test how close the microscopic system comes to the ideal exchange picture, we compare an energy-level construction with a local-operator construction.

In Majorana language, the target exchange operation used in this thesis is

$$B_{\gamma_2\gamma_3} = \exp\left(-\frac{\pi}{4}\gamma_2\gamma_3\right). \quad (2.17)$$

The resulting braid unitary is later compared with this target gate, both through its action on Majorana operators and through its projected parity-block representation.

The local-operator braid is built from microscopic endpoint operators rather than from idealized Majorana operators. In principle, one can project either the real or imaginary local Majorana component, corresponding to $c^\dagger + c$ and $i(c^\dagger - c)$, respectively. In the calculations presented here, the physical braid uses the projected minus component, $P i(c^\dagger - c) P$, for the relevant B and C subsystem sites. After projection into an N -dimensional low-energy subspace, these operators are normalized and used as the effective braid channels that replace γ_2 and γ_3 . The driven Hamiltonian therefore takes the schematic form

$$\begin{aligned} H_{\text{local}} = & \Delta_{01} i\gamma_0\gamma_1 + \\ & \Delta_{02} i\gamma_0\Gamma_B^P + \\ & \Delta_{03} i\gamma_0\Gamma_C^P, \end{aligned} \quad (2.18)$$

with

$$\Gamma_B^P \propto P i(c_B^\dagger - c_B) P, \quad \Gamma_C^P \propto P i(c_C^\dagger - c_C) P. \quad (2.19)$$

Mention that it is physically meaningful to check both the X and Y local Majorana components.

A local fermionic operator has two Majorana quadratures,

$$\chi_x = c^\dagger + c, \quad \chi_y = i(c^\dagger - c).$$

Choosing only χ_y assumes that the physically relevant Majorana is aligned with that quadrature. But in an interacting/projected low-energy system, the effective Majorana can be rotated in this local Majorana plane,

$$\chi(\theta) = \cos\theta \chi_x + \sin\theta \chi_y.$$

Here P is constructed by diagonalizing the full Hamiltonian and selecting the N lowest-energy eigenstates. The local-operator braid therefore tests whether the physically projected microscopic endpoint operators access the same exchange channel as the ideal target gate.

The energy-level construction starts from the optimized subsystem Hamiltonian. For clarity, this is the n -dot extension of the effective two-dot model 2.10,

$$H = \sum_i^n \epsilon_i n_i + \sum_j^{n-1} t_j (c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j) + \sum_j^{n-1} \Delta_j (c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j) + \sum_j^{n-1} U n_j n_{j+1}, \quad (2.20)$$

Diagonalizing this Hamiltonian 2.20, we obtain the energy eigenstates, which can be classified by parity. By pairing even and odd states that are close in energy, we can construct effective Majorana operators as combinations of these parity partners. In its simplest form, this construction looks like:

$$\begin{aligned} \gamma_+ &= \sum_i^n (|e_i\rangle \langle o_i| + |o_i\rangle \langle e_i|), \\ \gamma_- &= i \sum_i^n (|e_i\rangle \langle o_i| - |o_i\rangle \langle e_i|). \end{aligned} \quad (2.21)$$

Here n is the number of even-odd energy pairs included in the construction <https://arxiv.org/pdf/2510.20538>. For $n = 1$, only the ground-state parity pair contributes. Larger values of n retain more of the low-energy spectrum and allow the braid construction to be tested in enlarged projected spaces. The paired-state construction is designed to realize the Majorana algebra within the selected parity-pair subspace; after embedding and projection into the full retained low-energy space, the resulting operators are checked numerically for normalization and anticommutation errors.

In the higher-energy analysis, the relevant projected Majorana channel is not assumed to be purely γ_+ or purely γ_- . Instead, the implementation constructs the accumulated projected γ_+ and γ_- components from the chosen parity pairs, and then fits a global linear combination to the projected local reference operator:

$$\gamma_{\text{fit}} = a \gamma_+^P + b \gamma_-^P. \quad (2.22)$$

The coefficients a and b are chosen to minimize the mismatch with the projected microscopic local operator in the retained subspace. This fitting procedure can be interpreted as selecting the effective Majorana direction that best aligns the energy-level construction with the physically projected local channel.

The comparison between these two constructions is one of the central checks in this thesis. If the energy-level and local-operator braids agree, then the optimized spectrum and the physical endpoint couplings select the same Majorana exchange channel. If they differ, the spectrum may still contain Majorana-like parity pairs, but the local microscopic operators do not access the ideal braid cleanly. This distinction is what allows the braiding analysis to go beyond asking whether Majorana-like states exist, and instead test whether they can be used to implement the intended exchange operation.

Chapter 3

Method

This chapter describes the methods used to optimize the Majorana modes in the junction setup and to analyze their properties. We first discuss the parameter optimization process, including the model parameterization, loss function design based on Majorana criteria, and the optimization strategy. Next, we detail the post-optimization analysis of the setup, covering Hamiltonian reconstruction, parity classification, and various diagnostics. Finally, we describe the braiding protocol implemented in an effective Majorana model and how we validate the results using gate-validation checks.

3.1 Numerical PMM Hamiltonian and Basis

The numerical calculations are built from the spinless n -dot PMM Hamiltonian introduced in the theory chapter. This section specifies how that model is represented in the code: which Hamiltonian is diagonalized, which basis is used, how the fermionic operators are constructed, and how the model parameters are organized before optimization and braiding calculations. The goal here is not to rederive the PMM model, but to define the concrete numerical object used throughout the rest of the method chapter.

3.1.1 Single-chain Hamiltonian

For a single n -dot PMM chain, the Hamiltonian used in the optimization and diagnostic calculations is

$$H_{\text{PMM}} = \sum_{i=1}^n \epsilon_i n_i + \sum_{i=1}^{n-1} \left[-t_i (c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i) + \Delta_i (c_i c_{i+1} + c_{i+1}^\dagger c_i^\dagger) + U_i n_i n_{i+1} \right]. \quad (3.1)$$

Here $c_i^\dagger$ and c_i create and annihilate a spinless fermion on dot i , and $n_i = c_i^\dagger c_i$ is the corresponding number operator. The onsite energies ϵ_i play the role of local chemical potentials, t_i are elastic cotunneling amplitudes, Δ_i are crossed-Andreev pairing amplitudes, and U_i are nearest-neighbor density-density interaction strengths. This is the same operator structure used both when searching for optimized PMM chains and when reconstructing selected chains for later braiding calculations.

3.1.2 Many-body Fock basis

All operators are represented in the full many-body Fock basis. For a chain with n dots, a basis state is specified by the occupation pattern

$$|n_1 n_2 \cdots n_n\rangle, \quad n_i \in \{0, 1\}, \quad (3.2)$$

so the Hilbert-space dimension of a single chain is 2^n . Diagonalizing Eq. 3.1 in this basis gives the full many-body spectrum and eigenstates. This is important because the diagnostics used later, such as parity classification, charge expectation values, and Majorana-polarization matrix elements, are all many-body quantities.

For braiding calculations, several optimized chains are combined into a larger Hilbert space. If D identical PMM subsystems are used, each containing n dots, the total number of fermionic sites is $N_{\text{tot}} = Dn$ and the full Hilbert-space dimension is $2^{N_{\text{tot}}}$. In the three-arm braiding geometry used later, $D = 3$, corresponding to subsystems A , B , and C .

3.1.3 Jordan-Wigner construction of fermionic operators

The fermionic creation and annihilation operators must satisfy the canonical anticommutation relations

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \quad (3.3)$$

In the code, these operators are constructed with Jordan-Wigner strings. Starting from local spin raising and lowering operators on site j , the fermionic operators are represented schematically as

$$c_j^\dagger = \left[\prod_{k < j} (-\sigma_k^z) \right] \sigma_j^+, \quad c_j = \left[\prod_{k < j} (-\sigma_k^z) \right] \sigma_j^-. \quad (3.4)$$

The string of σ^z operators keeps track of the fermionic sign picked up when operators on different sites are exchanged. This same construction is used for single-chain calculations and for the enlarged three-subsystem Hilbert space. In the latter case, the site ordering fixes the Jordan-Wigner strings across subsystems, ensuring that operators belonging to different PMM chains still anticommute correctly.

Once the creation and annihilation operators have been constructed, the code precomputes the basic operator products needed repeatedly:

$$n_i = c_i^\dagger c_i, \quad h_{ij} = c_i^\dagger c_j + c_j^\dagger c_i, \quad p_{ij} = c_i^\dagger c_j^\dagger + c_j c_i, \quad d_{ij} = n_i n_j. \quad (3.5)$$

These building blocks are then combined with the parameter values in Eq. 3.1 to assemble the Hamiltonian matrix. The same Jordan-Wigner ordering is also used when several PMM chains are embedded into the combined braiding Hilbert space, so that the fermionic operators of different subsystems anticommute correctly. For the three-subsystem geometry, sites are ordered as

$$A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_n, C_1, C_2, \dots, C_n.$$

Operators acting on subsystem B therefore carry a Jordan-Wigner string over all sites in subsystem A, while all operators acting on subsystem C carry a string over all sites in both A and B. This ensures that the fermionic algebra is preserved across the entire device, which is crucial for correctly defining the Majorana operators and their braiding properties later on.

3.1.4 Parameter conventions

The numerical implementation allows each parameter family in Eq. 3.1 to be fixed, homogeneous, or inhomogeneous. A fixed parameter is held at a prescribed value and is not included in the optimization vector. A homogeneous parameter has the same value on every site or bond, while an inhomogeneous parameter is allowed to vary independently from site to site or bond to bond. This convention makes it possible to compare restricted ansätze with more flexible parameter searches using the same Hamiltonian-building routine.

In the main optimization runs used in this thesis, the hopping amplitudes are fixed to $t_i = 1$, and the interaction strength is fixed to selected values of U . The onsite energies ϵ_i and pairing amplitudes Δ_i are optimized. Fixing the hopping scale prevents the optimizer from lowering the loss through a trivial global rescaling of the kinetic energy, and instead forces the search to find nontrivial onsite and pairing profiles for each interaction regime. This also rules out the possibility of blowing up all other parameters in order to suppress the relative effect of interactions, which would be an unphysical way to achieve a low loss in the strongly interacting regime. The optimized parameters are stored in a reduced vector form, where only the free parameters are included. After optimization, the full set of physical parameters is reconstructed by reinserting fixed values and expanding homogeneous parameters as needed.

After optimization, the selected physical parameters are stored and later reloaded for post-optimization diagnostics and braiding calculations. The same parameter convention is also used when constructing the three-subsystem braiding geometry: the optimized single-chain parameters define each subsystem, while additional inter-subsystem couplings are introduced only in the braiding-model sections below.

3.2 Parameter Optimization and Loss Function

The purpose of the optimization is to find parameter regimes where the finite PMM chain has the spectral and local signatures of Majorana-like states. The theory chapter defined the physical criteria; here they are turned into a numerical objective. For each trial parameter vector, the Hamiltonian in Eq. 3.1 is constructed, diagonalized, separated into parity sectors, and assigned a scalar loss. The optimizer then updates only the free parameters, while fixed parameters are reinserted when the physical Hamiltonian is rebuilt.

3.2.1 Optimization vector and physical parameters

The optimizer does not act directly on a full dictionary of physical parameters. Instead, it acts on a reduced real vector θ containing only the entries that are not fixed by the chosen parameter configuration. For example, if t_i is fixed and U_i is fixed to a selected interaction strength, then those entries are omitted from θ . If a parameter is homogeneous, only one number is included; if it is inhomogeneous, one number is included for each site or bond.

The reduced vector is converted back into a parameter dictionary before the Hamiltonian is evaluated. Schematically,

$$\theta \longrightarrow \{t_i, U_i, \epsilon_i, \Delta_i\} \longrightarrow H_{\text{PMM}}(\theta). \quad (3.6)$$

During this conversion, fixed values are reinserted, homogeneous values are expanded across all sites or bonds when needed, and positive couplings such as t_i , U_i , and Δ_i are

constrained to be non-negative by taking their absolute values. This separation between the optimization vector and the physical parameter dictionary makes it possible to use the same code for different ansätze.

In the main optimization runs, the hopping amplitudes are fixed to $t_i = 1$, while the interaction strength U is fixed to the value being studied. The remaining parameters, primarily the onsite energies ϵ_i and pairing amplitudes Δ_i , are optimized. For each system size n and interaction strength U , the lowest-loss physical parameter set is stored and later reused in the diagnostic and braiding calculations.

3.2.2 Parity classification inside the loss

For each trial Hamiltonian, the full many-body spectrum is obtained by exact diagonalization. The eigenstates are then classified by the total fermion parity operator,

$$P_{\text{tot}} = \prod_{i=1}^n (1 - 2n_i). \quad (3.7)$$

This gives two ordered energy lists, one in the even sector and one in the odd sector,

$$\{E_k^{(e)}\}, \quad \{E_k^{(o)}\}. \quad (3.8)$$

The loss function compares these parity partners level by level. If the diagonalization produces an invalid parity structure, for example if one sector is missing or if the two sectors cannot be paired consistently, the trial point is discarded and the optimizer proceeds to another restart or candidate point.

3.2.3 Even-odd degeneracy term

The first loss contribution penalizes energy splitting between even- and odd-parity partners,

$$\delta_k = |E_k^{(e)} - E_k^{(o)}|. \quad (3.9)$$

These splittings are weighted more strongly at low energy, because the ground-state degeneracy is the most important Majorana signature. In the implementation, the weights decrease from a maximum value for the lowest parity pair to a smaller value for higher levels. The degeneracy contribution therefore has the schematic form

$$\mathcal{L}_{\text{deg}} = \sum_k w_k |E_k^{(e)} - E_k^{(o)}|, \quad w_0 > w_1 > \dots \quad (3.10)$$

This term encourages the optimizer to produce an approximately parity-paired spectrum, with particular emphasis on the lowest-energy states.

3.2.4 Excitation-gap term

Degeneracy alone is not sufficient: the parity-paired states must also be separated from nearby levels. The second loss contribution penalizes small gaps within the parity-resolved spectra. For adjacent levels, the relevant gap is taken as the smaller of the even-sector and odd-sector spacings,

$$\Delta_k = \min(E_{k+1}^{(e)} - E_k^{(e)}, E_{k+1}^{(o)} - E_k^{(o)}). \quad (3.11)$$

Small gaps are penalized using a smooth threshold function,

$$\mathcal{L}_{\text{gap}} = \sum_k \tilde{w}_k \text{softplus}(\Delta_{\text{target}} - \Delta_k), \quad (3.12)$$

where Δ_{target} is the desired minimum gap scale. This term favors spectra where the relevant parity-paired states are not only close to each other, but also isolated from the rest of the spectrum.

3.2.5 Charge and Majorana-polarization terms

The loss also includes local diagnostics that test whether the parity partners resemble Majorana states rather than accidental degeneracies. The charge penalty measures how differently the even and odd partners occupy each site,

$$\mathcal{L}_{\text{MP}} = \sum_k \sum_j |\langle e_k | n_j | e_k \rangle - \langle o_k | n_j | o_k \rangle|. \quad (3.13)$$

A small value means that the parity partners have similar local charge distributions, consistent with non-local encoding.

The Majorana-polarization penalty compares the site-resolved Majorana-polarization profile with the expected edge-localized pattern. In the target configuration, the polarization is concentrated on the two ends of the chain with opposite sign, while the bulk sites carry little weight. Numerically, the penalty is evaluated from matrix elements of local Majorana operators between the even and odd parity partners. The sign of the target pattern is allowed to flip, since exchanging the labels of the two Majorana components changes the overall sign but not the physical localization pattern.

3.2.6 Total normalized loss

The full loss is a weighted sum of the spectral and local contributions. In the implementation, the degeneracy and gap terms are first collected into a single weighted spectral penalty array, after which the charge and Majorana-polarization penalties are added. Written in terms of the contributions above, the loss has the schematic form

$$\mathcal{L} = \frac{\mathcal{L}_{\text{deg}} + \mathcal{L}_{\text{gap}} + \mathcal{L}_Q + \mathcal{L}_{\text{MP}}}{|\overline{E}| + \eta}, \quad (3.14)$$

where $|\overline{E}|$ is the mean absolute energy scale of the spectrum and η is a small numerical offset. This normalization makes losses from different parameter sets more comparable and reduces the tendency to favor configurations only because the full spectrum has been rescaled.

3.2.7 Search strategy

The resulting loss landscape is non-convex, especially for interacting and inhomogeneous chains. The numerical search therefore combines local gradient-based optimization with repeated restarts and basin hopping. Local optimization is performed with Adam. At each iteration, the reduced vector θ is converted into physical parameters, the Hamiltonian is built and diagonalized, the loss is evaluated within the PyTorch optimization loop and used to update the free parameters with Adam.

Because a single local run can converge to a poor local minimum, the local optimization is repeated from randomly perturbed initial vectors. The lowest-loss result

among these restarts is retained as the representative of that region of parameter space. Basin hopping is then used to explore different regions: a random displacement is applied to the current parameter vector, local optimization with restarts is performed again, and the new candidate is accepted if it improves the loss. Candidates with slightly worse loss can also be accepted with a Metropolis probability, allowing the search to escape shallow local minima.

For each pair (n, U) , the best configuration found by this procedure is stored together with its loss, parameter configuration, and reconstructed physical parameters. These stored configurations form the input to the post-optimization diagnostics described in the next section.

3.3 Post-Optimization Diagnostics

The optimization procedure returns a scalar loss and a set of optimized parameters, but the loss alone is not sufficient to decide whether the result has a useful Majorana interpretation. After each optimization run, the best parameter set is therefore reconstructed and analyzed with a separate diagnostic workflow. The purpose of this step is to translate the numerical minimum into physical quantities that can be compared across system sizes, interaction strengths, and later braiding calculations.

3.3.1 Reconstruction of optimized Hamiltonians

The saved optimization output contains the raw optimization vector, the parameter configuration, the reconstructed physical parameters, and the final loss. For a chosen system size n and interaction strength U , the diagnostic routine selects the lowest-loss stored configuration. The raw vector is then converted back into the parameter dictionary described in Sec. 3.2, fixed values are reinserted, and the physical Hamiltonian

$$H_{\text{PMM}}(\theta_{\text{opt}}) \quad (3.15)$$

is rebuilt using the same Hamiltonian construction as in the optimization.

This reconstruction step is important because the diagnostics are not evaluated directly from the optimizer state. Instead, the optimized Hamiltonian is rebuilt from the stored physical parameters and diagonalized again. This makes the diagnostic plots and tables reproducible from the saved configuration files and ensures that the reported quantities correspond to the same Hamiltonian that is later used as input for the braiding calculations.

3.3.2 Parity-resolved spectrum

The reconstructed Hamiltonian is diagonalized exactly in the many-body Fock basis. The eigenstates are classified with the total fermion parity operator,

$$P_{\text{tot}} = \prod_{i=1}^n (1 - 2n_i), \quad (3.16)$$

giving ordered even- and odd-parity spectra. For plotting and comparison, all energies are shifted so that the lowest state in either parity sector has energy zero. The shifted spectra make it possible to inspect whether the optimized configuration produces the expected near-degenerate parity pairs.

The most important scalar spectral diagnostic is the splitting of the lowest even-odd pair,

$$\delta_0 = |E_0^{(e)} - E_0^{(o)}|. \quad (3.17)$$

A small value of δ_0 indicates that the lowest even and odd states are approximately degenerate. However, this degeneracy is only useful if the pair is also separated from excited states. For this reason, the low-energy gap is defined as

$$\Delta_{\text{gap}} = \min(E_1^{(e)}, E_1^{(o)}) - \max(E_0^{(e)}, E_0^{(o)}), \quad (3.18)$$

after the common ground-state energy shift. This quantity measures the distance from the lowest parity pair to the nearest excited state in either parity sector. The combination of small δ_0 and large Δ_{gap} is the main spectral criterion used to identify promising optimized chains.

3.3.3 Local charge difference

Spectral degeneracy can occur for reasons unrelated to Majorana physics, so the post-optimization analysis also evaluates local observables in the even and odd states. The first local diagnostic is the site-resolved charge difference,

$$q_i = |\langle e_0 | n_i | e_0 \rangle - \langle o_0 | n_i | o_0 \rangle|. \quad (3.19)$$

The total charge difference reported in the results is

$$Q_0 = \sum_{i=1}^n q_i. \quad (3.20)$$

If the parity information is stored non-locally, the two parity partners should have nearly identical local charge distributions. A small Q_0 therefore indicates that local charge measurements cannot easily distinguish the even and odd states. In the diagnostic figures, the individual q_i values are plotted on top of the Majorana-polarization profile so that charge visibility and spatial localization can be compared site by site.

3.3.4 Majorana-polarization profile

The Majorana-polarization profile is computed from matrix elements of local Majorana operators between parity-partner states. In the summary diagnostics reported here, M_i denotes the profile of the lowest even-odd pair and measures where the parity-changing Majorana character is located. For an ideal finite PMM chain, the profile should be concentrated near the two ends, with opposite signs on the left and right edges and little weight in the bulk.

To summarize this profile in the results, two scalar quantities are extracted. The edge polarization is defined as the mean absolute polarization on the two end sites,

$$\overline{|M_{\text{edge}}|} = \frac{1}{2} (|M_1| + |M_n|), \quad (3.21)$$

while the bulk polarization is defined as the largest absolute value on the interior sites,

$$\max |M_{\text{bulk}}| = \max_{2 \leq i \leq n-1} |M_i|. \quad (3.22)$$

For two-dot systems there are no interior sites, so the bulk contribution is set to zero. A good Majorana-like configuration should have $\overline{|M_{\text{edge}}|}$ close to one and $\max |M_{\text{bulk}}|$ close to zero.

3.3.5 Selection for braiding calculations

The post-optimization diagnostics provide the bridge between the optimization results and the braiding part of the thesis. The parity-resolved spectra identify whether the optimized chain has a low-energy manifold suitable for projection. The charge and Majorana-polarization diagnostics then test whether this manifold has the expected non-local and edge-localized structure.

Although Majorana physics is often discussed in terms of the lowest parity pair, the optimization used here was deliberately applied to the full parity-resolved spectrum of the finite many-body Hamiltonian. This choice was motivated by the later braiding analysis, where projected spaces of increasing dimension, up to the full $2^9 = 512$ dimensional three-subsystem Hilbert space, are tested. The goal was therefore not only to obtain a clean ground-state Majorana signature, but also to find parameter sets whose parity pairing, excitation gaps, charge structure, and Majorana-polarization properties remain well behaved beyond the lowest states.

The post-optimization diagnostics therefore serve two purposes. First, they identify whether the lowest parity pair has the expected Majorana-like structure. Second, they indicate whether the optimized configuration remains suitable when larger projected subspaces are retained in the braiding calculations. Configurations that satisfy both requirements are selected as candidates for the effective Majorana construction described next.

3.4 Effective Majorana Operator Construction

The braiding calculations require explicit matrix representations of the Majorana operators associated with the optimized PMM chains. The theory chapter described why these operators should connect even and odd parity states. Here the focus is narrower: how the operators are constructed from the optimized eigenstates, embedded into the three-subsystem Hilbert space, projected to a chosen low-energy space, and checked before they are used in the braid.

3.4.1 Energy-level Majoranas from parity pairs

For each optimized subsystem, the Hamiltonian is diagonalized and the eigenstates are separated into even and odd parity sectors. The states are then paired by their order in energy. From the j th even-odd pair, two parity-changing operators are formed,

$$\gamma_{+,j} = |e_j\rangle \langle o_j| + |o_j\rangle \langle e_j|, \quad \gamma_{-,j} = i(|e_j\rangle \langle o_j| - |o_j\rangle \langle e_j|). \quad (3.23)$$

The effective Majorana operators used in the projected calculations are built by summing these contributions over a selected number of parity pairs,

$$\gamma_+^{(m)} = \sum_{j=0}^{m-1} \gamma_{+,j}, \quad \gamma_-^{(m)} = \sum_{j=0}^{m-1} \gamma_{-,j}. \quad (3.24)$$

Here m is the number of even-odd pairs included in the operator construction. In the code this is controlled separately from the total projection dimension: the projection dimension determines how much of the full three-subsystem Hilbert space is retained, while m determines how many subsystem parity pairs are used to build the Majorana operator itself.

3.4.2 Embedding into the three-subsystem Hilbert space

The braiding geometry contains three optimized PMM subsystems, labelled A , B , and C . The full basis is ordered as described in Sec. 3.1, with all A sites first, followed by all B sites and then all C sites. After an operator has been constructed in the Hilbert space of one subsystem, it is embedded into the full Hilbert space using the same Jordan-Wigner ordering as the microscopic fermion operators.

For an operator γ_α acting on subsystem α , the embedding is schematically

$$\gamma_A \mapsto \gamma_A \otimes I_B \otimes I_C, \quad (3.25)$$

$$\gamma_B \mapsto JW_A \otimes \gamma_B \otimes I_C, \quad (3.26)$$

and

$$\gamma_C \mapsto JW_{AB} \otimes \gamma_C, \quad (3.27)$$

where JW_A is the Jordan-Wigner string over subsystem A , and JW_{AB} is the corresponding string over subsystems A and B . These strings are essential because the subsystem Majoranas must anticommute as fermionic operators after they are placed in the same Hilbert space.

3.4.3 Projection to the retained low-energy space

The full three-subsystem Hamiltonian is diagonalized, and a projection basis V is formed from the lowest d_P eigenvectors. Here d_P is the chosen projection dimension. Any full-space operator O is then represented in the retained space by

$$O_P = V^\dagger O V. \quad (3.28)$$

This projection is applied to the embedded Majorana operators before they are used in the projected braid models. The same projection basis is also used when projecting microscopic junction operators later, so that all operators act on the same retained low-energy Hilbert space.

3.4.4 Projected local components

In addition to the energy-level construction in Eq. 3.24, the code also constructs projected local Majorana components from the microscopic fermion operators on the relevant subsystem sites,

$$\Gamma_{+,P} = V^\dagger (c^\dagger + c) V, \quad \Gamma_{-,P} = V^\dagger i(c^\dagger - c) V. \quad (3.29)$$

These projected local components provide a more microscopic reference for the Majorana operator at a chosen end of a subsystem. In the implementation, the energy-level operators are compared with these projected local components by minimizing a matrix-norm mismatch. When both components are allowed, the fitted operator has the form

$$\gamma_{\text{fit}} = a \gamma_+^{(m)} + b \gamma_-^{(m)}, \quad (3.30)$$

where the coefficients a and b are chosen from a least-squares fit in the projected operator space. This step is used to quantify how well the energy-level Majorana extracted from parity pairs matches the local microscopic Majorana component used in the projected-local braid comparison.

3.4.5 Normalization and algebra checks

After projection, the Majorana operators are no longer assumed to satisfy the ideal algebra exactly. The projected operators are therefore checked numerically before being used in the braid. The diagnostics test Hermiticity, normalization,

$$\gamma_i^2 \approx I, \quad (3.31)$$

and mutual anticommutation,

$$\{\gamma_i, \gamma_j\} \approx 0 \quad (i \neq j). \quad (3.32)$$

The matrix mismatch between the fitted energy-level operator and the projected local component is also stored. These checks are important because the later braid errors should not be interpreted only as failures of the pulse protocol: they can also reflect imperfect Majorana operators after projection.

3.5 Braiding Models and Pulse Protocol

The braiding calculations use the optimized PMM chains as building blocks for a three-subsystem junction. The purpose of this section is to define the Hamiltonians that are driven during the braid and the common pulse sequence used for all model variants. The following section then describes how the resulting time-dependent evolution is transported and compared with the target exchange gate.

3.5.1 Three-subsystem geometry

The junction consists of three optimized PMM subsystems, labelled A , B , and C . Subsystem A acts as the reference arm and supplies two Majorana operators, denoted γ_0 and γ_1 . The Majorana operators to be exchanged are associated with subsystems B and C , and are denoted γ_2 and γ_3 . In this notation, the intended exchange is generated by moving the coupling of γ_0 from γ_1 to γ_2 , then to γ_3 , and finally back to γ_1 .

All braiding models are built in the projected Hilbert space introduced in Sec. 3.4. The projection dimension determines how many low-energy states of the full three-subsystem Hamiltonian are retained. Within this common projected space, different choices of coupling operators can be compared directly.

In the projection scans, the driven braid terms can also be supplemented by the projected static Hamiltonian of the uncoupled subsystem background,

$$H_{\text{static}} = V^\dagger (H_B + H_C) V. \quad (3.33)$$

This term keeps the internal optimized-chain structure of subsystems B and C while the time-dependent junction couplings are varied. Since it is included in the same way for the model comparisons, the differences between the braid results come from the choice of driven coupling operators rather than from a different static background.

3.5.2 Ideal four-Majorana model

The simplest braid model uses only four Majorana operators and hand-chosen bilinear couplings. In the projected space, the time-dependent Hamiltonian is

$$H_{\text{ideal}}(t) = \lambda_1(t) i\gamma_0\gamma_1 + \lambda_2(t) i\gamma_0\gamma_2 + \lambda_3(t) i\gamma_0\gamma_3. \quad (3.34)$$

This model is used as a reference calculation. Since the couplings are written directly as Majorana bilinears, it isolates the braid protocol itself from the additional complications of microscopic junction operators and imperfect projected Majoranas.

3.5.3 Projected microscopic junction model

To connect the braid more directly to the optimized dot-chain Hamiltonian, the ideal bilinears can be replaced by microscopic junction operators. The junction between subsystems A and B is constructed from ordinary hopping and pairing terms between the endpoint dots,

$$H_{AB}^{\text{junc}} = -t_j (c_A^\dagger c_B + c_B^\dagger c_A) + \Delta_j (c_A c_B + c_B^\dagger c_A^\dagger), \quad (3.35)$$

with an analogous operator H_{AC}^{junc} for the A - C junction. These full-space junction operators are projected into the retained low-energy space,

$$T_{AB} = V^\dagger H_{AB}^{\text{junc}} V, \quad T_{AC} = V^\dagger H_{AC}^{\text{junc}} V. \quad (3.36)$$

The corresponding projected microscopic braid Hamiltonian is then

$$H_{\text{junc}}(t) = \lambda_1(t) T_A + \lambda_2(t) T_{AB} + \lambda_3(t) T_{AC}, \quad (3.37)$$

where $T_A = i\gamma_0\gamma_1$ fixes the initial pairing inside subsystem A . This model tests whether the physical junction couplings reproduce the desired Majorana exchange after projection.

3.5.4 Projected local-operator model

A second microscopic comparison keeps the bilinear braid structure but replaces the ideal exchanged Majoranas by projected local Majorana components. For example, on the relevant endpoint of subsystem B one constructs

$$\Gamma_{B,-} = V^\dagger i(c_B^\dagger - c_B) V, \quad (3.38)$$

and similarly for subsystem C . After normalization, these local projected operators define $\gamma_{2,\text{loc}}$ and $\gamma_{3,\text{loc}}$. The local-operator braid Hamiltonian is

$$H_{\text{loc}}(t) = \lambda_1(t) i\gamma_0\gamma_1 + \lambda_2(t) i\gamma_0\gamma_{2,\text{loc}} + \lambda_3(t) i\gamma_0\gamma_{3,\text{loc}}. \quad (3.39)$$

This model is useful because it sits between the ideal and fully microscopic descriptions. The projected microscopic model first projects the full junction operator, while the projected local-operator model first projects local Majorana components and then forms bilinears from them. It therefore keeps a clear Majorana-bilinear form, but the operators being coupled are obtained from projected local fermionic operators rather than only from energy-level parity pairs.

3.5.5 Pulse sequence

The same pulse sequence is used for each braid model. To avoid confusing the braid amplitudes with the superconducting pairing parameters Δ_i , the time-dependent braid

couplings are denoted by $\lambda_i(t)$. The coupling envelope is a smooth pulse constructed from two sigmoid functions,

$$\lambda(t; T_p) = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \frac{1}{1 + \exp[-s(t - T_p + w/2)]} \frac{1}{1 + \exp[s(t - T_p - w/2)]}, \quad (3.40)$$

where T_p is the pulse center, w is the pulse width, and s controls the steepness of the rise and fall. The three braid couplings are then chosen as

$$\lambda_1(t) = \lambda(t; 0) + \lambda(t; T) - \lambda_{\min}, \quad \lambda_2(t) = \lambda(t; T/3), \quad \lambda_3(t) = \lambda(t; 2T/3). \quad (3.41)$$

Thus the A -arm coupling is active at the beginning and end of the cycle, while the A - B and A - C couplings are activated in sequence during the middle of the protocol. In the numerical calculations used here, the total braid time is set to $T = 1$, with $\lambda_{\max} = 1$, $\lambda_{\min} = 0$, pulse width $w = T/3$, and steepness $s = 20/w$.

At each time step, the Hamiltonian for the chosen model is assembled from the same three time-dependent coefficients. The instantaneous spectrum is monitored along the path, and the isolated low-energy band is then transported using the Kato procedure described in the next section.

3.6 Kato Transport and Braid Validation

After a braid Hamiltonian has been defined, the remaining task is to compute the unitary generated by the adiabatic path and to check whether it implements the desired Majorana exchange. The calculation is performed in the projected Hilbert space, but the transported subspace can be smaller than the full projection space. This section describes how that transported subspace is selected, how the Kato evolution is computed, and which diagnostics are used to compare the resulting unitary with the target braid.

3.6.1 Choice of transported subspace

For a given projected Hamiltonian $H(t)$, the instantaneous spectrum is sampled along the braid path. Candidate transported dimensions are tested by looking at the minimum spectral gap above each candidate band throughout the protocol. If the transported dimension is d_T , the relevant gap is

$$\Delta_T = \min_t [E_{d_T}(t) - E_{d_T-1}(t)], \quad (3.42)$$

where the eigenvalues are ordered increasingly and the index convention starts at zero. The numerical procedure selects the candidate d_T with the largest persistent gap. This choice avoids fixing the transported manifold by hand when the projection dimension is increased, and instead follows the most clearly isolated low-energy band available in the chosen projected model.

3.6.2 Numerical Kato transport

The braid path is discretized into time points $t_0, \dots, t_N$. At each time step, the Hamiltonian is diagonalized and the projector onto the transported band is constructed,

$$P(t_\ell) = \sum_{a=1}^{d_T} |\psi_a(t_\ell)\rangle \langle \psi_a(t_\ell)|. \quad (3.43)$$

The Kato generator is approximated from consecutive projectors,

$$K(t_\ell) = P(t_\ell)\dot{P}(t_\ell) - \dot{P}(t_\ell)P(t_\ell), \quad \dot{P}(t_\ell) \approx \frac{P(t_{\ell+1}) - P(t_\ell)}{\Delta t}. \quad (3.44)$$

The transport unitary is updated as

$$U_K(t_{\ell+1}) = \exp[-\Delta t K(t_\ell)]U_K(t_\ell), \quad U_K(t_0) = I. \quad (3.45)$$

This convention matches the numerical implementation and gives the geometric transport of the selected subspace along the braid path. The instantaneous energies and pulse amplitudes are stored at the same time, so that the quality of the transported band can be checked together with the final braid unitary.

3.6.3 Target exchange gate

The target unitary for exchanging γ_2 and γ_3 is taken to be

$$U_{\text{target}} = \exp\left(-\frac{\pi}{4}\gamma_2\gamma_3\right). \quad (3.46)$$

This convention corresponds to the exchange action

$$\gamma_2 \mapsto -\gamma_3, \quad \gamma_3 \mapsto \gamma_2, \quad (3.47)$$

up to the overall sign convention chosen for the Majorana operators. Since a global phase of the transported unitary is physically irrelevant, comparisons with the target gate are made after aligning the overall phase. In practice, the reported gate error is a Frobenius-norm distance between the transported unitary and the phase-aligned target unitary, restricted to the initial transported basis.

3.6.4 Operator-action checks

The first validation step checks the induced action on the Majorana operators directly. For each model, the transported unitary is applied as

$$\gamma'_i = U_K^\dagger \gamma_i U_K. \quad (3.48)$$

The resulting operators are compared with the expected single-exchange map,

$$\gamma'_2 \approx -\gamma_3, \quad \gamma'_3 \approx \gamma_2, \quad \gamma'_0 \approx \gamma_0, \quad \gamma'_1 \approx \gamma_1. \quad (3.49)$$

The mismatch is measured using a normalized Frobenius norm,

$$\|A\|_{\text{norm}} = \frac{\|A\|_F}{\sqrt{d}}, \quad (3.50)$$

where d is the dimension of the matrix being compared. These checks are useful because they test the braid as an operator transformation, rather than only as a matrix acting on a chosen basis.

3.6.5 Parity-resolved gate checks

The transported unitary is also analyzed in a parity basis. The total parity operator is projected into the initial transported space and diagonalized, separating the unitary into odd- and even-parity blocks. The off-block leakage is defined from the norm of the odd-to-even and even-to-odd blocks,

$$\mathcal{L}_{\text{parity}} = \|U_{oe}\| + \|U_{eo}\|. \quad (3.51)$$

A small value indicates that the braid approximately preserves fermion parity inside the transported manifold.

After the parity blocks have been separated, each block is compared with the corresponding block of the target exchange gate in Eq. 3.46. This produces separate odd- and even-sector target-gate errors. Reporting the parity blocks separately is useful because the same Majorana exchange can appear with different matrix representations in the two parity sectors, while still describing the same physical braid.

3.6.6 Energy-level and projected-local targets

The final comparison distinguishes between two possible choices of target Majoranas. The energy-level target uses the Majorana operators constructed from even-odd parity pairs, as described in Sec. 3.4. The projected-local target instead uses the normalized projected local operators obtained from microscopic combinations such as $i(c^\dagger - c)$. These two targets need not be identical after projection, especially when more excited states are retained or when the optimized Majorana operators are not perfectly local.

For this reason, the physical braid is compared in several ways: against the energy-level target, against the projected-local target, and against the ideal reference braid. The mismatch between the two target constructions is also recorded. This separation is important for interpreting the results. If the physical braid disagrees with the energy-level target but agrees with the projected-local target, the issue may be the identification of the effective Majorana operators rather than the adiabatic pulse itself. Conversely, if both target comparisons fail, the projected braid Hamiltonian is not implementing the desired exchange within the chosen transported subspace.

Chapter 4

Results

In this chapter, I will present the main numerical results of the thesis. The first part of the chapter will cover the optimized parameters found by the gradient-based optimization procedure, how these parameters depend on system size and interaction strength, whether the optimized Hamiltonians display Majorana-like spectral and what trends are observed in the local diagnostics. The second part of the chapter will focus on the braiding results. The results are solely based on the optimized parameter configurations, after picking the best candidates from the optimization scheme. The braiding analysis is divided into two parts: first, I will compare the ideal four-Majorana braid, the projected microscopic junction braid, and the projected local-operator braid in the low-energy projected setting. This, along with the well known analytical non-interacting sweet spot braid, serves as a controlled reference calculation to test whether the optimized microscopic system reproduces the exchange behavior expected from the ideal effective model. Then, I will extend the braiding analysis to larger projected Hilbert spaces up to the full $2^9 = 512$ dimensional three-subsystem system. This tests how much of the agreement with the ideal reference survives once higher sectors are retained, and how the energy-level and local-operator braid constructions compare in larger projection spaces.

4.1 Optimized PMM Chains

In this section, we will review the parameters found by the basin-hopping optimization procedure, and how these parameters depend on the system size n and the fixed interaction strength U . As mentioned in the methods section, the optimization is performed for chain lengths $n = 2, 3$, and 4 , and for interaction strengths $U = 0, 0.1, 0.5, 1.0, 2.0$, and 5.0 , with a fixed hopping amplitude $t_i = 1$ for all i . The purpose of the fixed hopping amplitude was to make sure that the hopping amplitude t_i and the pairing amplitude Δ_i do not make the implication of the interaction strength vanish, by scaling the other parameters up. The fixed hopping amplitude also makes it easier to compare the optimized pairing amplitudes Δ_i across different interaction strengths and system sizes, since they are all measured relative to the same fixed scale. The main purpose of this section is therefore to summarize the best parameter sets before applying the more detailed diagnostics in the following section, and to identify any trends in the optimized parameters as the interaction strength and system size are varied.

4.1.1 Lowest-loss configurations

Table 5.1 summarizes the lowest-loss configuration found for each combination of system size n and fixed interaction strength U . Since the hopping amplitudes were fixed to $t_i = 1$ in these runs, only the optimized pairing amplitudes and onsite energies are listed. Empty entries indicate parameters that do not exist for that system size.

n	U	Loss	Δ_1	Δ_2	Δ_3	ϵ_1	ϵ_2	ϵ_3	ϵ_4
2	0.0	5.657e-03	1.000	—	—	-0.001	0.002	—	—
2	0.1	2.617e-01	1.047	—	—	-0.050	-0.047	—	—
2	0.5	1.083e+00	1.249	—	—	-0.255	-0.249	—	—
2	1.0	1.940e+00	1.498	—	—	-0.499	-0.496	—	—
2	2.0	3.203e+00	2.000	—	—	-1.006	-1.004	—	—
2	5.0	4.004e+00	3.499	—	—	-2.508	-2.506	—	—
3	0.0	3.442e-01	1.010	1.348	—	0.019	1.774	-0.466	—
3	0.1	5.603e-01	1.001	1.002	—	-0.101	-0.607	-0.003	—
3	0.5	9.359e-01	1.004	1.012	—	-0.046	-0.889	-0.454	—
3	1.0	1.452e+00	1.025	1.130	—	-0.195	-1.530	-0.802	—
3	2.0	2.326e+00	1.584	0.978	—	-1.511	-3.077	-0.484	—
3	5.0	3.142e+00	0.846	1.746	—	-4.814	1.009	-0.184	—
4	0.0	3.161e-01	2.698	5.332	1.004	-0.059	-4.252	0.236	-0.003
4	0.1	4.840e-01	1.790	5.002	1.013	0.156	2.376	-1.966	-0.050
4	0.5	1.006e+00	1.212	3.490	1.206	-0.246	-0.499	-0.503	-0.250
4	1.0	5.670e-01	2.238	18.807	1.072	-0.500	-1.028	-1.025	-0.496
4	2.0	1.824e+00	1.492	5.537	1.588	-0.869	3.526	-6.508	-1.220
4	5.0	2.600e+00	3.389	9.513	3.336	-2.645	5.032	-15.431	-2.293

Table 4.1: Lowest-loss optimized configurations extracted from `configuration.json` for each combination of system size n and fixed interaction strength U .

For the two-dot system, the optimizer reproduces the expected non-interacting sweet spot at $U = 0$, where $\Delta_1 \approx t_1 = 1$ and the onsite energies remain close to zero. As the interaction strength is increased, the optimized pairing amplitude grows above the fixed hopping scale, while the onsite energies move approximately toward $\epsilon_i \approx -U/2$. This indicates that the optimizer deforms the known two-dot sweet spot in a smooth and physically interpretable way.

For the three-dot system, the weak- and intermediate-interaction configurations remain relatively close to the homogeneous pairing limit, with $\Delta_1 \approx \Delta_2 \approx 1$ for $U = 0.1$ and $U = 0.5$. At the same time, the middle-site energy becomes more negative than the edge-site energies. This suggests that the optimizer favors a differentiated central region even when the pairing amplitudes remain nearly uniform.

The four-dot system shows the clearest sign that the optimization landscape becomes more complicated as the chain length increases. In several of the best configurations, the optimizer strongly enhances the central pairing amplitude while leaving the outer pairing amplitudes closer to unity. The optimized onsite energies also become highly inhomogeneous, particularly at stronger interactions. Taken together, these trends suggest that longer PMM chains do not simply reproduce the two-dot sweet spot, but instead require more structured inhomogeneous parameter profiles.

4.1.2 Representative optimized spectra

The two-dot spectra in Figs. 5.1 and 5.2 illustrate how the optimized spectrum changes as interactions are increased. At $U = 0$, both parity pairs are degenerate, while at $U = 0.1$ the excited pair begins to split. By $U = 2.0$, the splitting is much more pronounced. This is consistent with the fact that the interacting two-dot system can retain a low-energy Majorana-like signature while failing to maintain full parity-pair degeneracy across the entire finite spectrum.

The three-dot system at $U = 0.1$ is especially important because it becomes the main candidate for the braiding analysis. Figure 5.3 shows both the parity-resolved spectrum and the Majorana-polarization profile for this configuration. The parity pairs are nearly degenerate, and the Majorana polarization has the expected structure: strong edge weight with opposite signs on the two ends and strongly suppressed weight on the middle dot.

4.2 Post-Optimization Diagnostics

The optimized parameters alone do not determine whether a configuration is useful for Majorana braiding. The next step is therefore to inspect the parity-resolved spectra, the low-energy gaps, the charge difference between parity partners, and the spatial Majorana-polarization profile. These diagnostics test whether the optimized states are not only low-loss numerical solutions, but also physically interpretable Majorana-like states.

4.2.1 Parity-resolved spectra and low-energy gaps

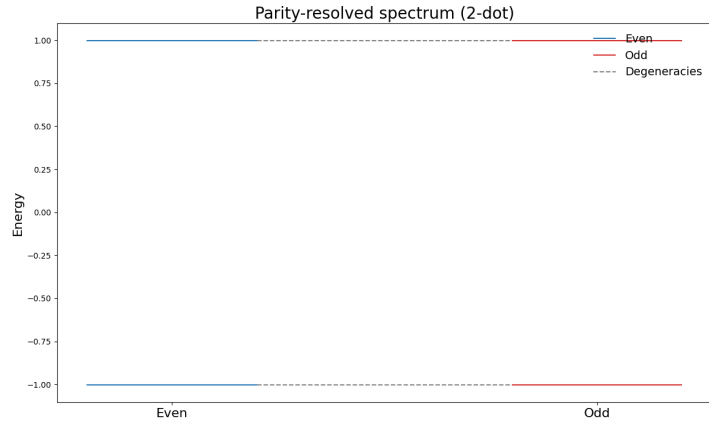
Figure 5.4 compares the parity-resolved spectra for $n = 2, 3$, and 4 at selected interaction strengths. The non-interacting cases provide the cleanest spectra, as expected. For the two-dot system, the excited-state parity splitting becomes visible already at weak interaction and grows at stronger interaction. For the four-dot system, the weakly interacting cases still show good parity pairing, but the stronger-interaction case is less clean and develops a noticeable low-energy splitting.

The three-dot system is the most stable across the interaction strengths shown. Even at stronger interaction, the optimizer finds configurations with reasonably well-paired parity sectors, although the gap structure becomes less favorable than in the weakly interacting case. This makes the three-dot system, especially at $U = 0.1$, the most promising compromise between interaction, spectral degeneracy, and later braiding feasibility.

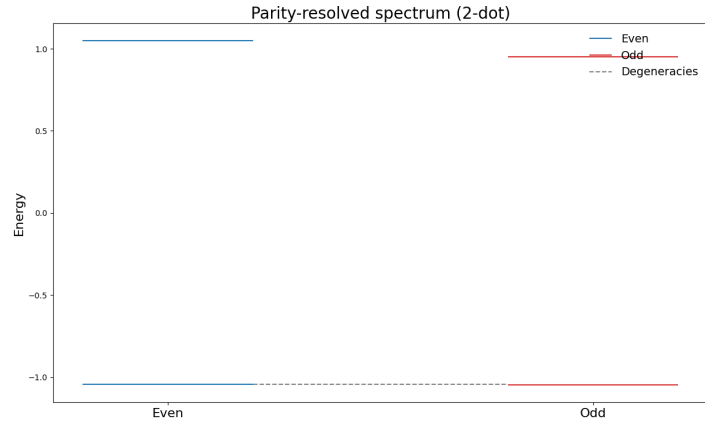
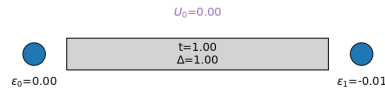
4.2.2 Local diagnostics and Majorana character

The parity-resolved spectrum shows whether the optimizer has found approximate even-odd degeneracy, but degeneracy alone is not enough to identify Majorana-like states. To test the spatial structure of the optimized states, I compare the site-resolved Majorana polarization and the local charge difference of the lowest even-odd pair. A convincing Majorana-like configuration should display opposite polarization on the two ends of the chain, weak polarization in the middle, and only a small charge difference between the even and odd states.

Figure 5.5 compares the spatial profiles directly, while Table 5.2 summarizes the corresponding scalar diagnostics. For the two-dot system, the edge polarization is



Optimized QD-SC-QD setup



Optimized QD-SC-QD setup

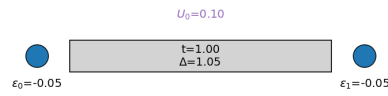


Figure 4.1: Parity-resolved spectra after optimization for the two-dot system at $U = 0$ and $U = 0.1$. The optimized parameters are shown schematically below each spectrum.

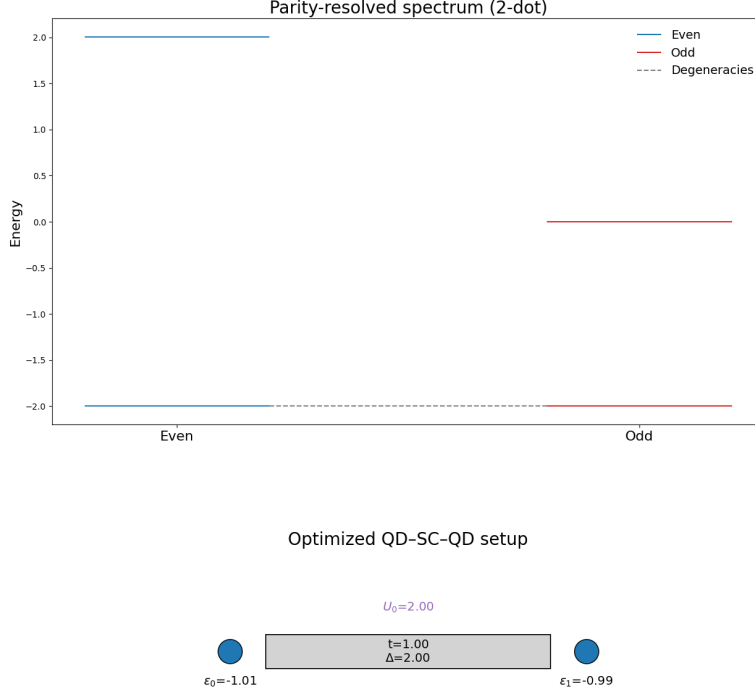


Figure 4.2: Parity-resolved spectrum after optimization for the two-dot system at $U = 2.0$. The optimized parameters are shown schematically below the spectrum.

n	U	δ_0	Δ_{gap}	Q_0	$\overline{ M_{\text{edge}} }$	$\max M_{\text{bulk}} $
2	0	7.64e-05	2	0.00139	1	0
2	0.1	0.00344	2	0.00142	1	0
3	0.1	9.39e-05	1.77	0.0492	0.998	0.000202
4	0.1	0.00423	1.21	0.007	0.986	0.0233

Table 4.2: Summary of local diagnostics for the representative configurations shown in Fig. 5.5. Here δ_0 is the splitting of the lowest even-odd pair, Δ_{gap} is the gap separating this pair from the next excited states, Q_0 is the total charge difference between the parity partners, $\overline{|M_{\text{edge}}|}$ is the mean absolute Majorana polarization on the first and last sites, and $\max |M_{\text{bulk}}|$ is the largest absolute polarization on the interior sites.

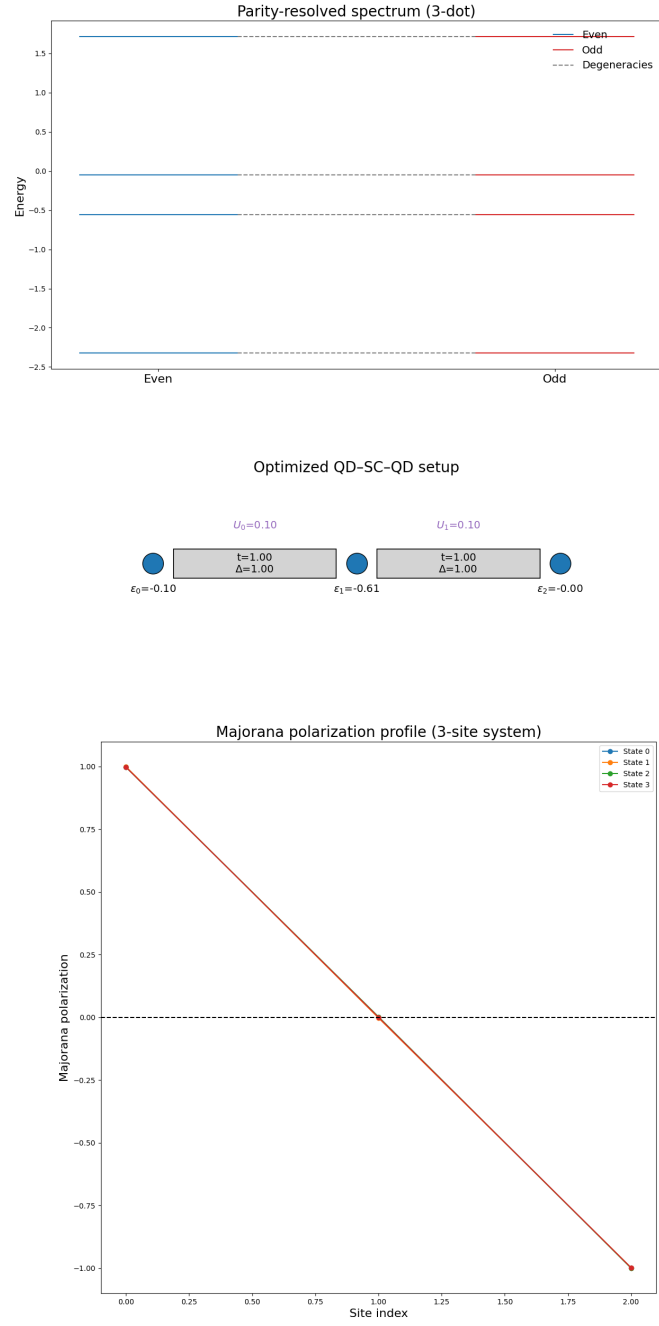


Figure 4.3: Parity-resolved spectrum and Majorana-polarization profile for the optimized three-dot system at $U = 0.1$. The left panel shows the optimized spectrum and parameters, while the right panel shows the Majorana polarization of the lowest even-odd pair.

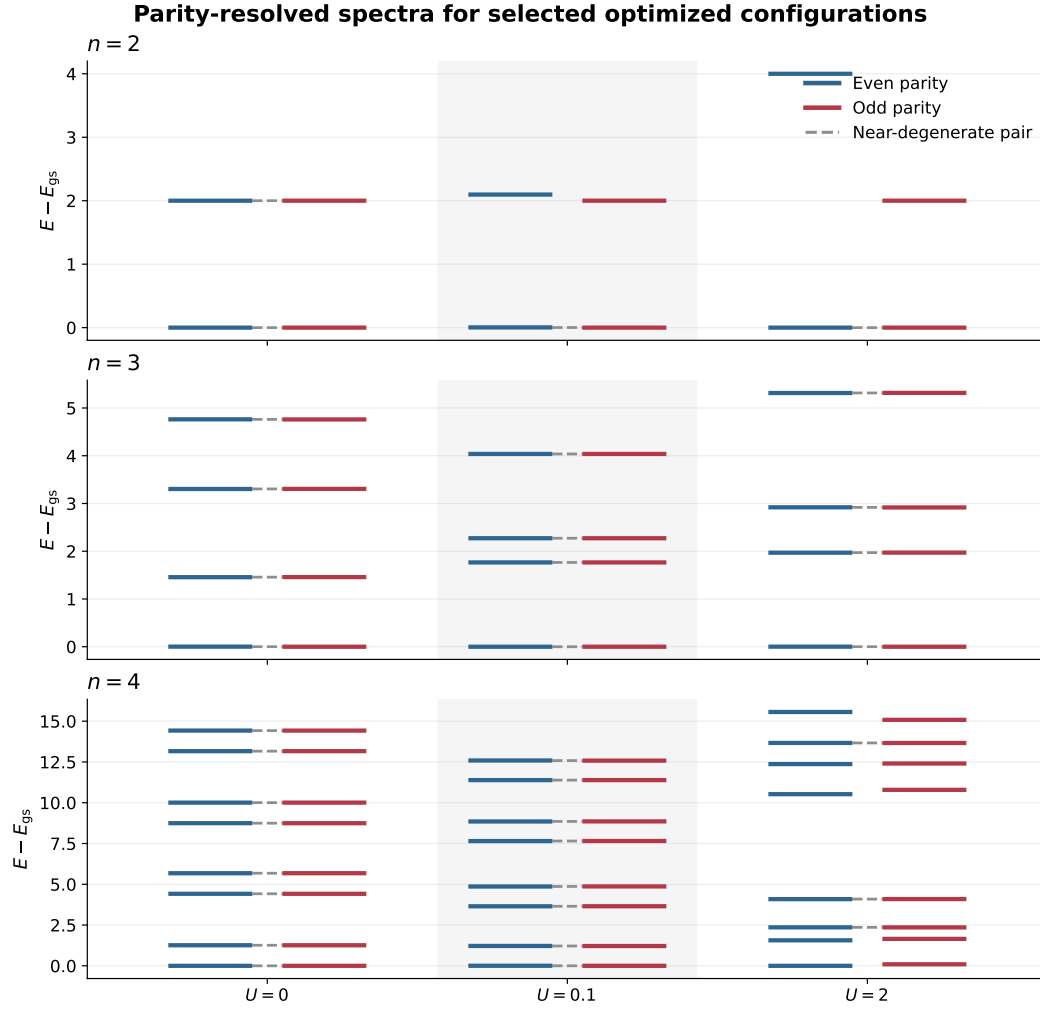


Figure 4.4: Parity-resolved spectra for the best optimized configurations at selected interaction strengths. Each panel corresponds to a different system size n , and energies are shifted so that the ground state is at zero.

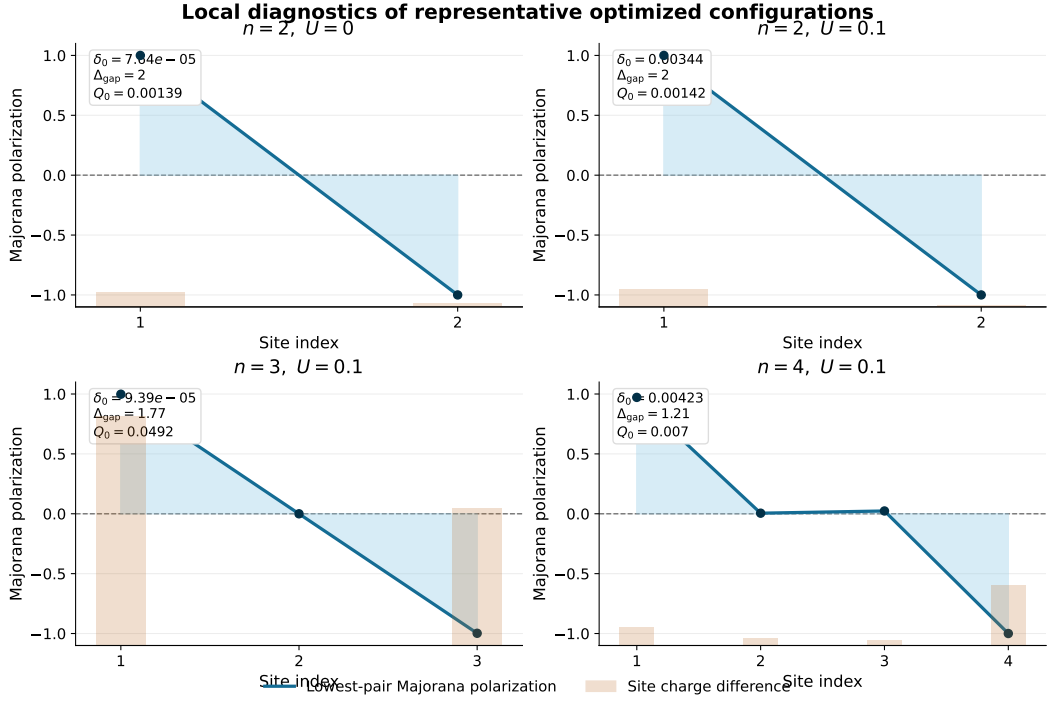


Figure 4.5: Local diagnostics for representative optimized configurations. The blue curve shows the Majorana polarization profile of the lowest even-odd pair, while the orange bars show the local charge difference $|\langle e|n_i|e\rangle - \langle o|n_i|o\rangle|$ on each site. The inset in each panel reports the ground-state splitting δ_0 , the low-energy gap Δ_{gap} , and the total charge difference Q_0 .

essentially ideal, with $|\overline{M_{\text{edge}}}| \approx 1$ and no bulk contribution by construction. The total charge difference is also very small, $Q_0 \approx 1.4 \times 10^{-3}$, both at $U = 0$ and at $U = 0.1$. These two-dot cases are useful references, but because they have no interior sites they do not test suppression of Majorana weight in the bulk.

The three-dot configuration at $U = 0.1$ is the clearest interacting example of the expected Majorana pattern. The ground-state splitting is only $\delta_0 = 9.39 \times 10^{-5}$, the excitation gap remains large, $\Delta_{\text{gap}} = 1.77$, and the bulk polarization is almost completely suppressed, with $\max |M_{\text{bulk}}| = 2.02 \times 10^{-4}$. In contrast, the four-dot configuration at $U = 0.1$ has comparably strong edge polarization, $|\overline{M_{\text{edge}}}| \approx 0.986$, but the splitting increases to $\delta_0 = 4.23 \times 10^{-3}$ and the bulk polarization rises to $\max |M_{\text{bulk}}| = 2.33 \times 10^{-2}$.

Taken together, these diagnostics show that different system sizes optimize different aspects of the Majorana criteria. The four-dot case performs well in terms of charge neutrality, while the three-dot case performs better in terms of spectral near-degeneracy and suppression of bulk Majorana weight. Since the braiding analysis requires a low-energy subspace that is both nearly degenerate and spatially well localized, the interacting three-dot system remains the most convincing candidate for the rest of the thesis.

4.3 Selection of Braiding Candidate

The braiding calculations are performed using the optimized three-dot configuration at $U = 0.1$. This choice is based on the combined diagnostics above: the configuration has a small ground-state splitting, a large gap to the next states, a clear edge-localized

Majorana-polarization profile, and strongly suppressed bulk polarization. The optimized parameters used for the braiding setup are

$$\Delta_1 = 1.001, \quad \Delta_2 = 1.002, \quad \epsilon_1 = -0.101, \quad \epsilon_2 = -0.607, \quad \epsilon_3 = -0.003, \quad (4.1)$$

with fixed hopping amplitudes $t_i = 1$ and fixed interaction strength $U = 0.1$.

For a three-dot chain, the single-subsystem Hilbert space has dimension $2^3 = 8$. In the braiding geometry, three such subsystems are combined into a total Hilbert space of dimension $2^9 = 512$. This makes the three-dot system large enough to test braiding in nontrivial projected spaces, while still being small enough for exact diagonalization and systematic projection scans.

4.4 Braiding in Projected Majorana Models

Before studying large projection dimensions, it is useful to establish a controlled reference calculation. I first compare the ideal four-Majorana braid, the projected microscopic junction braid, and the projected local-operator braid in the low-energy projected setting. This tests whether the optimized microscopic system reproduces the exchange behavior expected from the ideal effective model.

4.4.1 Ideal four-Majorana reference

Figure 5.7 shows the reference braid in the effective four-Majorana model. The three couplings are turned on sequentially, so that the dominant pairing is first between γ_0 and γ_1 , then between γ_0 and γ_2 , and finally between γ_0 and γ_3 before returning to the initial configuration. Throughout this cycle, the low-energy manifold remains essentially four-fold degenerate. The largest instantaneous splitting within the ground-state manifold is only $\max(E_3 - E_0) = 2.28 \times 10^{-12}$, while the minimum gap to the excited states remains open at $\min(E_4 - E_3) = 1.414$.

Using Kato transport, the resulting unitary reproduces the expected single-exchange action on the Majorana operators. The largest error in the transformations $\gamma_2 \rightarrow -\gamma_3$ and $\gamma_3 \rightarrow \gamma_2$ is 1.33×10^{-5} . In the parity-resolved gate blocks, the odd and even 2×2 blocks agree with the target exchange gate to within 6.53×10^{-8} , while the off-block leakage is only 1.02×10^{-11} . The effective model therefore provides a clean benchmark for the more microscopic projected calculations.

4.4.2 Projected microscopic and local-operator braids

After replacing the hand-chosen couplings of the ideal model by projected microscopic junction operators, the braid remains close to the reference result. Figure 5.8 shows that the four-dimensional low-energy manifold is still well separated from higher states throughout the cycle. The maximum splitting inside the ground-state manifold increases only to $\max(E_3 - E_0) = 7.18 \times 10^{-7}$, while the minimum gap remains large, $\min(E_4 - E_3) = 1.413$.

The junction decomposition in Fig. 5.9 explains why the projected microscopic braid agrees so well with the ideal reference. The projected AB coupling is dominated by the bilinear $i\gamma_{A1}\gamma_{B2}$ with coefficient -0.999 , and the projected AC coupling is dominated by $i\gamma_{A1}\gamma_{C2}$ with the same coefficient. This identifies γ_{B2} and γ_{C2} as the Majoranas that participate in the microscopic braid, while γ_{A1} acts as the shared Majorana through which the exchange is mediated.

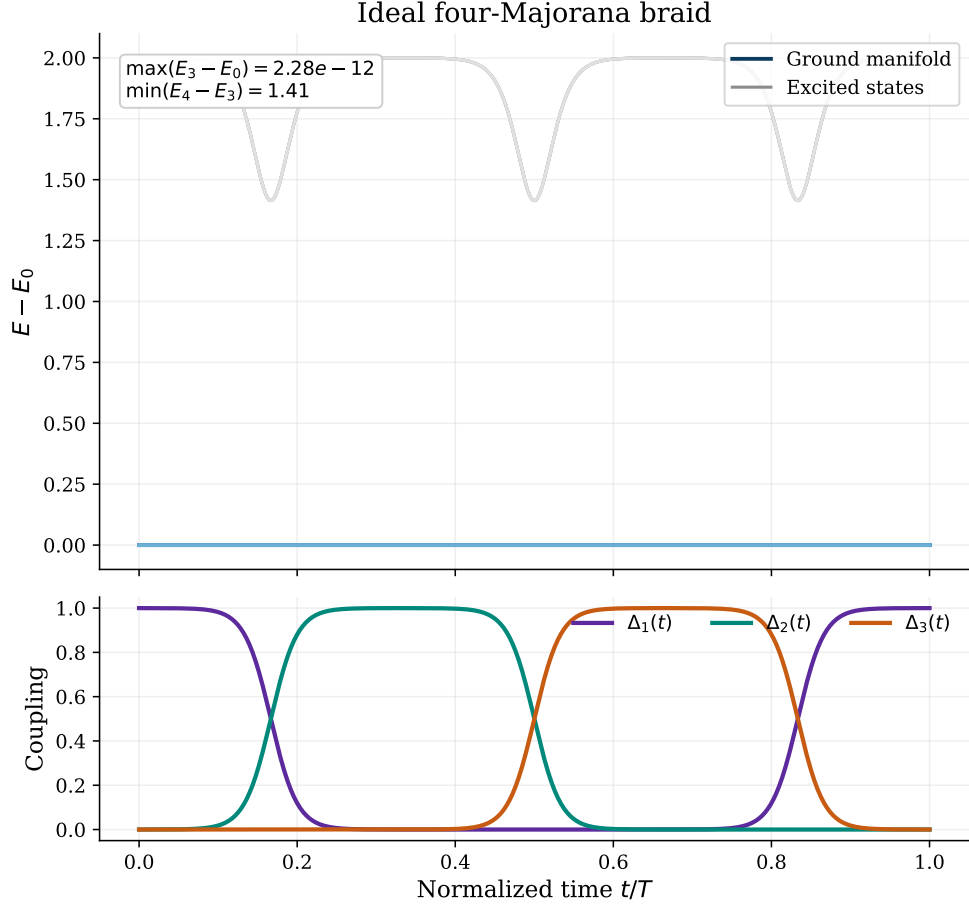


Figure 4.6: Idealized four-Majorana braiding protocol. The top panel shows the instantaneous spectrum of the effective Hamiltonian during the braid, with all energies shifted so that the lowest state remains at zero. The bottom panel shows the three smooth pulse couplings that move the dominant Majorana pairing through the network.

Model	$\max(E_3 - E_0)$	$\min(E_4 - E_3)$	Max 1x err.	Leakage	Odd err.	Even err.
Ideal effective model	2.275e-12	1.414	1.334e-05	1.016e-11	6.535e-08	6.535e-08
Projected microscopic junction model	7.175e-07	1.413	1.334e-05	8.844e-16	6.538e-08	6.538e-08
Projected local-operator model	4.663e-15	1.414	1.334e-05	7.287e-16	6.537e-08	6.537e-08

Table 4.3: Comparison of the main braid diagnostics for the ideal effective model, the projected microscopic junction model, and the alternative projected local-operator model. Here $\max(E_3 - E_0)$ measures the largest instantaneous splitting inside the four-dimensional low-energy manifold, $\min(E_4 - E_3)$ is the minimum gap to the excited sector, “Max 1x err.” is the largest deviation in the expected single-exchange Majorana map, “Leakage” is the off-block norm in the parity basis, and the last two columns compare the odd and even parity blocks with the target exchange gate.

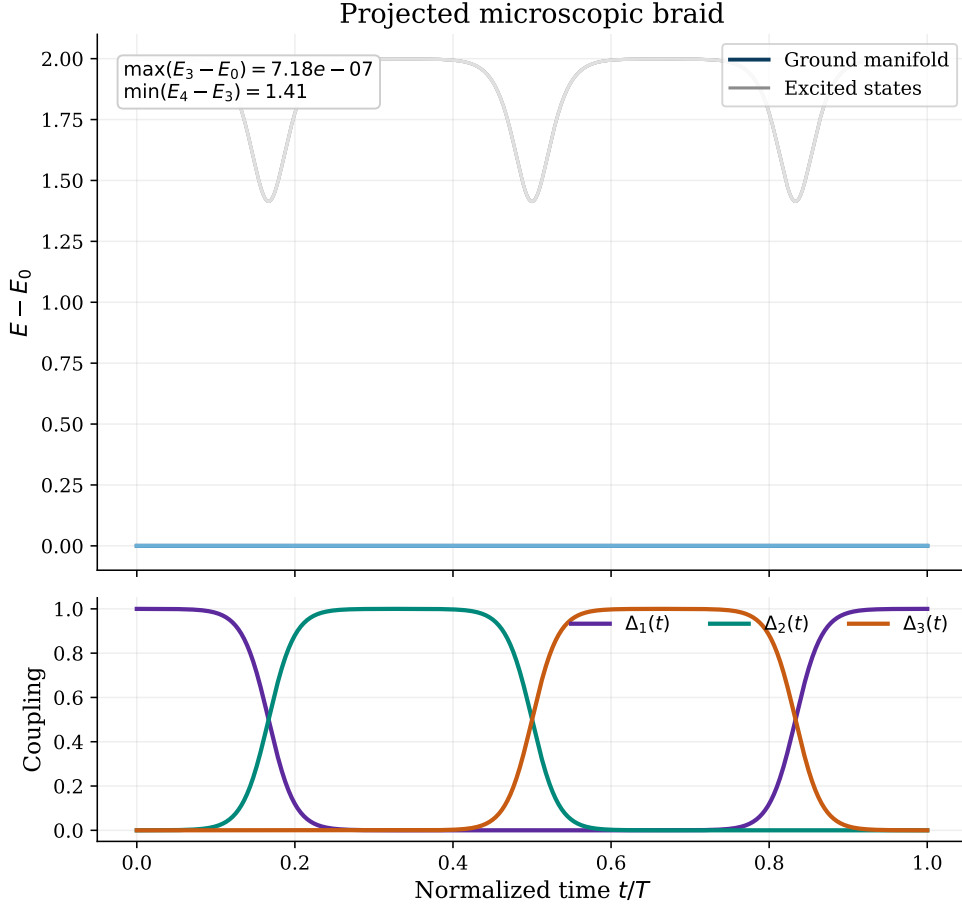


Figure 4.7: Braiding protocol after replacing the idealized couplings by projected microscopic junction operators. The spectral evolution remains almost identical to the effective-model benchmark, with a well-isolated low-energy manifold throughout the cycle.

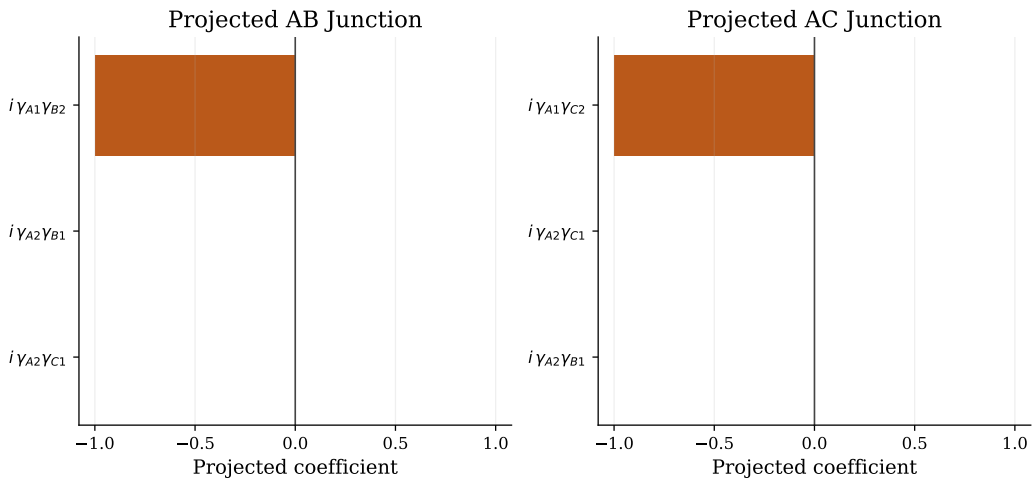


Figure 4.8: Dominant Majorana-bilinear components of the projected microscopic junction operators. The AB junction is almost entirely $i\gamma_{A1}\gamma_{B2}$, while the AC junction is almost entirely $i\gamma_{A1}\gamma_{C2}$. All subleading bilinears are strongly suppressed.

Projected operator	Dominant Majorana	Coefficient
$\chi_{B1,y}$	γ_{B2}	-1.000
$\chi_{B2,y}$	γ_{B2}	1.000
$\chi_{C1,y}$	γ_{C2}	-1.000
$\chi_{C2,y}$	γ_{C2}	1.000
Pairwise relation errors		
$\chi_{B1,y} \approx -\chi_{B2,y}$	opposite sign	8.969e-14
$\chi_{C1,y} \approx -\chi_{C2,y}$	opposite sign	5.787e-14

Table 4.4: Projected local-site operators used in the alternative braid model. The table shows that the y -type local operators on the first and last sites of subsystems B and C collapse onto the same low-energy Majorana channels, γ_{B2} and γ_{C2} , up to an overall sign.

Table 5.3 shows that the microscopic braid reproduces the ideal benchmark extremely well. The largest single-exchange error stays at 1.33×10^{-5} in all three models, and the odd and even parity blocks continue to match the target gate to approximately 6.54×10^{-8} . The projected local-operator version produces the same conclusion, which makes it a useful consistency check rather than an independent competing protocol.

Table 5.4 explains why the local-site version appears insensitive to whether the first or last site of subsystem B or C is used. After projection, both $\chi_{B1,y}$ and $\chi_{B2,y}$ reduce to γ_{B2} up to opposite sign, and similarly both $\chi_{C1,y}$ and $\chi_{C2,y}$ reduce to γ_{C2} . The different microscopic site choices therefore feed into the same low-energy braid channel rather than producing genuinely different braiding operations.

4.5 Braiding in Larger Projection Spaces

The previous braiding checks focused on the lowest projected manifold. To test how much of that agreement survives once higher sectors are retained, I repeated the braid diagnostics for projection dimensions $\dim P = 8, 32, 56, 80, 256$, and 512, and for interaction strengths $U = 0, 0.1$, and 2.0. For each projected Hamiltonian, the transported band is chosen from the largest persistent spectral gap along the braid path. This separates the projection dimension, which controls how much Hilbert space is retained, from the transported dimension $\dim T$, which controls which isolated band is actually braided.

There are two effective target constructions in this scan. In the energy-level braid, the Majorana operators are built from paired even and odd eigenstates of the subsystem Hamiltonians. In the local-operator braid, the relevant operators are the projected local endpoint operators $Pi(c^\dagger - c)P$. The energy-level construction tests whether the many-body spectrum supports a clean exchange channel, while the local-operator construction tests whether that channel is accessed by a local microscopic coupling.

Table 5.5 shows that the corrected diagonal energy-level construction gives a very clean braid target. The important comparison is therefore not simply whether the braid unitary is accurate, but which effective operator it is accurate with respect to. The energy-level construction is adapted to the many-body spectrum, while the local-operator construction is constrained to the projected microscopic endpoint operator.

Figures 5.10 to 5.14 show the main distinction between the two braid constructions. The energy-level braid is the cleaner effective operation because it is built directly from paired eigenstates of the optimized Hamiltonians. The local-operator braid is more physically constrained because it asks what is generated by the projected microscopic endpoint operator. The fact that the energy-level braid performs better should therefore

U	$\dim P$	$\dim T$	Phys. vs. γ target	Phys. vs. Γ^P target	Target mismatch	Γ^P algebra
0	8	4	1.15e-06	1.15e-06	2.54e-16	1.65e-15
0	32	8	1.15e-06	9.80e-05	9.86e-05	1.08e-04
0	56	8	1.15e-06	1.36e-04	1.37e-04	7.57e-05
0	80	12	1.15e-06	1.10e-04	1.11e-04	1.06e-04
0	256	16	1.15e-06	1.32e-04	1.32e-04	8.44e-05
0	512	16	1.15e-06	1.52e-04	1.53e-04	2.99e-15
0.1	8	4	9.46e-07	9.46e-07	7.85e-17	3.20e-15
0.1	32	8	1.59e-04	1.59e-04	3.68e-07	3.67e-04
0.1	56	12	1.23e-04	2.67e-04	2.90e-04	5.90e-04
0.1	80	12	4.18e-05	2.84e-04	2.73e-04	5.82e-04
0.1	256	16	1.58e-04	4.24e-04	5.16e-04	5.99e-04
0.1	512	16	4.20e-04	5.93e-04	9.37e-04	3.22e-15
2	8	4	1.02e-06	1.02e-06	2.62e-16	1.95e-15
2	32	8	2.35e-02	2.10e-02	5.43e-03	5.77e-02
2	56	12	8.34e-02	5.10e-02	3.43e-02	1.75e-01
2	80	12	7.81e-02	5.37e-02	2.69e-02	1.53e-01
2	256	16	1.76e-01	4.09e-02	1.38e-01	1.50e-01
2	512	16	3.09e-01	2.04e-02	3.21e-01	3.15e-15

Table 4.5: Target diagnostic for the corrected higher-energy braid scan with $\dim P = 8, 32, 56, 80, 256, 512$. All errors are normalized in the transported band. “Phys. vs. γ target” compares the local-operator braid to $\exp(-\pi\gamma_2\gamma_3/4)$, where γ_2 and γ_3 are constructed from diagonal even-odd energy-level pairs. “Phys. vs. Γ^P target” compares the same braid to $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$, where Γ_2^P and Γ_3^P are projected local Majorana operators. “Target mismatch” directly compares these two effective target gates.

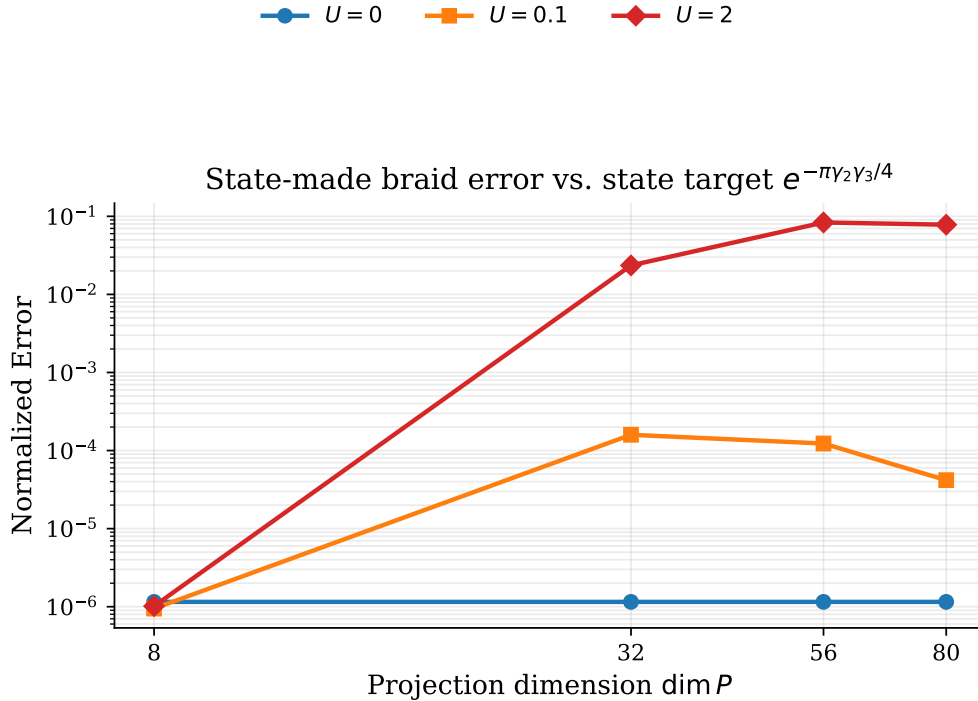


Figure 4.9: Energy-level braid error up to $\dim P = 80$. The braid operators are constructed from diagonal even-odd energy-level pairs of the subsystem Hamiltonians and are compared with the corresponding energy-level target $\exp(-\pi\gamma_2\gamma_3/4)$.

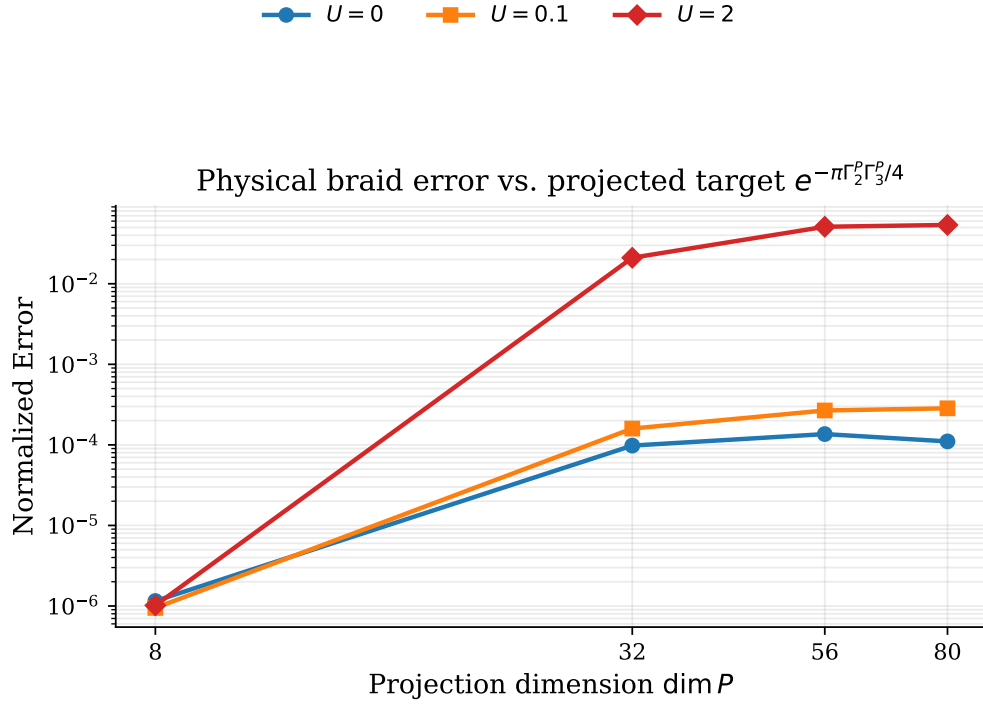


Figure 4.10: Local-operator braid error up to $\dim P = 80$. Here the braid is generated and judged using the projected local endpoint operators $Pi(c^\dagger - c)P$.

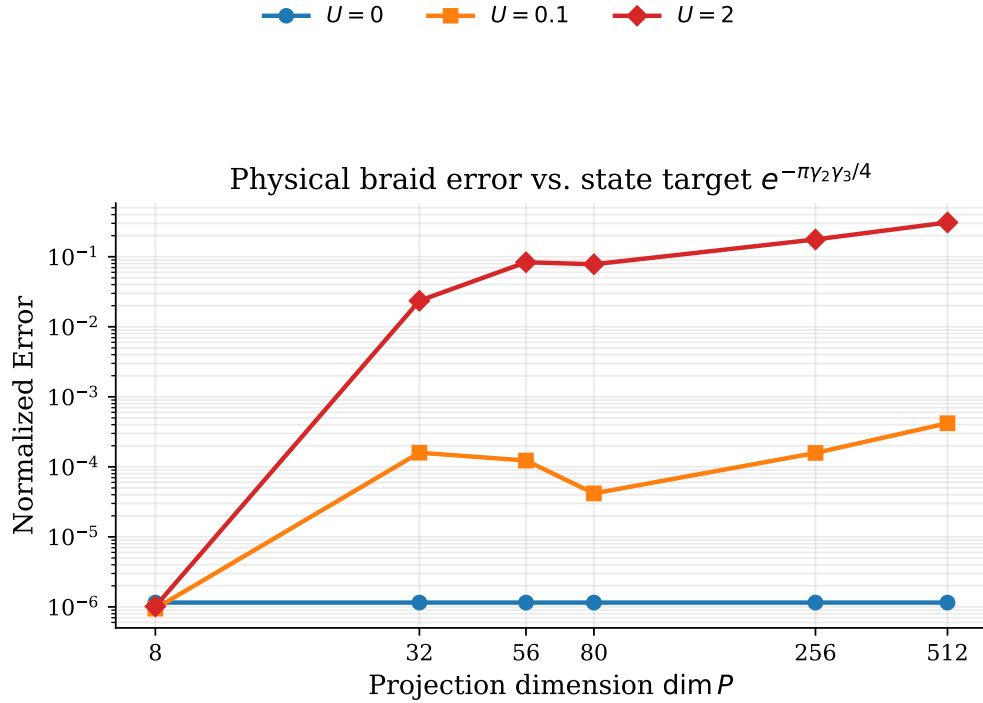


Figure 4.11: Energy-level braid error up to $\dim P = 512$. Extending the projection dimension tests whether the same eigenstate-pair construction remains stable once higher-energy sectors are retained.

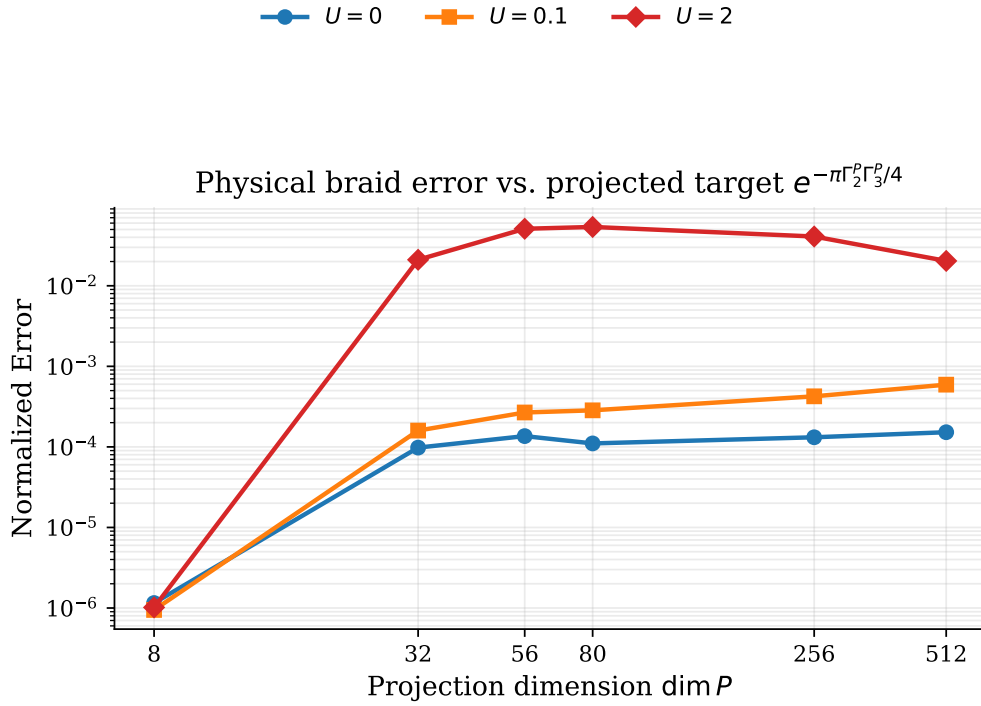


Figure 4.12: Local-operator braid error up to $\dim P = 512$, evaluated against the projected local target $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$.

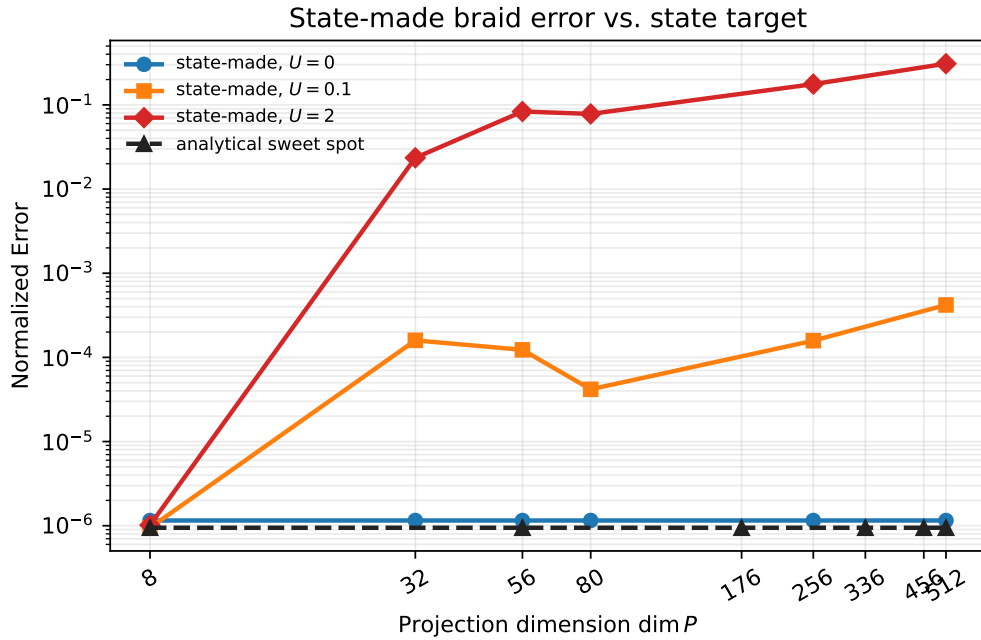


Figure 4.13: Energy-level braid error up to $\dim P = 512$, with the analytical non-interacting sweet-spot braid included as a reference. The analytical curve is evaluated only at projection dimensions that complete exact degeneracy blocks.

be interpreted as evidence that the many-body spectrum contains a clean exchange channel, not as automatic proof that the corresponding operator is perfectly local. The local-operator results provide the complementary localization and implementability check.

4.6 Summary of Main Findings

The optimization results show that the known two-dot sweet spot is recovered in the non-interacting limit, while interactions and larger chain lengths lead to increasingly inhomogeneous optimized parameter profiles. The post-optimization diagnostics identify the three-dot system at $U = 0.1$ as the strongest interacting candidate, because it combines small ground-state splitting, a large low-energy gap, strong edge Majorana polarization, and strongly suppressed bulk polarization.

The low-energy braiding results show that this optimized configuration reproduces the ideal Majorana exchange very accurately after projection. The projected microscopic junction operators are dominated by the intended Majorana bilinears, and the ideal, microscopic, and local-operator braid models give nearly identical gate-validation errors in the lowest projected manifold.

The larger projection scans show that the distinction between energy-level and local-operator braiding becomes important beyond the lowest projected space. The energy-level construction identifies a clean exchange channel in the many-body spectrum, while the local-operator construction tests how well that channel is accessed by projected microscopic endpoint operators. This comparison is central to interpreting the higher-dimensional braid results.

Chapter 5

Results OLD

This section presents the main numerical results of the thesis. I first report the outcome of the parameter optimization, then analyze the spectral and local signatures of the optimized configurations, and finally study whether the resulting low-energy states support the intended braiding behavior. The broader interpretation of these findings and their main limitations are deferred to the discussion section.

5.1 Parameter Optimization Results

To obtain an overview of the optimization landscape, I first compare the lowest-loss configuration found for each combination of system size n and fixed interaction strength U . This separates the global trends of the optimization procedure from the more detailed diagnostics of individual configurations. For the two-dot system, the optimizer reproduces the known non-interacting sweet spot at $U = 0$, while increasing interaction strength gradually shifts the optimal on-site energies and pairing amplitudes away from that limit. For three and four dots, the optimization landscape becomes more complex, and the best solutions show stronger inhomogeneity in both the onsite energies and the pairing amplitudes.

5.1.1 Dependence on interaction strength and system size

Table 5.1 summarizes the lowest-loss configuration found for each combination of system size n and fixed interaction strength U . Since the hopping amplitudes were fixed to $t_i = 1$ in these runs, only the optimized pairing amplitudes and on-site energies are listed. Empty entries indicate parameters that do not exist for that system size.

The reason for a fixed $t_i = 1$ is that if we do not lock the hopping amplitudes, the optimizer can simply increase t_i in order to replicate a non-interacting sweet spot even at strong interactions, which is not the intended behavior. By fixing t_i , we force the optimizer to find nontrivial parameter profiles that satisfy the Majorana-based loss criteria without relying on trivial rescaling. Several trends are immediately visible. For the two-dot system, the optimizer reproduces the expected non-interacting sweet spot at $U = 0$, where $\Delta_1 \approx t_1$ and the on-site energies remain very close to zero. As the interaction strength is increased, the loss grows steadily, the optimized pairing amplitude increases above the fixed hopping, and the on-site energies move approximately toward $\epsilon_i \approx -U/2$. This indicates that the optimizer is deforming the well-known two-dot sweet spot in a smooth and physically interpretable way.

When the interaction strength is increased further, the loss continues to grow, and the

n	U	Loss	Δ_1	Δ_2	Δ_3	ϵ_1	ϵ_2	ϵ_3	ϵ_4
2	0.0	5.657e-03	1.000	—	—	-0.001	0.002	—	—
2	0.1	2.617e-01	1.047	—	—	-0.050	-0.047	—	—
2	0.5	1.083e+00	1.249	—	—	-0.255	-0.249	—	—
2	1.0	1.940e+00	1.498	—	—	-0.499	-0.496	—	—
2	2.0	3.203e+00	2.000	—	—	-1.006	-1.004	—	—
2	5.0	4.004e+00	3.499	—	—	-2.508	-2.506	—	—
3	0.0	3.442e-01	1.010	1.348	—	0.019	1.774	-0.466	—
3	0.1	5.603e-01	1.001	1.002	—	-0.101	-0.607	-0.003	—
3	0.5	9.359e-01	1.004	1.012	—	-0.046	-0.889	-0.454	—
3	1.0	1.452e+00	1.025	1.130	—	-0.195	-1.530	-0.802	—
3	2.0	2.326e+00	1.584	0.978	—	-1.511	-3.077	-0.484	—
3	5.0	3.142e+00	0.846	1.746	—	-4.814	1.009	-0.184	—
4	0.0	3.161e-01	2.698	5.332	1.004	-0.059	-4.252	0.236	-0.003
4	0.1	4.840e-01	1.790	5.002	1.013	0.156	2.376	-1.966	-0.050
4	0.5	1.006e+00	1.212	3.490	1.206	-0.246	-0.499	-0.503	-0.250
4	1.0	5.670e-01	2.238	18.807	1.072	-0.500	-1.028	-1.025	-0.496
4	2.0	1.824e+00	1.492	5.537	1.588	-0.869	3.526	-6.508	-1.220
4	5.0	2.600e+00	3.389	9.513	3.336	-2.645	5.032	-15.431	-2.293

Table 5.1: Lowest-loss optimized configurations extracted from `configuration.json` for each combination of system size n and fixed interaction strength U .

optimized parameters deviate more strongly from the non-interacting limit. This can be due to the fact that the system would need stronger hopping and pairing amplitudes to maintain the Majorana-like low-energy structure in the presence of stronger interactions, but since t_i is fixed, the optimizer can only increase Δ_i and adjust ϵ_i to partially compensate. The structure of the loss function could also suggest that it favors lower parameter values, because when the parameters become too large, the energy scales of the system increase which could lead to numerically larger losses, even though the Majorana-like physics might still be present. This is a subtle point that could be clarified by analyzing the loss landscape in more detail, but the overall trend is that stronger interactions push the system away from the simple non-interacting sweet spot and require more significant parameter adjustments to maintain low loss.

5.1.2 Trends in the optimized parameters

For all three system sizes in 5.1, the best configurations are found where the interaction strength is relatively weak. For all three cases the best solution is found at $U = 0$, which is not surprising since we have well known non-interacting sweet spots that the optimizer can easily find.

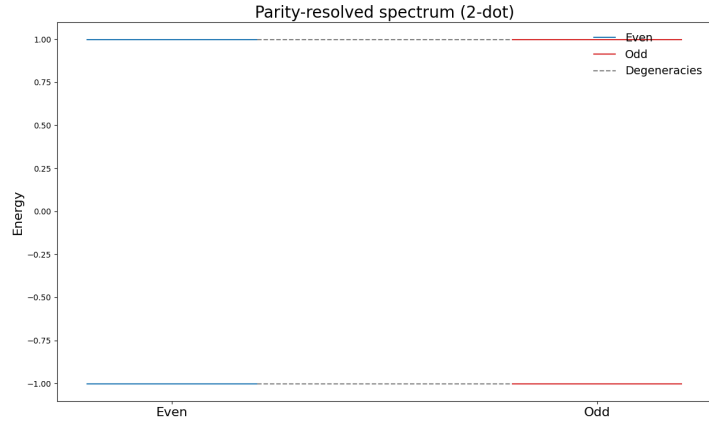
For the two-dot system, the optimized parameters at $U = 0$ are very close to the ideal non-interacting sweet spot, with $\Delta_1 \approx t_1 = 1$ and $\epsilon_i \approx 0$. This is also the analytically known solution for the two-dot system, so it serves as a sanity check that the optimizer can reproduce this limit, which is a good indicator that the optimization procedure is working correctly. With a small interaction of $U = 0.1$, the optimized parameters deviate slightly from the non-interacting sweet spot, with Δ_1 increasing to about 1.05. Figure 5.1 and 5.2 show how for when increasing the interaction strength to $U = 0.1$ and $U = 2.0$, the parity-resolved spectrum changes, and the excited state becomes more

and more split for higher interaction strengths, which is consistent with the fact that the two-dot sweet spot only guarantees degeneracy of the ground states, and not of the excited states.

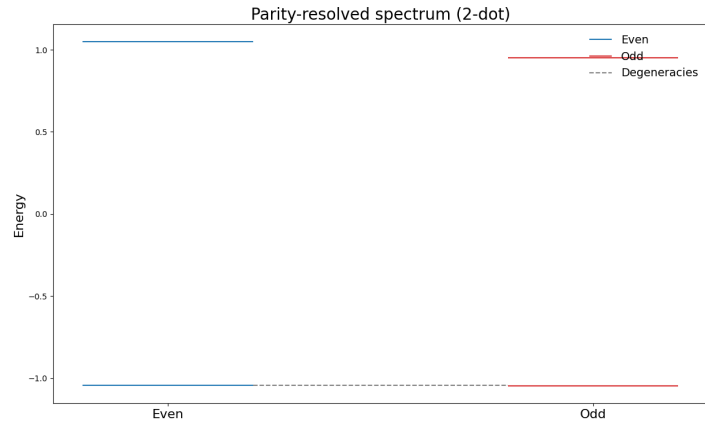
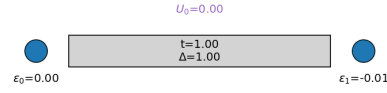
For the three-dot system, the best configurations at weak and intermediate interaction strengths still remain relatively close to the homogeneous limit, with $\Delta_1 \approx \Delta_2 \approx 1$ for $U = 0.1$ and $U = 0.5$. At the same time, the middle-site energy becomes substantially more negative than the edge-site energies, suggesting that the optimization favors a differentiated central region even when the pairing amplitudes remain nearly uniform. At larger interaction strengths, both the loss and the parameter asymmetry increase, indicating that the system is moving away from the simplest weak-coupling Majorana picture. In figure 5.3 we can see the parity resolved spectrum and the Majorana polarization profile for the three-dot system at $U = 0.1$. We can see that the parameter optimizer found a set of parameters where all 4 energy levels are degenerate, even when the system is interacting. Looking at the plot showing the Majorana polarization, we can see that the three dots have the expected MP values, where the left quantum dot has a MP close to +1, the right quantum dot has a MP close to -1, and the middle dot has a MP close to zero.

The four-dot system shows the clearest sign that the optimization landscape becomes more complicated as the chain length increases. In several of the best configurations, the optimizer strongly enhances the central pairing amplitudes while leaving the outer pairing amplitudes much smaller. This is especially visible for $U = 0$, $U = 0.1$, and $U = 1.0$, where the middle bond is favored much more strongly than the edge bonds. The optimized on-site energies also become highly inhomogeneous, particularly at stronger interactions. Taken together, these results suggest that for longer PMM chains the optimizer does not simply preserve the two-dot sweet-spot structure, but instead searches for more nontrivial inhomogeneous parameter profiles that still satisfy the Majorana-based loss criteria.

An interesting repeated occurrence in the optimized parameters is that the optimizer tends to favor larger chemical potentials for the middle sites, $\epsilon_m \gg \epsilon_e$, where m and e refer to middle and edge sites respectively. This is especially visible for the three-dot system, where the middle-site energy becomes substantially more negative than the edge-site energies even at weak interactions 5.1. Meaning in order to create a Majorana-like structure at the edge dots, the optimizer finds it favorable to create a strong potential well in the middle of the chain. This could be interpreted as a way for the system to effectively decouple the edge sites from each other, which is a key requirement for realizing Majorana modes. By creating a strong potential barrier in the middle, the optimizer may be trying to localize the low-energy states at the edges and prevent them from hybridizing with each other through the bulk of the chain. This trend is also visible in the four-dot system, where the optimized on-site energies become highly inhomogeneous, with larger chemical potentials for the middle sites compared to the edge sites. This suggests that even in longer chains, the optimization procedure may favor parameter profiles that enhance localization at the edges by creating potential barriers in the middle of the chain.



Optimized QD-SC-QD setup



Optimized QD-SC-QD setup

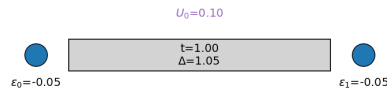


Figure 5.1: The two plots show the parity resolved spectrum after optimization for the two-dot system at $U = 0$ and $U = 0.1$ respectively. The optimized parameters are shown schematically below each plot. We can see that for $U = 0$, both energy levels are exactly degenerate, while for $U = 0.1$ the degeneracy of the excited state is slightly lifted.

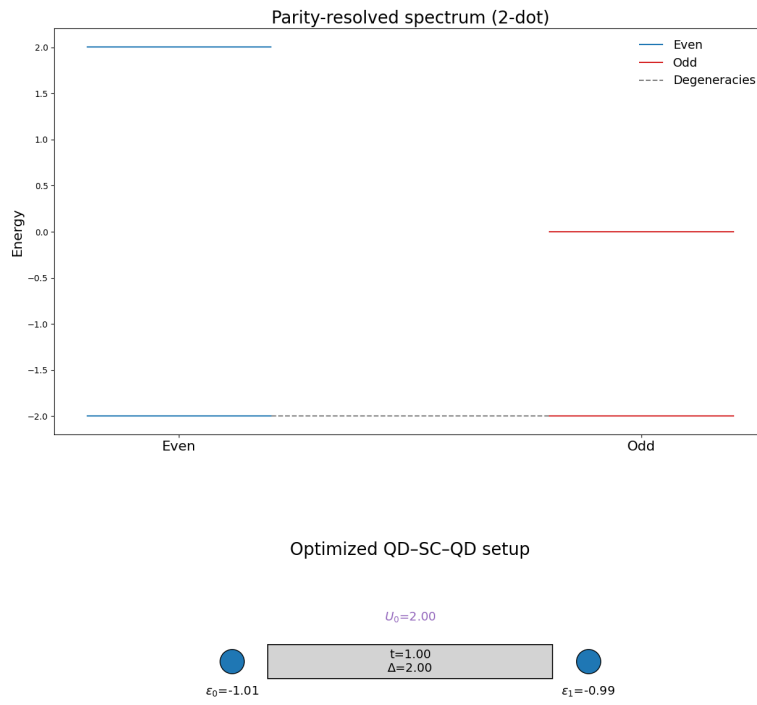


Figure 5.2: The plot shows the parity resolved spectrum after optimization for the two-dot system at $U = 2.0$. The optimized parameters are shown schematically below the plot. We can see that for $U = 2.0$, the degeneracy of both the ground state and the excited state is lifted

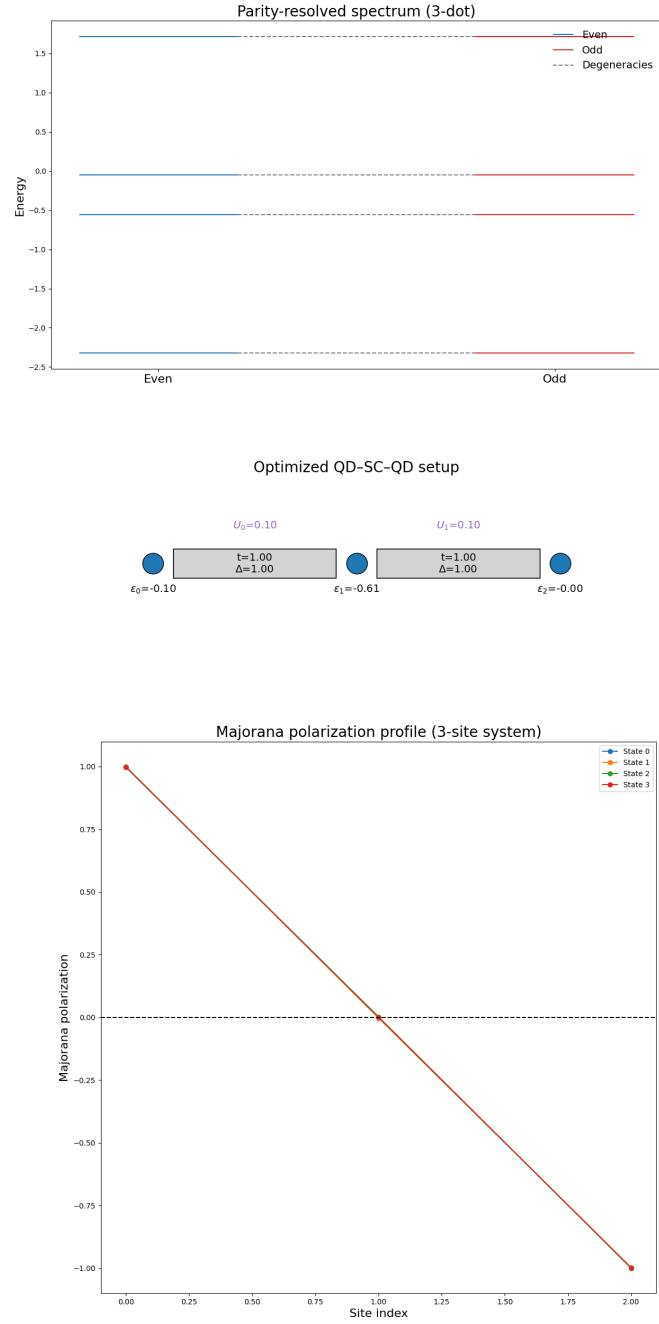


Figure 5.3: The two plots show the parity resolved spectrum after optimization for the three dot system, with interaction strength $U = 0.1$. The optimized parameters are shown schematically below the left plot, and the Majorana polarization profile of the lowest-energy states is shown in the right plot.

5.2 Post-Optimization Diagnostics

After presenting the optimized parameters themselves, the next step is to test whether the corresponding Hamiltonians display the expected signatures of Majorana-like physics. This section should therefore focus on the parity-resolved spectrum and on the local observables introduced in the method chapter. After finding the optimized parameters, we can use them to set up the Hamiltonian for the desired number of quantum dots. After the parameter optimizing scheme, we found promising parameters for the interacting 3-dot system. Using the parameters found in the optimization scheme for $U = 0.1$, we can diagonalize the Hamiltonian and look at the parity-resolved spectrum to create Majorana-like states.

$$H = \sum_{i=1}^3 \epsilon_i n_i + \sum_{i=1}^2 t_i (c_i^\dagger c_{i+1} + h.c.) + \sum_{i=1}^2 \Delta_i (c_i c_{i+1} + h.c.) + U \sum_{i=1}^3 n_i n_{i+1} \quad (5.1)$$

Diagonalizing we will obtain $2^3 = 8$ energy levels, which can be grouped into two parity sectors with four levels each. The parameters found are optimized to make the four levels in each parity sector as close to degenerate as possible, which is a key signature of Majorana-like physics. In figure 5.3 we can see the parity resolved spectrum for the three-dot system at $U = 0.1$. Using these parameters we are able to create a set of Majorana-like states:

$$\begin{aligned} \gamma_1 &= \sum_{i=1}^{n=4} |e\rangle \langle o| + |o\rangle \langle e| \\ \gamma_2 &= i \sum_{i=1}^{n=4} (|e\rangle \langle o| - |o\rangle \langle e|) \end{aligned}$$

with these Majorana operators, we are able to choose the number of energy levels we want to include in the operators, and we are able to explore the properties of these operators and if they are stable under the braiding protocol.

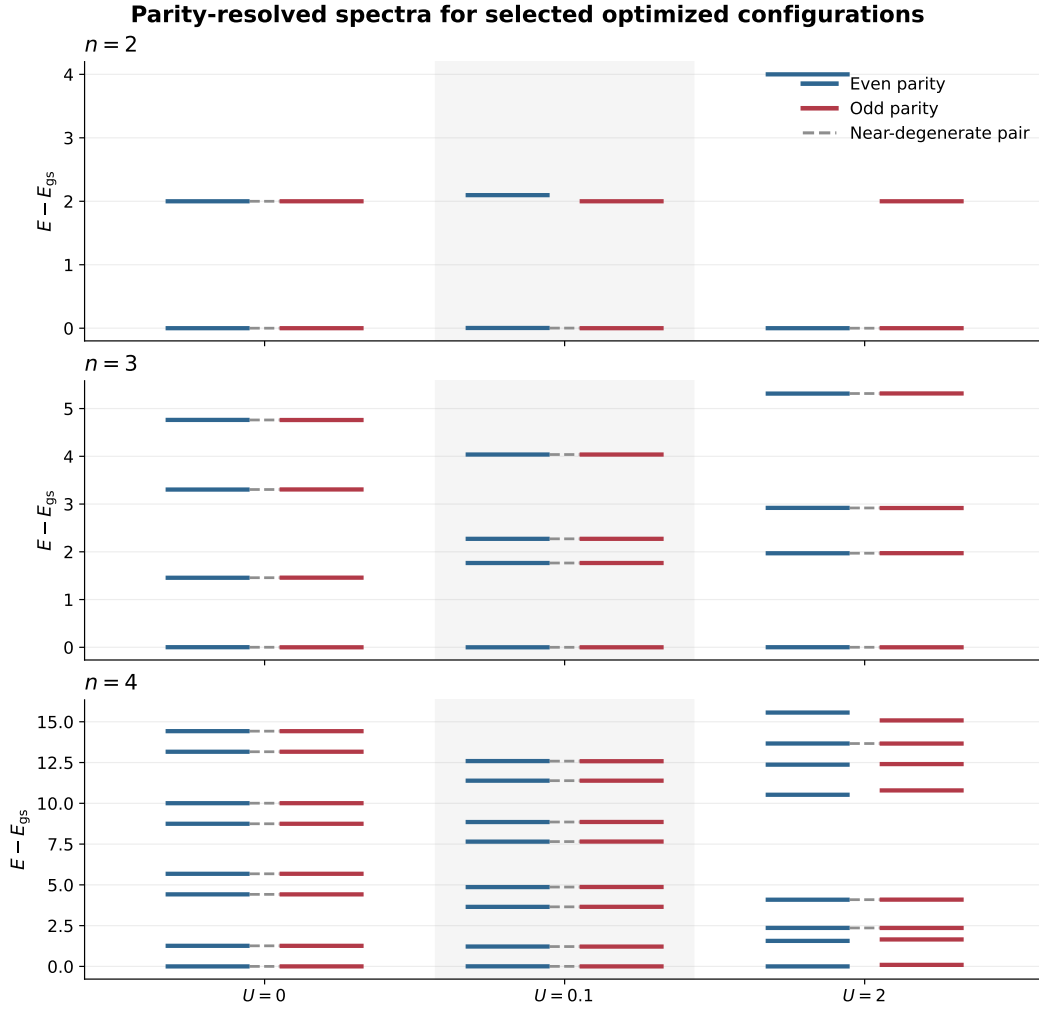


Figure 5.4: Parity-resolved spectra for the best optimized configurations at selected interaction strengths. Each panel corresponds to a different system size n , and energies are shifted so that the ground state is at zero.

5.2.1 Parity-resolved spectra and low-energy gaps

Figure 5.4 shows the parity-resolved spectra for the best optimized configurations at selected interaction strengths. Each panel corresponds to a different system size $n = 2, 3$ and 4 , with interaction strengths set at $U = 0$, $U = 0.1$, and $U = 2$ and energies are shifted so that the ground state is at zero.

We can see that for all system sizes, the optimizer managed to find a set of parameters where the four energy levels in each parity sector are close to degenerate in the non-interacting system. The first sign of instability appears at $U = 0.1$, for the two-dot system, where the degeneracy of the excited state is slightly lifted. This effect becomes more pronounced at stronger interactions, where the excited states are completely split for $U = 2.0$.

The four dot system shows a different trend. The optimizer finds a stable set of parameters for both $U = 0$ and $U = 0.1$, where the four energy levels in each parity sector are close to degenerate and the energy gap is relatively large. However, with stronger interaction strengths, $U = 2.0$, we can see that the optimizer struggled. Even though

n	U	δ_0	Δ_{gap}	Q_0	$\overline{ M_{\text{edge}} }$	$\max M_{\text{bulk}} $
2	0	7.64e-05	2	0.00139	1	0
2	0.1	0.00344	2	0.00142	1	0
3	0.1	9.39e-05	1.77	0.0492	0.998	0.000202
4	0.1	0.00423	1.21	0.007	0.986	0.0233

Table 5.2: Summary of local diagnostics for the representative configurations shown in Fig. 5.5. Here δ_0 is the splitting of the lowest even-odd pair, Δ_{gap} is the gap separating this pair from the next excited states, Q_0 is the total charge difference between the parity partners, $\overline{|M_{\text{edge}}|}$ is the mean absolute Majorana polarization on the first and last sites, and $\max |M_{\text{bulk}}|$ is the largest absolute polarization on the interior sites.

ground state degeneracy is highly weighted in the loss function, it did not manage to find a set of parameters where the ground states are close to degenerate. The two lowest states show splitting, and thus these parameters are not ideal for idealizing Majorana modes. This could be due to the fact that the optimization landscape becomes more complex at stronger interactions, and the optimizer may have difficulty finding good solutions in that regime.

Interestingly, the model found a stable set of parameters for all interaction strengths for the three-dot system. Even at $U = 2.0$, the optimizer found a set of parameters where the four energy levels in each parity sector are close to degenerate, although the gap to excited states is smaller than in the non-interacting case. This suggests that for the three-dot system, there may be more flexibility in finding Majorana-like configurations even at stronger interactions, compared to the two-dot and four-dot systems. Due to this, the three-dot system was chosen as the most promising candidate for the braiding analysis in the next part of the thesis.

5.2.2 Local diagnostics and Majorana character

The parity-resolved spectrum shows whether the optimizer has found a low-energy manifold with approximate even-odd degeneracy, but this is not sufficient on its own to identify Majorana-like states. To test whether the optimized configurations also have the expected spatial structure, I therefore examine local diagnostics of the lowest even-odd pair. The two most useful quantities are the site-resolved Majorana polarization and the local charge difference between the parity partners. A convincing Majorana-like configuration should display opposite polarization on the two ends of the chain, weak polarization in the middle, and only a small charge difference between the even and odd states.

Figure 5.5 compares the spatial profiles directly, while Table 5.2 summarizes the corresponding scalar diagnostics. For the two-dot system, the edge polarization is essentially ideal, with $\overline{|M_{\text{edge}}|} \approx 1$ and no bulk contribution by construction. The total charge difference is also very small, $Q_0 \approx 1.4 \times 10^{-3}$, both at $U = 0$ and at $U = 0.1$. These two-dot cases are therefore useful reference points, but because there are no interior sites they do not probe how well the optimizer suppresses Majorana weight in the bulk.

The three-dot configuration at $U = 0.1$ is the clearest interacting example of the expected Majorana pattern. The ground-state splitting is only $\delta_0 = 9.39 \times 10^{-5}$, the excitation gap remains large, $\Delta_{\text{gap}} = 1.77$, and the bulk polarization is almost completely suppressed, with $\max |M_{\text{bulk}}| = 2.02 \times 10^{-4}$. This means that even though the total charge difference is somewhat larger than in the two-dot reference cases, the low-energy pair is still extremely well localized at the ends of the chain. In contrast, the four-dot

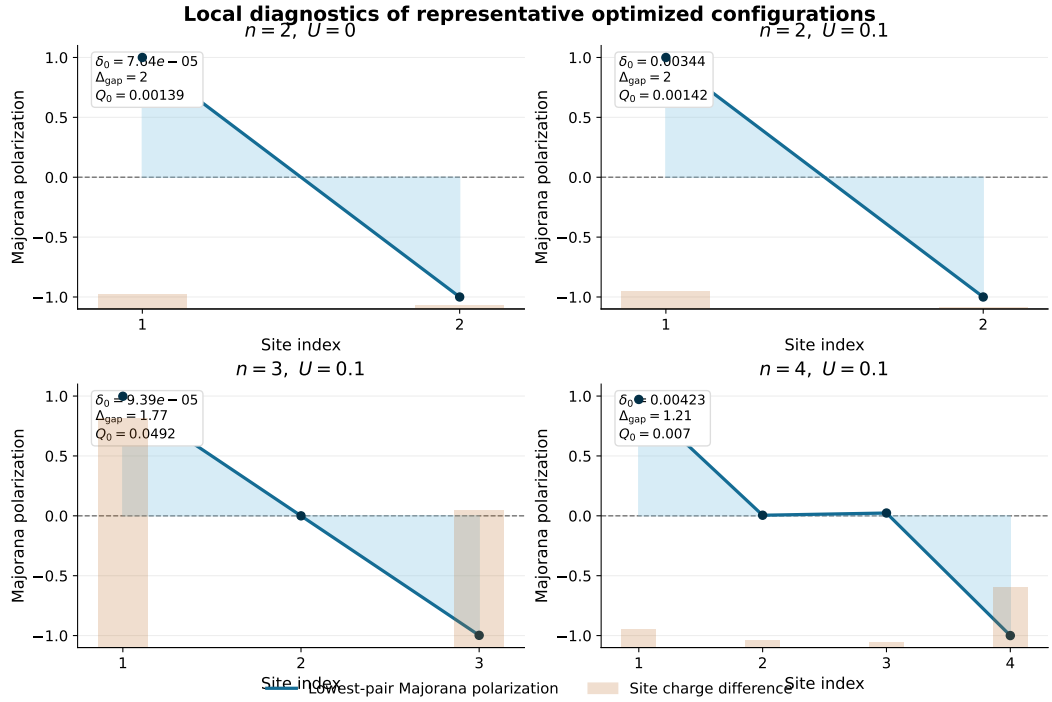


Figure 5.5: Local diagnostics for representative optimized configurations. The blue curve shows the Majorana polarization profile of the lowest even-odd pair, while the orange bars show the local charge difference $|\langle e|n_i|e\rangle - \langle o|n_i|o\rangle|$ on each site. The inset in each panel reports the ground-state splitting δ_0 , the low-energy gap Δ_{gap} , and the total charge difference Q_0 . A clean Majorana-like configuration is expected to have opposite edge polarization, minimal bulk weight, and a small charge imbalance between the parity partners.

configuration at $U = 0.1$ has a comparably strong edge polarization, $|\overline{M_{\text{edge}}}| \approx 0.986$, but the splitting increases to $\delta_0 = 4.23 \times 10^{-3}$ and the bulk polarization rises to $\max |M_{\text{bulk}}| = 2.33 \times 10^{-2}$. The mode is therefore less cleanly separated from the interior of the chain, even though the charge difference is smaller than in the three-dot case.

Taken together, the figure and table show that different diagnostics emphasize different aspects of the optimized states. The four-dot case performs well in terms of charge neutrality, while the three-dot case performs much better in terms of spectral near-degeneracy and suppression of bulk Majorana weight. Since the subsequent braiding analysis requires a low-energy subspace that is both nearly degenerate and spatially well localized, the interacting three-dot system remains the most convincing candidate for the rest of the thesis.

5.3 Braiding Results in the Effective Model

Before interpreting the results of the full microscopic setup, it is useful to summarize the idealized braiding calculation. This section should therefore act as a reference point for what the intended exchange looks like in the simplified four-Majorana model.

Prior to this section we have found a few different optimized n-dot configurations for the Poor Man's Majorana chain, and found a couple of candidates for the braiding analysis. From 5.5, 5.4 and 5.2 we can see that the three-dot system at $U = 0.1$ is the most promising candidate for the braiding analysis, since it has a very small ground-state splitting, a large gap to excited states, and a very clean Majorana polarization profile.

Using the optimized parameters for the three-dot system at $U = 0.1$, we can set up the Hamiltonian and perform the idealized braiding protocol. The hamiltonian is given by equation 5.1, and the optimized parameters are $\Delta_1 = 1.001$, $\Delta_2 = 1.002$, $\epsilon_1 = -0.101$, $\epsilon_2 = -0.607$, and $\epsilon_3 = -0.003$. The hopping amplitudes are fixed to $t_i = 1$ from table 5.1. With these parameters, we diagonalize the Hamiltonian, and sort the 8 different eigenvalues and vectors by energy and by parity. This allows us to construct the Majorana operators γ_1 and γ_2 with any desired number of energy levels included in the sum.

There are three different ways we tested the braiding protocol on our Majorana operators. First was simply to set up the effective Hamiltonian for defined for sweet spot Majorana modes, 2.15.

This Hamiltonian is defined by 4 Majorana operators, and the couplings between them. Since we wanted to use the optimized parameters for the three-dot system, and we want to replicate the setup of 2.7, we define it so that in refrence to the figure 2.7 we get γ_0 and γ_1 from subsystem A, γ_2 and $\tilde{\gamma}_2$ from subsystem B, and γ_3 and $\tilde{\gamma}_3$ from subsystem C. The couplings in between them are created by some simple pulses, which are constructed as a combination of sigmoid functions, which are smooth and have a well defined time derivative. This braiding model assumes a perfect four-Majorana structure, with the parameters at their respective sweet spots, and no additional couplings or perturbations. The purpose of this calculation is to confirm that the intended braiding protocol produces the expected unitary transformation in the idealized limit, and to see if the Majorana operators constructed from the optimized parameters can reproduce this behavior when projected onto the low-energy subspace, $P_g s$, where $P_g s$ is the projection operator onto the ground-state of the full Hamiltonian $H_{full} = H_A + H_B + H_C$.

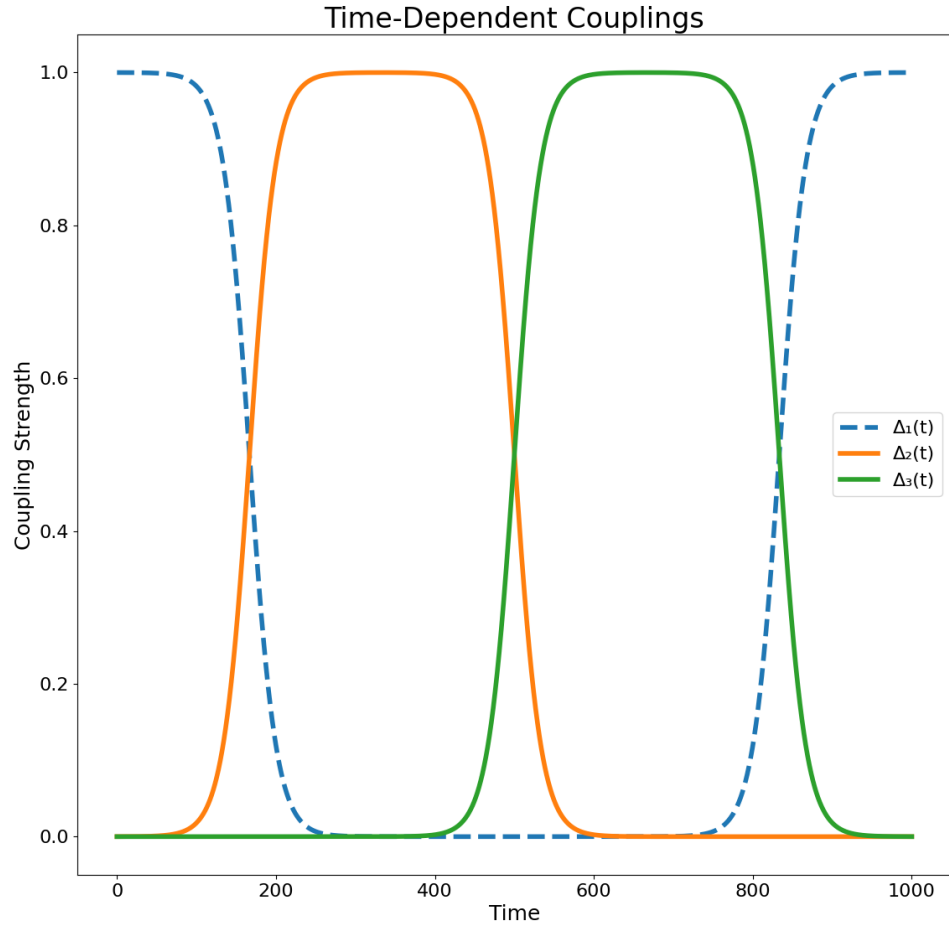


Figure 5.6: Time-dependent couplings used in the idealized braiding protocol. The left panel shows the couplings $\Delta_{ij}(t)$ between the Majorana modes, while the right panel shows the corresponding instantaneous spectrum of the effective Hamiltonian. The low-energy manifold remains approximately degenerate and well separated from excited states throughout the protocol.

The next step is to implement the Junction Hamiltonian, ?? & 2.16, which is defined by the microscopic hopping and pairing terms that couple the different subsystems. This allows us to test whether the optimized microscopic setup can reproduce the intended braiding behavior in its low-energy sector, and whether there is any leakage between different sites during the protocol. This also tells us which one of the γ_2 and $\tilde{\gamma}_2$, as well as which one of the γ_3 and $\tilde{\gamma}_3$ are the ones that are actually participating in the braiding, since the junction Hamiltonian is defined by the microscopic hopping and pairing terms, and not by hand-chosen Majorana bilinears. This method also allows us to explore the weights of the Majoranas during the braiding protocol. What this means is that we can detect if γ_2 and $\tilde{\gamma}_2$ are equally weighted or not. Ideally the weights should heavily favor one of the two, which would indicate that the braiding is happening in a well-defined low-energy subspace. During the protocol, we detected that the Majoranas that were participating in the braiding were $\tilde{\gamma}_2$ and $\tilde{\gamma}_3$, which are the ones that have the largest weight during the protocol. There was little to no leakage between different sites, which is a good sign that the braiding is happening in a well-defined low-energy subspace.

The last check we performed in order is to substitute γ_2 and γ_3 , from 2.15 with $c^\dagger$ and c for the different subsystems. So the effective Hamiltonian becomes:

$$H_{\text{eff}}(t) = i\Delta_{01}(t)\gamma_0\gamma_1 + i\Delta_{2c}(t)\gamma_0(c_B^\dagger + c_B) + i\Delta_{3c}(t)(c_C^\dagger + c_C) \quad (5.2)$$

This is also a more realistic model for the braiding protocol, since it does not assume a perfect four-Majorana structure, and it allows us to test whether the optimized parameters can reproduce the intended braiding behavior even when we include the full microscopic details of the junction couplings. This is also a good check to see if the Majorana operators constructed from the optimized parameters can reproduce the expected unitary transformation when projected onto the low-energy subspace, even when we include the full microscopic details of the junction couplings.

5.3.1 Evolution of the effective spectrum

Figure 5.7 shows the reference braid in the effective four-Majorana model. The three couplings are turned on sequentially, so that the dominant pairing is first between γ_0 and γ_1 , then between γ_0 and γ_2 , and finally between γ_0 and γ_3 before returning to the initial configuration. Throughout this cycle, the low-energy manifold remains essentially four-fold degenerate. The largest instantaneous splitting within the ground-state manifold is only $\max(E_3 - E_0) = 2.28 \times 10^{-12}$, while the minimum gap to the excited states remains open at $\min(E_4 - E_3) = 1.414$.

This is the behavior expected from an ideal adiabatic braid. The low-energy states stay well isolated from the rest of the spectrum, so the Kato transport can be applied without the path ever closing the many-body gap. In this sense, Fig. 5.7 provides a clean numerical benchmark for what the microscopic braid should reproduce as closely as possible.

5.3.2 Berry phase and ideal exchange behavior

Using the Kato formulation of adiabatic transport, I compute the unitary acting on the four-dimensional low-energy subspace after one braid cycle. In the effective model, the transported operator remains highly unitary, with $\|U^\dagger U - I\| = 3.96 \times 10^{-14}$, and the transported projector agrees with the final ground-state subspace up to a transport error of 6.67×10^{-6} . The resulting braid also reproduces the expected single-exchange action

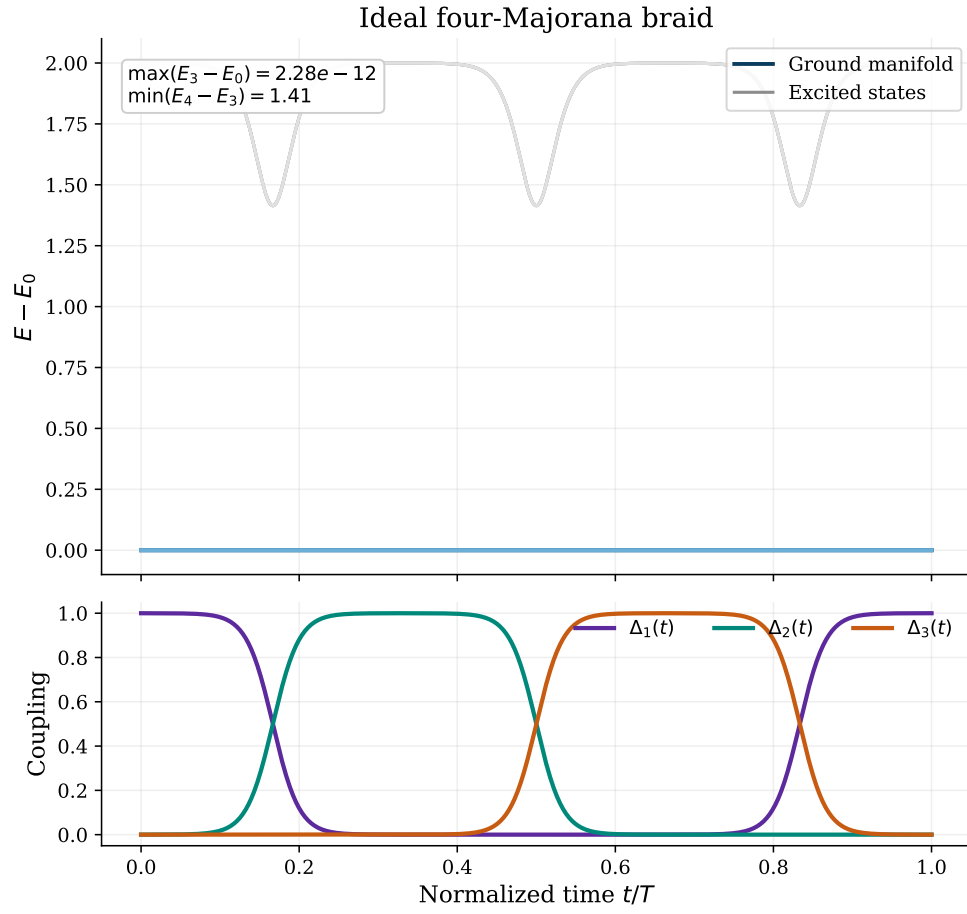


Figure 5.7: Idealized four-Majorana braiding protocol. The top panel shows the instantaneous spectrum of the effective Hamiltonian during the braid, with all energies shifted so that the lowest state remains at zero. The bottom panel shows the three smooth pulse couplings that move the dominant Majorana pairing through the network.

on the Majorana operators, with the largest error in the transformations $\gamma_2 \rightarrow -\gamma_3$ and $\gamma_3 \rightarrow \gamma_2$ equal to 1.33×10^{-5} .

The same conclusion is visible in the parity-resolved gate blocks. After diagonalizing total parity within the ground-state manifold, the odd and even 2×2 blocks agree with the target exchange gate to within 6.53×10^{-8} , while the off-block leakage is only 1.02×10^{-11} . The effective model therefore behaves as a nearly ideal non-Abelian exchange, and these numbers serve as a reference point for the more realistic projected calculations below.

5.4 Braiding Results for Optimized Majorana Modes

This section should present the main braiding results of the thesis: whether the optimized microscopic system can reproduce the exchange behavior suggested by the ideal model.

5.4.1 Projected microscopic braid

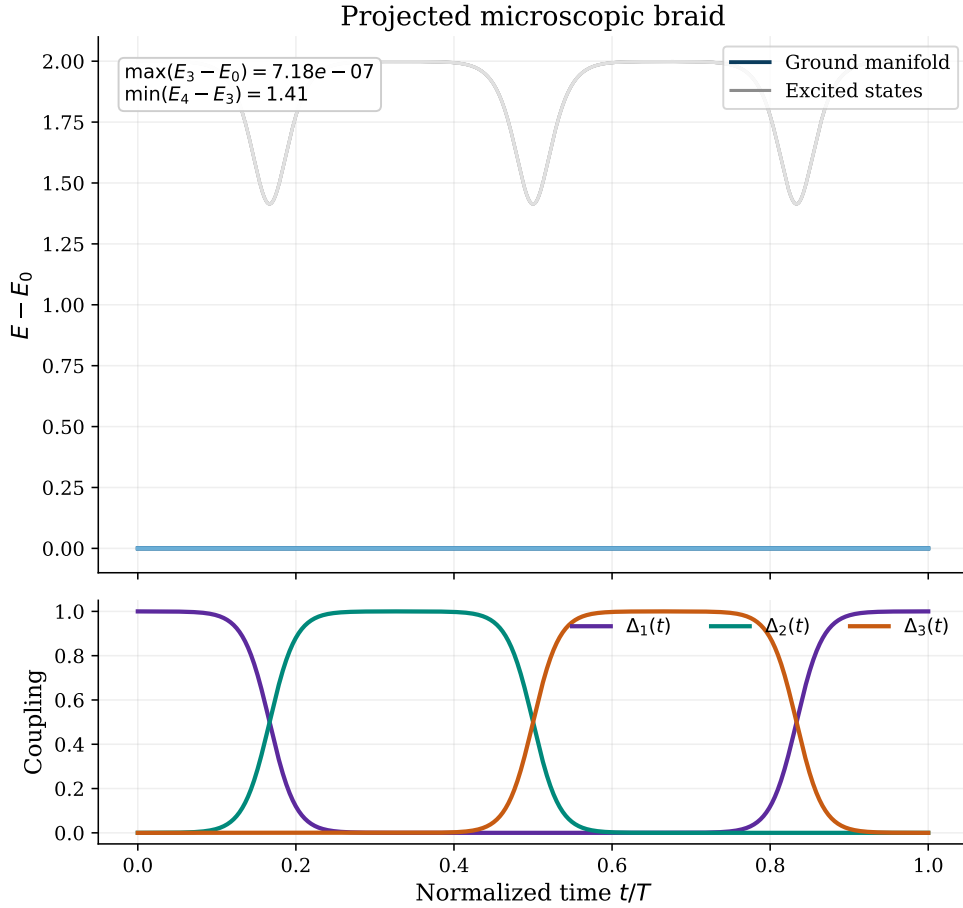


Figure 5.8: Braiding protocol after replacing the idealized couplings by projected microscopic junction operators. The spectral evolution remains almost identical to the effective-model benchmark, with a well-isolated low-energy manifold throughout the cycle.

After replacing the hand-chosen couplings of the toy model by projected microscopic junction operators, the braid remains remarkably close to the ideal reference. Figure 5.8

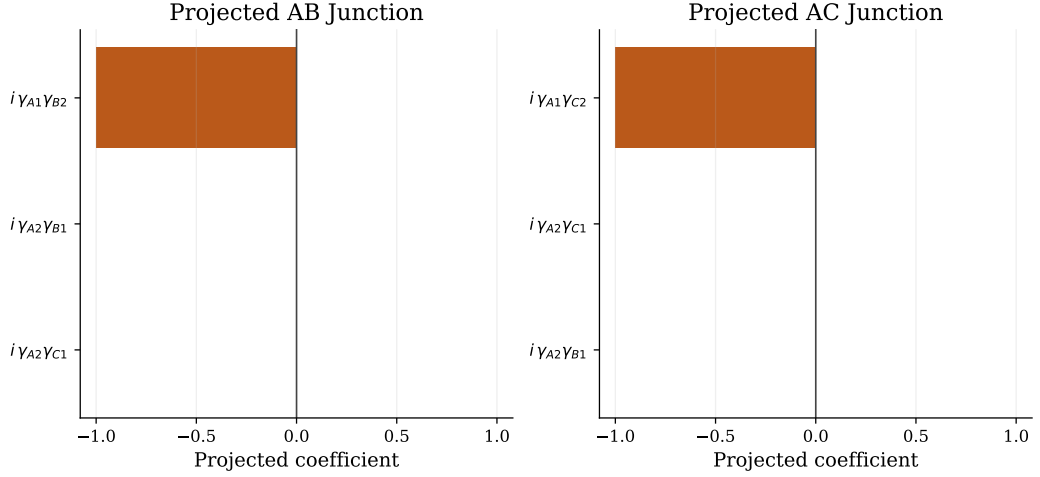


Figure 5.9: Dominant Majorana-bilinear components of the projected microscopic junction operators. The AB junction is almost entirely $i\gamma_{A1}\gamma_{B2}$, while the AC junction is almost entirely $i\gamma_{A1}\gamma_{C2}$. All subleading bilinears are strongly suppressed.

shows that the four-dimensional low-energy manifold is still well separated from higher states throughout the cycle. The maximum splitting inside the ground-state manifold increases only to $\max(E_3 - E_0) = 7.18 \times 10^{-7}$, while the minimum gap remains large, $\min(E_4 - E_3) = 1.413$. The projected microscopic protocol therefore preserves the same basic adiabatic structure as the idealized model.

The junction decomposition in Fig. 5.9 makes the physical meaning of this agreement more transparent. The projected AB coupling is dominated by the bilinear $i\gamma_{A1}\gamma_{B2}$ with coefficient -0.999 , and the projected AC coupling is dominated by $i\gamma_{A1}\gamma_{C2}$ with the same coefficient. This identifies γ_{B2} and γ_{C2} as the Majoranas that actually participate in the microscopic braid, while γ_{A1} acts as the shared Majorana through which the exchange is mediated. The microscopic model is therefore not just qualitatively similar to the toy model: after projection, the junction operators line up almost perfectly with the intended effective bilinears.

5.4.2 Gate-validation checks

Model	$\max(E_3 - E_0)$	$\min(E_4 - E_3)$	Max 1x err.	Leakage	Odd err.	Even err.
Ideal effective model	2.275e-12	1.414	1.334e-05	1.016e-11	6.535e-08	6.535e-08
Projected microscopic junction model	7.175e-07	1.413	1.334e-05	8.844e-16	6.538e-08	6.538e-08
Projected local-operator model	4.663e-15	1.414	1.334e-05	7.287e-16	6.537e-08	6.537e-08

Table 5.3: Comparison of the main braid diagnostics for the ideal effective model, the projected microscopic junction model, and the alternative projected local-operator model. Here $\max(E_3 - E_0)$ measures the largest instantaneous splitting inside the four-dimensional low-energy manifold, $\min(E_4 - E_3)$ is the minimum gap to the excited sector, “Max 1x err.” is the largest deviation in the expected single-exchange Majorana map, “Leakage” is the off-block norm in the parity basis, and the last two columns compare the odd and even parity blocks with the target exchange gate.

Table 5.3 shows that the microscopic braid reproduces the ideal benchmark extremely well. The largest single-exchange error stays at 1.33×10^{-5} in all three models, and the odd and even parity blocks continue to match the target gate to approximately 6.54×10^{-8} . In other words, once the microscopic junction operators are projected

Projected operator	Dominant Majorana	Coefficient
$\chi_{B1,y}$	γ_{B2}	-1.000
$\chi_{B2,y}$	γ_{B2}	1.000
$\chi_{C1,y}$	γ_{C2}	-1.000
$\chi_{C2,y}$	γ_{C2}	1.000
Pairwise relation errors		
$\chi_{B1,y} \approx -\chi_{B2,y}$	opposite sign	8.969e-14
$\chi_{C1,y} \approx -\chi_{C2,y}$	opposite sign	5.787e-14

Table 5.4: Projected local-site operators used in the alternative braid model. The table shows that the y -type local operators on the first and last sites of subsystems B and C collapse onto the same low-energy Majorana channels, γ_{B2} and γ_{C2} , up to an overall sign.

into the low-energy manifold, they generate essentially the same braid unitary as the ideal four-Majorana model. The projected local-operator version produces the same conclusion, which makes it a useful consistency check rather than an independent competing protocol.

Table 5.4 also explains why the local-site version in `braiding_model_ad2.py` appears insensitive to whether the first or last site of subsystem B or C is used. After projection, both $\chi_{B1,y}$ and $\chi_{B2,y}$ reduce to γ_{B2} up to opposite sign, and similarly both $\chi_{C1,y}$ and $\chi_{C2,y}$ reduce to γ_{C2} . The different microscopic site choices therefore feed into the same low-energy braid channel rather than producing genuinely different braiding operations. This makes the apparent insensitivity of the projected local model a physical consequence of the low-energy projection, not a numerical inconsistency in the implementation.

5.5 Braiding in Higher Energy Levels

The previous braiding checks focused on the lowest projected manifold. To test how much of that agreement survives once excited sectors are retained, I repeated the braid diagnostics for projection dimensions $\dim P = 8, 32, 56, 80, 256$, and 512, and for interaction strengths $U = 0, 0.1$, and 2.0. For each projected Hamiltonian I choose the transported band from the largest persistent spectral gap along the braid path. This separates the projection dimension, which controls how much Hilbert space is retained, from the transported dimension $\dim T$, which controls which isolated band is actually braided.

The physical junction is taken to be the projected local minus operator, $i(c^\dagger - c)$, on the inner B and C sites. A subtle point is that there are now two different effective braid constructions. In the first, which I call the energy-level braid, the operators are built from paired even and odd eigenstates of the subsystem Hamiltonians. In the second, which I call the local-operator braid, the junction is built from the projected microscopic operators $Pi(c^\dagger - c)P$. The former tests whether the spectrum supports a clean exchange channel, while the latter tests whether that channel is accessed by the expected local microscopic coupling.

Table 5.5 shows that the corrected diagonal energy-level construction gives a very clean braid target. The important comparison is therefore not simply whether the braid unitary is accurate, but which effective operator it is accurate with respect to. The energy-level construction is adapted to the many-body spectrum, while the local-operator construction is constrained to the projected microscopic endpoint operator.

Figures 5.10 to 5.14 show the main distinction between the two braid constructions. The energy-level braid is the cleaner effective operation, because it is built directly from

U	$\dim P$	$\dim T$	Phys. vs. γ target	Phys. vs. Γ^P target	Target mismatch	Γ^P algebra
0	8	4	1.15e-06	1.15e-06	2.54e-16	1.65e-15
0	32	8	1.15e-06	9.80e-05	9.86e-05	1.08e-04
0	56	8	1.15e-06	1.36e-04	1.37e-04	7.57e-05
0	80	12	1.15e-06	1.10e-04	1.11e-04	1.06e-04
0	256	16	1.15e-06	1.32e-04	1.32e-04	8.44e-05
0	512	16	1.15e-06	1.52e-04	1.53e-04	2.99e-15
0.1	8	4	9.46e-07	9.46e-07	7.85e-17	3.20e-15
0.1	32	8	1.59e-04	1.59e-04	3.68e-07	3.67e-04
0.1	56	12	1.23e-04	2.67e-04	2.90e-04	5.90e-04
0.1	80	12	4.18e-05	2.84e-04	2.73e-04	5.82e-04
0.1	256	16	1.58e-04	4.24e-04	5.16e-04	5.99e-04
0.1	512	16	4.20e-04	5.93e-04	9.37e-04	3.22e-15
2	8	4	1.02e-06	1.02e-06	2.62e-16	1.95e-15
2	32	8	2.35e-02	2.10e-02	5.43e-03	5.77e-02
2	56	12	8.34e-02	5.10e-02	3.43e-02	1.75e-01
2	80	12	7.81e-02	5.37e-02	2.69e-02	1.53e-01
2	256	16	1.76e-01	4.09e-02	1.38e-01	1.50e-01
2	512	16	3.09e-01	2.04e-02	3.21e-01	3.15e-15

Table 5.5: Target diagnostic for the corrected higher-energy braid scan with $\dim P = 8, 32, 56, 80, 256, 512$. All errors are normalized in the transported band. “Phys. vs. γ target” compares the local-operator braid to $\exp(-\pi\gamma_2\gamma_3/4)$, where γ_2 and γ_3 are constructed from diagonal even-odd energy-level pairs. “Phys. vs. Γ^P target” compares the same braid to $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$, where Γ_2^P and Γ_3^P are projected local Majorana operators. “Target mismatch” directly compares these two effective target gates.

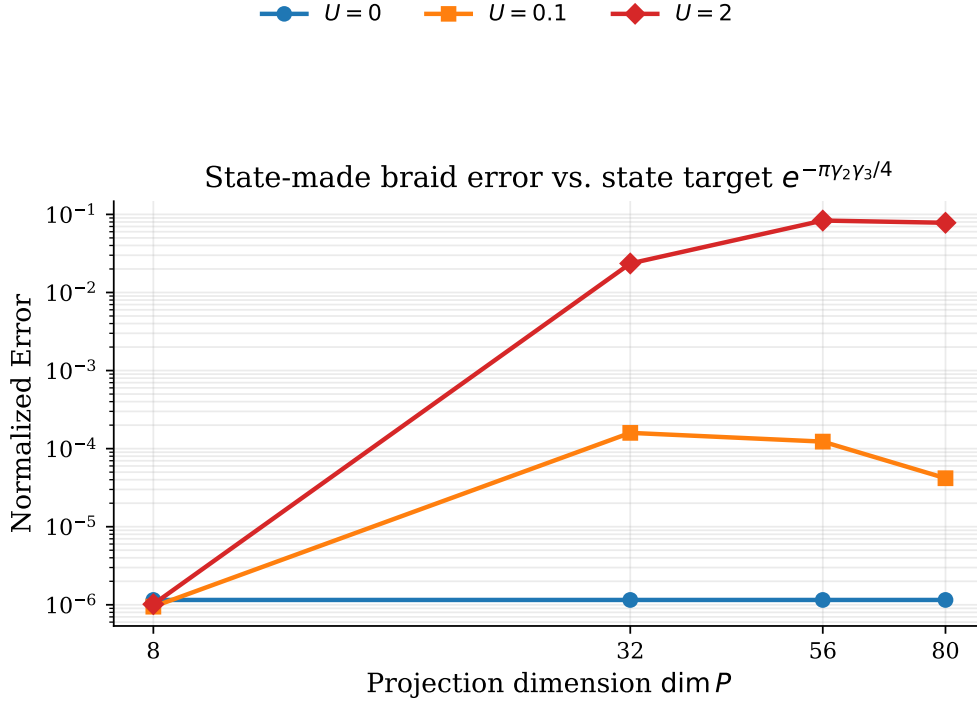


Figure 5.10: Energy-level braid error up to $\dim P = 80$. The braid operators are constructed from diagonal even-odd energy-level pairs of the subsystem Hamiltonians and are compared with the corresponding energy-level target $\exp(-\pi\gamma_2\gamma_3/4)$. This diagnostic tests whether the retained many-body spectrum supports a clean effective exchange channel.

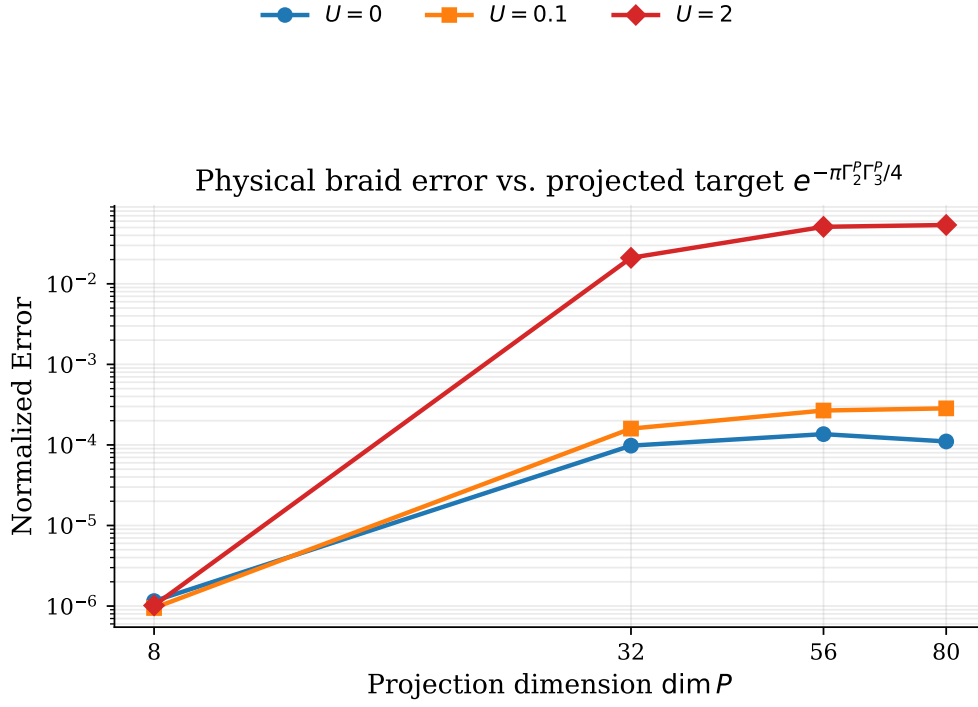


Figure 5.11: Local-operator braid error up to $\dim P = 80$. Here the braid is generated and judged using the projected local endpoint operators $Pi(c^\dagger - c)P$. This is the more microscopic diagnostic, because it tests how well the exchange channel is accessed by a local physical coupling.

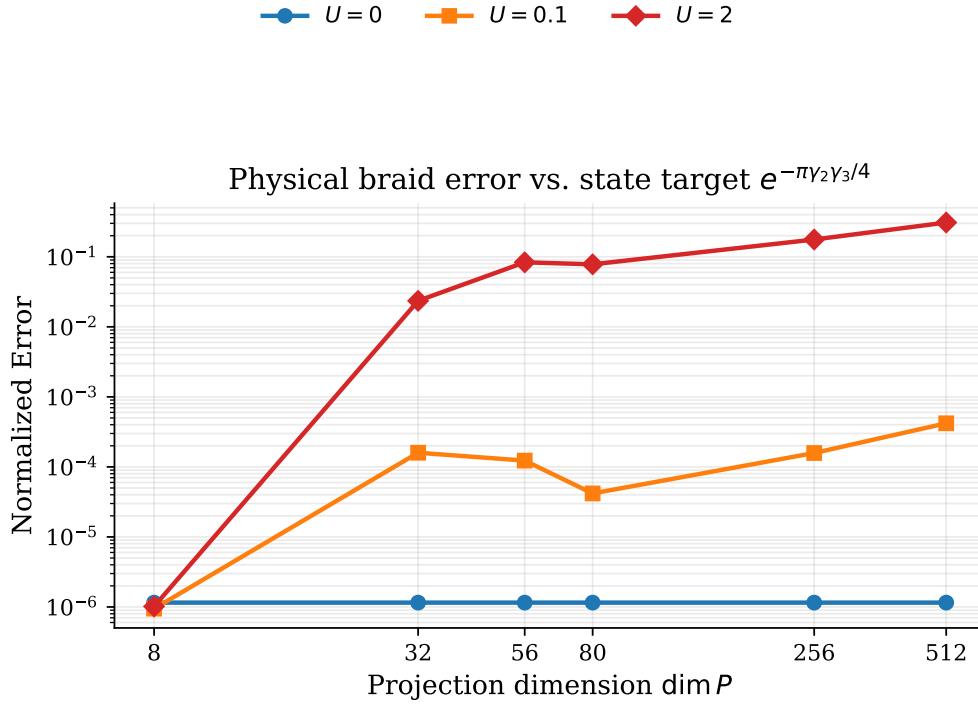


Figure 5.12: Energy-level braid error up to $\dim P = 512$. Extending the projection dimension tests whether the same eigenstate-pair construction remains stable once higher-energy sectors are retained. The low non-interacting error shows that the energy-level construction continues to capture a clean exchange channel even in the full projected Hilbert space.

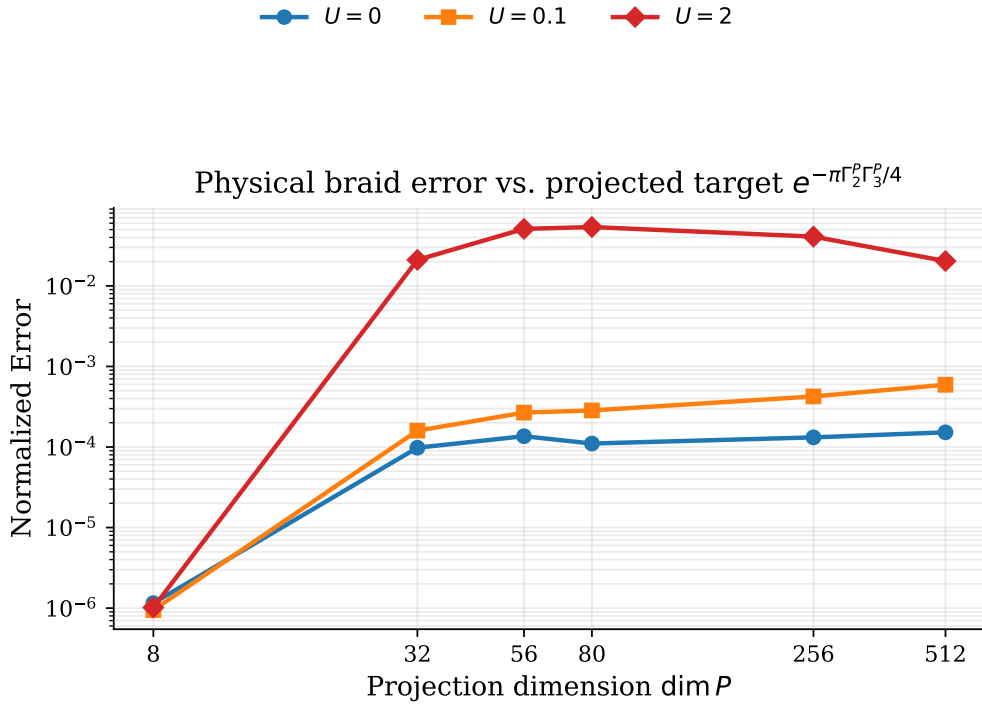


Figure 5.13: Local-operator braid error up to $\dim P = 512$, evaluated against the projected local target $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$. The local-operator braid remains accurate, but it is generally less clean than the energy-level braid, indicating that the best effective exchange channel in the spectrum is not identical to the channel selected by the strictly local projected operator.

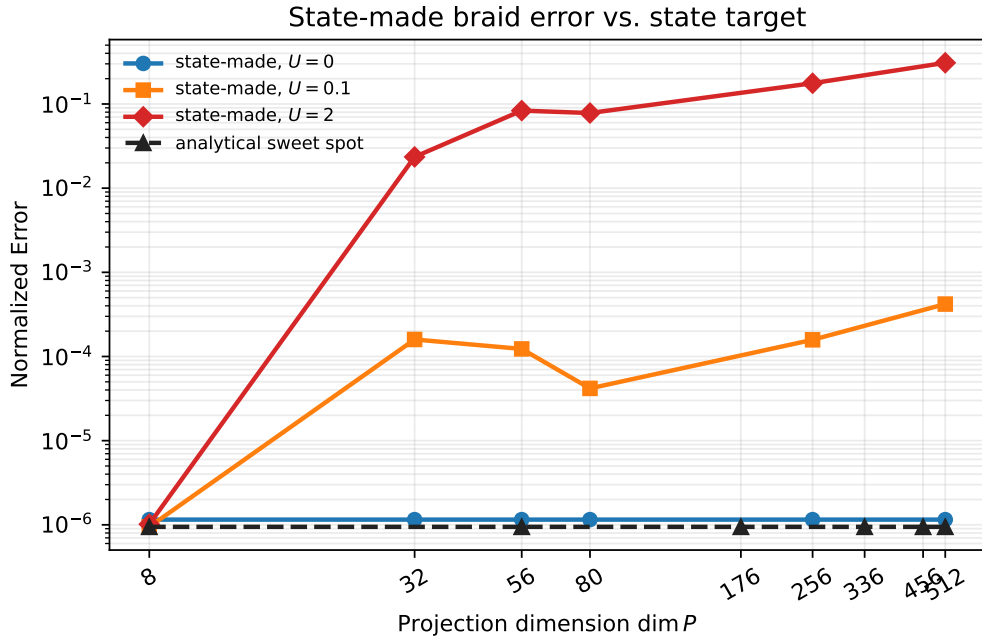


Figure 5.14: Energy-level braid error up to $\dim P = 512$, with the analytical non-interacting sweet-spot braid included as a reference. The analytical curve is evaluated only at projection dimensions that complete exact degeneracy blocks. The close agreement confirms that the diagonal energy-level construction reproduces the expected non-interacting exchange channel when the projection is chosen consistently.

the paired eigenstates of the optimized Hamiltonians. The local-operator braid is more physically constrained, because it asks what is generated by the projected microscopic endpoint operator. The fact that the energy-level braid performs better should therefore be interpreted as evidence that the many-body spectrum contains a very clean exchange channel, not as automatic proof that the corresponding operator is perfectly localized. The local-operator results provide the complementary localization and implementability check.

Chapter 6

Discussion

This section interprets the numerical results in relation to the main goal of the thesis: to test whether optimized interacting quantum-dot chains can support Majorana-like low-energy physics and whether that physics remains useful for braiding. The results suggest that the optimization procedure does more than recover known non-interacting limits. It also identifies nontrivial inhomogeneous parameter profiles that still preserve the low-energy signatures needed for a Majorana interpretation.

6.1 What the optimized parameter profiles imply

One of the clearest trends in the optimization results is that the best interacting configurations do not remain close to a simple homogeneous sweet-spot picture. For the two-dot system, the optimizer reproduces the expected non-interacting limit at $U = 0$, which is an important sanity check. As the interaction strength increases, however, the loss grows and the optimized parameters are pushed away from that point. This should not be interpreted as meaning that interacting two-dot systems cannot retain any useful degeneracy. Rather, the more precise conclusion is that the loss function used here targets approximate degeneracy across all parity-resolved sectors, while the familiar interacting two-dot picture only guarantees the ground-state degeneracy.

For the three- and four-dot systems, the optimizer repeatedly favors strong inhomogeneity in the middle of the chain. In particular, the middle-site chemical potentials become much larger in magnitude than those on the edges, and in the four-dot system the central pairing amplitudes are often enhanced as well. A natural interpretation is that the optimizer is trying to reduce hybridization between the edge modes by engineering an effective barrier or filtering region in the center of the chain. This interpretation is consistent with the Majorana-polarization profiles and with the parity-resolved spectra, both of which indicate that low-energy weight remains concentrated near the outer sites in the best weakly interacting configurations.

At the same time, this trend should be treated with some caution. Because the hopping amplitudes were fixed to $t_i = 1$, the optimizer could only respond to stronger interactions by reshaping Δ_i and ϵ_i . The repeated appearance of large middle-site potentials may therefore be a genuine physical feature, but it may also partly reflect the restrictions built into the ansatz. A useful next step would be to test whether the same tendency appears when the hopping amplitudes are allowed to vary under a constraint that prevents trivial global rescaling.

6.2 What the braiding results imply

The braiding analysis shows a clear separation between low-energy success and higher-energy breakdown. In the lowest projected manifold, the effective Majorana braid and the projected microscopic junction braid agree extremely well. This is a strong result, because it shows that the optimized microscopic model is not merely producing a vaguely similar spectrum. After projection, the relevant junction operators line up with the intended effective Majorana bilinears closely enough to reproduce the expected exchange gate with very small error.

The higher-energy projection scan sharpens this conclusion. For $U = 0$ and $U = 0.1$, the physical projected-junction braid remains close to the ideal control even as additional excited blocks are included. For $U = 2$, by contrast, the target-gate and exchange-channel errors increase rapidly once the projection space extends beyond the ground-state manifold. The important point is that this deterioration is not uniform across every projected operator. The dominant breakdown occurs in the exchanged channels, while the spectator channels remain much more stable. Physically, this suggests that strong interactions first destroy the clean exchange structure rather than the entire low-energy algebra at once.

The basis-diagnostic plots are also important for interpreting the braiding data correctly. Some of the very large error values that appear when the physical evolution is scored against the ideal projected-Majorana basis do not correspond to equally large physical gate failure. They mainly indicate that, in enlarged projection spaces, the projected microscopic operators are no longer identical to the ideal projected Majoranas used in the control calculation. This distinction matters for the interpretation of the results: the low-energy braid can remain physically meaningful even when comparisons to the ideal basis become misleadingly large.

Taken together, the braiding results support a practical design principle for the optimized PMM chains. Useful braiding behavior appears to require that the tunneling and pairing scales remain large compared with the interaction scale, so that the exchange is protected within a well-isolated low-energy manifold. In that sense, the results do not show that interactions are incompatible with braiding. Instead, they show that interactions strongly constrain the regime in which the microscopic model still realizes the intended Majorana exchange.

6.3 Limitations and outlook

The main limitation of the present study is that the optimization procedure does not locate exact sweet spots; it finds numerically favorable minima of the chosen loss function. In a complicated loss landscape, a value such as 10^{-3} may look essentially optimal even when a better solution exists elsewhere. Because all later diagnostics build on the optimized parameter sets, the braiding results should therefore be interpreted as evidence for approximate and practically useful Majorana-like behavior, not as proof of exact topological protection.

A second limitation is that the present braiding analysis is still highly controlled. The clean agreement found in the lowest manifold relies on projection to a small low-energy space and on an idealized, disorder-free setting. The study does not yet include disorder, quasiparticle poisoning, nonadiabatic driving errors, or explicit coupling to thermal environments. Those effects are precisely the kinds of ingredients that would determine

whether the optimized configurations remain useful in a more realistic device-level setting.

Several natural extensions follow from this. On the optimization side, it would be valuable to test more global search strategies, alternative loss functions, and less restrictive parameter ansätze. On the braiding side, it would be interesting to study how disorder and noise affect the projected junction operators, and whether the weakly interacting configurations remain robust when the protocol is pushed beyond the ideal adiabatic limit. The higher-energy results also point to a more specific open question: to what extent can one deliberately optimize not only the ground-state manifold, but also nearby excited manifolds, in order to make the exchange protocol more tolerant to thermal occupation?

Chapter 7

Conclusion

This thesis investigated whether interacting quantum-dot chains can be optimized to realize Majorana-like low-energy states and whether those optimized configurations support a useful braiding protocol. The optimization reproduces the known two-dot non-interacting limit and, for longer chains, finds more inhomogeneous parameter profiles with strong central structure in the on-site energies and pairing amplitudes. These configurations still display the main low-energy signatures targeted in the study, including near even-odd degeneracy, parity-resolved low-energy structure, and edge-weighted Majorana polarization.

The braiding analysis shows that the agreement between the ideal effective model and the microscopic projected model is excellent inside the lowest projected manifold. For weak interactions, especially at $U = 0.1$, this agreement remains good even when additional low-energy blocks are included. For stronger interactions, the braid deteriorates first in the exchanged channels and then more broadly as higher-energy sectors are added. The results therefore point to a clear physical conclusion: useful braiding is achievable in the optimized model, but only when the relevant pairing and tunneling scales remain sufficiently large compared with the interaction strength and the dynamics stay confined to a well-isolated low-energy subspace.

Overall, the thesis supports the view that optimized PMM chains can provide a meaningful microscopic route toward Majorana-based braiding, while also showing that the quality of the braid is highly sensitive to how interactions reshape the low-energy manifold. The most natural next steps are to improve the optimization strategy, test less restricted parameter families, and include disorder, noise, and finite-temperature effects in the braiding analysis.

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